

Dynamic Fluid Policy for Online Resource Allocation: Logarithmic Correction and Constant Optimality Gap

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Abstract

We study an online resource allocation problem with finitely many actions, random rewards, and bounded multidimensional random resource consumptions. We analyze the classic dynamic fluid policy, which re-solves a fluid relaxation at every state and uses the resulting optimal dual variable to approximate the continuation value. We show that, under standard non-degeneracy conditions, the value functions of the dynamic fluid policy and the optimal dynamic policy characterized by the Bellman equation admit the same asymptotic expansion: the fluid benchmark minus a common logarithmic correction term plus a uniformly bounded, policy-dependent remainder. Moreover, we explicitly characterize the coefficient of the logarithmic term. Consequently, the logarithmic term is a common correction to the fluid benchmark, and the dynamic fluid policy achieves a horizon-uniform constant optimality gap relative to the optimal online policy.

Subject classifications: Online resource allocation, dynamic fluid policy, logarithmic correction, constant optimality gap.

1 Introduction

In this paper, we study an online resource allocation problem, in which a decision maker manages multiple types of resources with limited inventory over a finite time horizon n . In each period, an opportunity arrives according to a known distribution, and the decision maker selects an action from a finite set. Each action consumes a combination of resources and generates a reward. The objective is to select a sequence of actions that maximizes the expected total reward subject to the resource constraints. This problem can be formulated as a finite-horizon discrete-time dynamic program, with the state given by the vector of remaining resources available.

Computing the optimal policy can be intractable due to the curse of dimensionality. Thus, much of the literature focuses on developing simple and computationally tractable policies with strong performance guarantees. A common and central heuristic is based on the deterministic fluid relaxation and the associated dual prices. The deterministic fluid relaxation simplifies the stochastic optimization problem by requiring that the capacity constraints hold only in expectation, and its optimal dual variables assign a price to each resource. In this paper, we focus on a classic policy, which we call the *dynamic fluid policy*. This policy re-solves the fluid relaxation and its optimal dual prices at each decision epoch, and selects the feasible action with the largest reward net of resource costs. This policy has been widely studied in the literature, including Lueker (1998), Bray (2025), and Chen and Wang (2025), and many other policies can be viewed as variants of it. Our question is how much reward the dynamic fluid policy loses relative to the optimal online policy characterized by the Bellman equation.

Instead of directly comparing a candidate policy with the optimal online policy, the literature primarily evaluates its performance against an optimal offline benchmark. The optimal offline benchmark allows the decision maker to observe the entire arrival sequence before selecting a sequence of actions and serves as an upper bound on the performance of the optimal online policy. For online resource allocation problems, when rewards and resource consumptions have finite support, a constant loss relative to this benchmark can be achieved (e.g., Arlotto and Gurvich 2019, Bumpensanti and Wang 2020, Vera and Banerjee 2021, and Vera et al. 2021). When rewards or resource consumptions are continuously distributed, under certain nondegeneracy conditions, the dynamic fluid policy and its variants typically incur a $\Theta(\ln n)$ loss relative to this benchmark. Moreover, this logarithmic loss is order-optimal, as no non-anticipative policy can achieve a loss of a smaller order (Bray 2025).

To our knowledge, the only exception is Wang and Wang (2022). Wang and Wang (2022) study

a different dynamic pricing problem and analyze a re-solving heuristic that is analogous to our dynamic fluid policy. Under their non-degeneracy conditions, the re-solving heuristic achieves a constant loss relative to the optimal online policy, whereas the fluid benchmark can exceed the optimal online value by $\Omega(\log T)$.

Like Wang and Wang (2022), we measure the performance loss of the dynamic fluid policy relative to the optimal online policy, for the resource allocation problem. However, our performance characterization is more refined. Under standard non-degeneracy conditions, we show that the value functions of both the dynamic fluid policy and the optimal online policy share the same expansion: fluid benchmark minus the same logarithmic correction term $\Gamma \log n$, up to a uniformly bounded remainder (Theorem 4.1). Consequently, the dynamic fluid policy incurs a constant optimality loss relative to the optimal online policy, in contrast to the order-optimal $\Theta(\ln n)$ loss relative to the offline optimal benchmark established by Bray (2025). The logarithmic correction $\Gamma \log n$ comes from the interaction between random resource consumption and the concavity of the fluid value function.

The proof of Theorem 4.1 is novel and is illustrated in Section 6. First, in Section 6.1, we motivate and construct the corrected fluid reference function, which subtracts the cumulative correction term $\Gamma \log n$ and a uniformly bounded remainder from the fluid benchmark. As a technical step, we also smooth the correction by convolution, altering its value by only a bounded amount, to ensure that its first- and second-order derivatives are suitably bounded for the subsequent analysis. Finally, we show that, within a local region containing the initial state, the corrected reference approximately satisfies the Bellman equation of the dynamic fluid policy (Proposition 6.1).

Then, in Section 6.2, we compare the value functions of the dynamic fluid policy and the optimal online policy with the corrected fluid reference and demonstrate that they both deviate from the corrected reference by at most a constant within a local region containing the initial state. The two comparisons use different arguments.

For the dynamic fluid policy, we show that the normalized state (i.e., the amount of inventory per remaining period) evolves as a martingale. Since the corrected reference approximately satisfies the Bellman equation of the dynamic fluid policy within the local region, the difference between the policy's value function and the corrected reference can be decomposed into the cumulative residuals up to the time when the normalized state exits the local region and a terminal error proportional to the remaining horizon at the time of exit. The cumulative residuals are uniformly bounded. Moreover, using a martingale concentration inequality, we show that the expected remaining horizon at the time of exit is also bounded by a constant. Consequently, the difference between the policy's

value function and the corrected reference is bounded by a constant (Proposition 6.2).

Unlike the dynamic fluid policy, the normalized state need not evolve as a martingale under the optimal online policy, so the same stopping-time and martingale-concentration argument is no longer available. For the optimal online policy, we instead use the corrected reference to construct a local supersolution to its Bellman equation (i.e., $T^*u_{n-1} \leq u_n$) within the local region. We then patch this local supersolution to obtain a global supersolution, which provides an upper bound on the optimal value function up to an additive constant. Finally, we show that, within the local region containing the initial state, the global supersolution exceeds the corrected reference by at most a constant. This leads to a constant upper bound on the difference between the optimal value function and the corrected reference (Proposition 6.3).

In Section 5, we apply our results to several examples. For the one-dimensional stochastic knapsack problem, Lueker (1998) derives the same performance expansion with an $o(\log n)$ remainder. As a direct implication of our results, we sharpen the remainder to $O(1)$ and extend the expansion to more general online resource allocation problems. Bray (2025) establishes a tight $\Theta(\ln n)$ performance bound relative to the offline optimum under standard non-degeneracy conditions for the online linear programming problem studied therein. Our model is more general. Under an additional local Hölder continuity of the second-order derivatives of the fluid relaxation value function, we show that the dynamic fluid policy and the optimal online policy admit the performance expansion up to an $O(1)$ remainder, and thus their performance gap is no larger than a constant.

The rest of the paper is organized as follows. Section 1.1 reviews the related literature. Section 2 formulates the problem and introduces the dynamic fluid policy. Section 3 states our non-degeneracy conditions. Section 4 presents the main theorem and interprets the coefficient of the logarithmic correction term. Section 5 illustrates several applications. Section 6 presents the proof of the main theorem. Finally, Section 7 concludes the paper.

1.1 Related Literature

There is a rich and well-developed literature on online resource allocation in the operations research and computer science communities, and numerous variants arise in a broad range of applications (Balseiro et al. 2023). In general, online resource allocation problems can be formulated as dynamic programs, with optimal policies characterized by the corresponding Bellman equation. However, computing these policies can be intractable due to the curse of dimensionality, motivating the development of simple and computationally tractable policies with strong performance guarantees.

Fluid Approximations and Re-solving A widely used approach to both policy design and performance analysis is based on deterministic fluid relaxations and the associated dual prices. Talluri and van Ryzin (1998) construct static bid-price controls based on deterministic fluid relaxation and establish an $O(\sqrt{T})$ loss relative to the fluid relaxation when demand and capacity grow proportionally. Much of the subsequent literature improves upon this baseline by periodically re-solving the fluid relaxation as time and remaining capacity evolve, thereby allowing the dual prices to adapt to the state. When rewards and resource consumptions have finite support, multiple papers (e.g., Arlotto and Gurvich 2019, Bumpensanti and Wang 2020, Vera and Banerjee 2021, and Vera et al. 2021) establish constant loss for such re-solving policies relative to the offline optimal benchmark.

Fluid Policies under Non-Degeneracy Conditions When rewards or consumptions are continuously distributed, a constant loss relative to the fluid or offline optimal benchmark is no longer attainable. Nevertheless, logarithmic regret relative to these benchmarks can be achieved under suitable non-degeneracy or stability conditions on the deterministic fluid relaxation. These conditions appear in various forms and typically require local smoothness, strict complementarity, stability of the binding resource constraints, or uniqueness of the optimal dual solution. Li and Ye (2021) establish $O(\ln n \ln \ln n)$ regret relative to the offline optimal benchmark under their non-degeneracy conditions. Under different sets of non-degeneracy assumptions, Balseiro et al. (2023) obtain an $O(\ln n)$ loss relative to the fluid relaxation, improving to $O(1)$ in locally affine cases, and Bray (2025) establish a tight $\Theta(\ln n)$ regret relative to the offline optimal benchmark. Finally, Chen and Wang (2025) impose regularity conditions on the request distributions rather than on the deterministic fluid relaxation. Their regularity conditions imply uniqueness of the optimal dual solution but allow primal degeneracy.

Our model is more general than the known distribution setting in Bray (2025). Moreover, their non-degeneracy assumptions imply ours, except that we additionally require local Hölder continuity of the second-order derivatives of the fluid value function. Under our non-degeneracy conditions, we show that the dynamic fluid policy and the optimal online policy admit the same fluid-minus-logarithmic expansion up to an $O(1)$ remainder. Consequently, the dynamic fluid policy incurs only a constant loss relative to the optimal online policy. See Section 5.3 for further discussion.

Comparison with Wang and Wang (2022) Our work is also closely related to Wang and Wang (2022), who study a classic dynamic pricing problem and analyze a re-solving heuristic that is

analogous to our dynamic fluid policy. Under their non-degeneracy conditions, they show that their re-solving heuristic achieves an $O(1)$ gap relative to the optimal online policy, whereas the fluid benchmark can exceed the optimal online value by $\Omega(\log T)$. We consider a distinct resource allocation problem and obtain a more refined performance characterization: the dynamic fluid policy and the optimal online policy admit the same expansion, $n\phi^F - \Gamma \log n + O(1)$, from which an $O(1)$ optimality gap follows directly.

Beyond studying different problems, the analytical approaches are also substantially different. Wang and Wang (2022) directly compare the stochastic inventory trajectories under the re-solving and optimal online policies using a stopping-time argument. In contrast, we compare the value functions of the dynamic fluid policy and the optimal online policy with a common corrected fluid reference, instead of coupling the trajectories of the two policies. For the dynamic fluid policy, we telescope summable one-period residuals up to a stopping time. For the optimal online policy, we construct a Bellman supersolution to bound the optimal online value.

Beyond Non-Degeneracy Conditions Recently, multiple papers on online resource allocation have relaxed various non-degeneracy assumptions. Besbes et al. (2024) allow gaps in the support and low probability mass around these gaps. Jiang et al. (2025) study the network revenue management problem with discrete resource consumption and continuously distributed rewards without imposing non-degeneracy. Zhang (2026a) consider settings in which both rewards and resource consumption are continuously distributed and allow both primal degeneracy and dual non-uniqueness. Zhang (2026b) develop a Bellman certificate approach to derive lower bounds on the gap between the optimal offline and online benchmarks in the multi-secretary setting without non-degeneracy assumptions. These papers typically use the offline optimum as the benchmark. Extending our analysis and performance guarantees to more general settings without non-degeneracy requirements is a promising direction for future research.

1.2 Notation and Terminology

For any integer $m \in \mathbb{N}_+$, let $[m] \triangleq \{1, 2, \dots, m\}$ denote the set of integers from 1 to m . For any real number $x \in \mathbb{R}$, let $(x)^+ \triangleq \max\{x, 0\}$. Given a vector $\mathbf{x} = (x_i)_{i \in [n]} \in \mathbb{R}^n$, we write $\mathbf{x} \geq \mathbf{0}$ if $x_i \geq 0$ for all $i \in [n]$. We define the harmonic numbers by $H_k \triangleq \sum_{j=1}^k j^{-1}$, with $H_0 \triangleq 0$. We use *a.s.* to denote almost surely. For two sets A and B , we write $A \Subset B$ if A is contained in B and has positive distance from the boundary of B ; equivalently, $A \subseteq B$ and $\text{dist}(A, B^c) > 0$. Unless otherwise specified, $\|\cdot\|$ denotes the Euclidean norm for vectors and the induced operator norm

(i.e., the spectral norm) for matrices.

2 Problem Formulation and Dynamic Fluid Policies

In this section, we introduce our online resource allocation problem (Section 2.1), present its fluid relaxation (Section 2.2), and describe the classic dynamic fluid policy analyzed in the paper (Section 2.3).

2.1 The Online Resource Allocation Problem with Finitely Many Actions

We consider an online resource allocation problem over a finite horizon of $n \in \mathbb{N}_+$ periods, during which a decision maker collects rewards subject to resource constraints. Specifically, there are $m \in \mathbb{N}_+$ resources. The decision maker is initially endowed with an inventory of $c_i > 0$ units of resource i for each $i \in [m]$. We let $\mathbf{c} = (c_i)_{i \in [m]} \in \mathbb{R}_+^m$ denote the vector of initial inventories.

There are $L + 1$ actions, indexed by $0, 1, \dots, L$, where action 0 represents rejection. We let $\mathcal{A} = \{0, 1, \dots, L\}$ denote the action set. In each period, an opportunity arises, and each opportunity is characterized by a vector of action-specific rewards and resource consumptions $\boldsymbol{\theta} = (R_a, B_a)_{a \in \mathcal{A}}$. Specifically, taking an action $a \in \mathcal{A}$ yields a reward $R_a \in \mathbb{R}$ and consumes resources $B_a = (B_{aj})_{j \in [m]} \in \mathbb{R}_+^m$, where B_{aj} represents the amount of resource j consumed by action a . We assume that $R_0 = 0$ and $B_0 = \mathbf{0}$, so that rejecting an opportunity yields no reward and consumes no resources. The opportunities are drawn independently over time from a distribution Φ . Finally, we assume that the positive part of rewards is integrable and that the resource consumptions are uniformly bounded, as stated in Assumption 2.1.

Assumption 2.1. There exist constants $\bar{r} > 0$ and $\bar{b} > 0$ such that

$$\mathbb{E} \left[\max_{a \in \mathcal{A}} (R_a)^+ \right] \leq \bar{r} \quad \text{and} \quad \max_{a \in \mathcal{A}} \|B_a\|_\infty \leq \bar{b} \quad a.s.$$

In each period, the decision maker first observes the arriving opportunity and then chooses an action. An action a is feasible only if its resource consumption do not exceed the remaining capacity. Therefore, when the remaining capacity is $\mathbf{c} \in \mathbb{R}_+^m$ and the arriving opportunity is $\boldsymbol{\theta}$, the set of feasible actions is

$$\mathcal{A}(\mathbf{c}, \boldsymbol{\theta}) \triangleq \{a \in \mathcal{A} : B_a \leq \mathbf{c}\}.$$

Let Π denote the set of all non-anticipating policies, that is, policies for which the action in period t depends only on the arriving opportunity in period t and the history $\mathcal{H}_{t-1}$ up to period $t - 1$,

which consists of all previously observed opportunities and actions. Let a_t^π denote the action taken in period t under a policy $\pi \in \Pi$. The decision maker's objective is to choose a policy $\pi \in \Pi$ that maximizes the expected total reward over the planning horizon, subject to the capacity constraints that the cumulative resource consumption does not exceed the initial inventory. The resulting stochastic optimization problem can be written as follows:

$$\begin{aligned} V_n^{\text{OPT}}(\mathbf{c}) &= \sup_{\pi \in \Pi} \mathbb{E} \left[\sum_{t=1}^n R_{a_t^\pi} \right] \\ \text{s.t.} \quad &\sum_{t=1}^n B_{a_t^\pi} \leq \mathbf{c}, \quad a.s. \end{aligned}$$

where $V_n^{\text{OPT}}(\mathbf{c})$ denotes the optimal expected reward with n periods remaining and remaining capacity $\mathbf{c} \in \mathbb{R}_+^m$. We remark that the optimal value function satisfies the Bellman equation

$$V_n^{\text{OPT}}(\mathbf{c}) = \mathbb{E}_{\theta \sim \Phi} \left[\max_{a \in \mathcal{A}(\mathbf{c}, \theta)} \left\{ R_a + V_{n-1}^{\text{OPT}}(\mathbf{c} - B_a) \right\} \right]$$

with boundary conditions $V_0^{\text{OPT}}(\mathbf{c}) = 0$ for any $\mathbf{c} \in \mathbb{R}_+^m$.

Finally, we show that the classic network revenue management and stochastic matching problems can both be formulated as instances of the the online resource allocation problem we study, thereby illustrating the generality of our formulation.

Remark 2.1 (Network Revenue Management). In the network revenue management problem, a decision maker dynamically allocates a limited amount of resources over a finite selling horizon. Customers arrive sequentially and belong to heterogeneous types $\theta \in \Theta$, where each type is characterized by a resource consumption vector $B_\theta \in \mathbb{R}_+^m$ and a fare $r_\theta \in \mathbb{R}$. The distribution of customer types is time-invariant. Upon the arrival of a customer of type $\theta \in \Theta$, the firm decides whether to accept or reject the request. If the customer is accepted, she consumes $(B_\theta)_i$ units of resource i for each $i \in [m]$ and pays fare r_θ . Otherwise, no revenue is collected and no resources are consumed. A customer can be accepted only if the remaining inventory is sufficient to satisfy the requested resource consumption. The objective is to maximize the expected total revenue over the selling horizon. This problem is a special case of our online resource allocation model with action set $\mathcal{A} = \{0, 1\}$, where action 1 corresponds to acceptance and action 0 to rejection. The random reward R_1 has the same distribution as r_θ , and the random resource consumption vector B_1 has the same distribution as B_θ . We do not require the customer type space Θ to be finite.

Remark 2.2 (Stochastic Matching). The stochastic matching problem is closely related to the network

revenue management problem, but now, instead of making accept/reject decisions, each arriving customer can be assigned to one of several resources, and the decision maker must determine the resource to which the customer is assigned. Specifically, there are $m \in \mathbb{N}_+$ resources. Customers arrive sequentially and belong to heterogeneous types $\theta \in \Theta$, where each type is characterized by a resource consumption vector $B_\theta \in \mathbb{R}_+^m$ and a fare vector $\mathbf{r}_\theta \in \mathbb{R}^m$. The distribution of customer types is time-invariant. Upon the arrival of a customer of type $\theta \in \Theta$, the decision maker either assigns the customer to a resource or rejects the request. If the customer is assigned to resource i , she consumes $(B_\theta)_i$ units of resource i and pays fare $(\mathbf{r}_\theta)_i$. Otherwise, no revenue is collected and no resources are consumed. A customer can be assigned to resource i only if the remaining inventory of resource i is sufficient to satisfy the requested consumption. To model infeasible assignments, we allow $(\mathbf{r}_\theta)_i$ to be negative. Since the decision maker can always reject a customer and obtain zero reward, a resource with $(\mathbf{r}_\theta)_i < 0$ will never be selected. Thus, a negative value of $(\mathbf{r}_\theta)_i$ represents the case in which assigning type- θ customers to resource i is infeasible. The objective is to maximize the expected total revenue over the selling horizon. This problem is a special case of our online resource allocation problem with action set $\mathcal{A} = \{0\} \cup [m]$, where action i corresponds to assigning the customer to resource i , and action 0 corresponds to rejection. For each action $a \in [m]$, the random reward R_a has the same distribution as $(\mathbf{r}_\theta)_a$, and the random resource consumption vector B_a has the same distribution as $(B_\theta)_a \mathbf{e}_a$, where $\mathbf{e}_a$ denotes the a -th standard basis vector in $\mathbb{R}^m$. We do not require the customer type space Θ to be finite.

2.2 Fluid Relaxation and Dual Price

In this section, we introduce the fluid relaxation of the online resource allocation problem. The fluid relaxation allows resource inventories to go negative and replaces the sample-path capacity constraints with expected capacity constraints; that is, it requires only that the expected total consumption of each resource not exceed its initial inventory.

Let $\boldsymbol{\rho} = \mathbf{c}/n \in \mathbb{R}_+^m$ denote the vector of normalized inventories, where each component represents the available inventory per remaining period. Let $\mathcal{P}$ denote the set of all opportunity-dependent randomized controls, $p : \Theta \rightarrow \Delta(\mathcal{A})$. The fluid relaxation is defined as

$$\begin{aligned} \phi^F(\boldsymbol{\rho}) = \sup_{p \in \mathcal{P}} \quad & \sum_{a \in \mathcal{A}} \mathbb{E}_{\boldsymbol{\theta} \sim \Phi} [R_a p(\boldsymbol{\theta}, a)] \\ \text{s.t.} \quad & \sum_{a \in \mathcal{A}} \mathbb{E}_{\boldsymbol{\theta} \sim \Phi} [B_a p(\boldsymbol{\theta}, a)] \leq \boldsymbol{\rho}. \end{aligned} \tag{1}$$

The optimal value of (1), denoted by $\phi^F(\boldsymbol{\rho})$, represents the maximum expected reward per period achievable under the fluid relaxation with normalized inventory $\boldsymbol{\rho}$. For a planning horizon of n periods with initial inventory $\mathbf{c} = n\boldsymbol{\rho}$, the following upper bound holds.

Lemma 2.1. *Let $\boldsymbol{\rho} = \mathbf{c}/n$ denote the vector of normalized inventories. Then*

$$V_n^{\text{OPT}}(\mathbf{c}) \leq V_n^{\text{OFF}}(\mathbf{c}) \leq n \cdot \phi^F(\boldsymbol{\rho}),$$

where $V_n^{\text{OFF}}(\mathbf{c})$ denotes the offline optimal upper bound with n periods remaining and remaining capacity $\mathbf{c} \in \mathbb{R}_+^m$.

We define the offline optimal upper bound and prove Lemma 2.1 in Section A.1. Lemma 2.1 shows that the expected reward under an optimal policy is upper bounded by the value of the fluid relaxation.

2.2.1 The Fluid Relaxation Dual Problem

We now derive the dual of the fluid relaxation (1) by dualizing the capacity constraints with a dual variable $\boldsymbol{\lambda} \in \mathbb{R}_+^m$. The dual problem of (1) is

$$\phi^F(\boldsymbol{\rho}) = \inf_{\boldsymbol{\lambda} \in \mathbb{R}_+^m} \left\{ \boldsymbol{\lambda}^\top \boldsymbol{\rho} + \mathcal{V}(\boldsymbol{\lambda}) \right\}, \quad (2)$$

where

$$\mathcal{V}(\boldsymbol{\lambda}) \triangleq \mathbb{E} \left[\max_{a \in \mathcal{A}} \left\{ R_a - \boldsymbol{\lambda}^\top B_a \right\} \right]$$

denotes the expected maximum net reward earned in a single period.

Since (1) is a linear program, strong duality results for linear optimization in vector spaces (see Theorem 1 in Section 8.6 of Luenberger 1997) imply that strong duality holds and that an optimal dual solution exists. Therefore, the optimal value of (2) is equal to $\phi^F(\boldsymbol{\rho})$, and we let $\boldsymbol{\lambda}^*(\boldsymbol{\rho})$ denote an optimal solution to (2).

Remark 2.3 (Use of Strong Duality). We remark that the use of strong duality requires Slater's condition, which in turn requires $\boldsymbol{\rho} > \mathbf{0}$. However, if $\rho_i = 0$ for some resource i , then resource i is depleted. In this case, the original resource allocation problem is equivalent to a reduced problem obtained by removing all depleted resources and prohibiting any action a that consumes them. Strong duality continues to hold and an optimal dual solution exists for this reduced problem. Although the corresponding dual variable has fewer than m components, the omitted components

correspond to the depleted resources, and they do not affect either the definition of the dynamic fluid policy in Section 2.3 or its subsequent analysis. We can extend the reduced dual variable to the original problem by setting the omitted components to zero.

2.3 The Dynamic Fluid Policy

In this section, we describe the dynamic fluid policy. At each state $(n, \mathbf{c})$, where n is the number of remaining periods and $\mathbf{c}$ is the remaining capacity, the policy solves the fluid relaxation dual problem (2) using the normalized remaining capacity $\boldsymbol{\rho} = \mathbf{c}/n$. It then uses the resulting optimal dual variable $\boldsymbol{\lambda}^*(\boldsymbol{\rho})$ as the shadow price of resource consumption. Formally, at state $(n, \mathbf{c})$, the dynamic fluid policy selects the action

$$a_n^F(\mathbf{c}) \triangleq \arg \max_{a \in \mathcal{A}(\mathbf{c}, \boldsymbol{\theta})} \left\{ R_a - \boldsymbol{\lambda}^*(\mathbf{c}/n)^\top B_a \right\} \quad (3)$$

with the smallest-index tie-breaking. We let $V_n^F(\mathbf{c})$ denote the expected total reward earned by the dynamic fluid policy starting from state $(n, \mathbf{c})$.

3 The Non-Degeneracy Conditions

In this section, we present the non-degeneracy conditions underlying our main result, Theorem 4.1 in Section 4, formalized in Assumptions 3.1 – 3.3. These assumptions are mild and, as we demonstrate in Section 5, are satisfied by many notable online resource allocation problems.

In the remainder of the paper, given a vector $\boldsymbol{\rho} = (\rho_i)_{i \in [m]} \in \mathbb{R}^m$ and a subset $\mathcal{B} \subseteq [m]$, we let

$$\boldsymbol{\rho}_{\mathcal{B}} = (\rho_i)_{i \in \mathcal{B}}$$

denote the restriction of $\boldsymbol{\rho}$ to the coordinates indexed by $\mathcal{B}$. Similarly, for a set $\mathcal{U} \subseteq \mathbb{R}^m$, we define

$$\mathcal{U}_{\mathcal{B}} \triangleq \left\{ \boldsymbol{\rho}_{\mathcal{B}} : \boldsymbol{\rho} \in \mathcal{U} \right\}$$

as the projection of $\mathcal{U}$ onto the coordinates indexed by $\mathcal{B}$.

We first impose Assumption 3.1, which requires that, in a neighborhood of the point of interest, the fluid value function admit a lower-dimensional representation (where the reduced coordinates will correspond to the binding resource constraints) and that this reduced function is sufficiently smooth.

Assumption 3.1 (Local Smoothness). There exist an open set $\mathcal{U} \subseteq \mathbb{R}_+^m$, a nonempty subset $\mathcal{B} \subseteq [m]$, and a function $\phi_{\mathcal{B}} : \mathbb{R}^{|\mathcal{B}|} \rightarrow \mathbb{R}$ such that:

1. For every $\boldsymbol{\rho} \in \mathcal{U}$, $\phi^{\text{F}}(\boldsymbol{\rho}) = \phi_{\mathcal{B}}(\boldsymbol{\rho}_{\mathcal{B}})$.
2. The function $\phi_{\mathcal{B}}$ belongs to $C^{2,\alpha}(\mathcal{U}_{\mathcal{B}})$ for some Hölder exponent $\alpha \in (0, 1]$; that is, $\phi_{\mathcal{B}}$ is twice continuously differentiable on $\mathcal{U}_{\mathcal{B}}$, and, for every compact set $\mathcal{K}_{\mathcal{B}} \Subset \mathcal{U}_{\mathcal{B}}$, there exists a finite constant L such that

$$\|\nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) - \nabla^2 \phi_{\mathcal{B}}(\mathbf{y})\| \leq L \|\mathbf{x} - \mathbf{y}\|^\alpha, \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{K}_{\mathcal{B}}.$$

Assumption 3.1 implies that, for every $\boldsymbol{\rho} \in \mathcal{U}$, the optimal dual variable $\boldsymbol{\lambda}^*(\boldsymbol{\rho})$ is unique and admits the following characterization in Lemma 3.1.

Lemma 3.1. *For every $\boldsymbol{\rho} \in \mathcal{U}$, the optimal dual variable $\boldsymbol{\lambda}^*(\boldsymbol{\rho})$ is unique. Moreover,*

$$\boldsymbol{\lambda}^*(\boldsymbol{\rho}) = (\boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}}), \mathbf{0}_S),$$

where $\boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}}) \triangleq \nabla \phi_{\mathcal{B}}(\boldsymbol{\rho}_{\mathcal{B}})$ and $S = \mathcal{B}^c$.

We prove Lemma 3.1 in Section A.2. Lemma 3.1 demonstrates that the optimal dual variables associated with the resource constraints indexed by $\mathcal{B}^c$ are all zero. Consequently, at each state $(n, \mathbf{c})$ with $\boldsymbol{\rho} = \mathbf{c}/n \in \mathcal{U}$, the dynamic fluid policy selects the action

$$a_n^{\text{F}}(\mathbf{c}) = \arg \max_{a \in \mathcal{A}(\mathbf{c}, \boldsymbol{\theta})} \left\{ R_a - \boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})^{\text{T}} B_{a, \mathcal{B}} \right\},$$

where $B_{a, \mathcal{B}} \triangleq (B_a)_{\mathcal{B}}$.

We next impose Assumption 3.2, which requires that the set of binding resource constraints in the fluid relaxation (1) remain unchanged throughout the neighborhood $\mathcal{U}$. Specifically, the constraints indexed by $\mathcal{B}$ are binding, whereas those indexed by $\mathcal{B}^c$ are strictly slack. We also assume that the expected consumption of the non-binding resources is a smooth function of $\boldsymbol{\rho}_{\mathcal{B}}$ on $\mathcal{U}_{\mathcal{B}}$.

Fix any $\boldsymbol{\rho} \in \mathcal{U}$. Consider a static fluid policy for the fluid relaxation (1) that always selects the action

$$a^{\text{F}} = \arg \max_{a \in \mathcal{A}} \left\{ R_a - \boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})^{\text{T}} B_{a, \mathcal{B}} \right\} \tag{4}$$

using the smallest-index tie-breaking rule. Unlike the dynamic fluid policy defined in Section 2.3, this policy keeps the dual variable $\boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})$ fixed throughout the horizon, rather than recomputing it

using the remaining resource inventories in each period. Recall also that the fluid relaxation (1) allows resource inventories to become negative. For each resource $j \in [m]$, define

$$\mu_j(\boldsymbol{\rho}_{\mathcal{B}}) \triangleq \mathbb{E}\left[(B_{a^{\text{F}}})_j\right],$$

as the expected per-period consumption of resource j in the fluid relaxation. Assumption 3.2 is stated as follows.

Assumption 3.2 (Fixed Binding Resources). For every $\boldsymbol{\rho} \in \mathcal{U}$, the corresponding static fluid policy evaluated at $\boldsymbol{\rho}_{\mathcal{B}}$ satisfies

1. $\mu_j(\boldsymbol{\rho}_{\mathcal{B}}) = \rho_j$ for all $j \in \mathcal{B}$,
2. $\mu_j(\boldsymbol{\rho}_{\mathcal{B}}) < \rho_j$ for all $j \in \mathcal{B}^c$.

Moreover, for every $j \in \mathcal{B}^c$, the consumption function $\mu_j(\boldsymbol{\rho}_{\mathcal{B}}) \in C^1(\mathcal{U}_{\mathcal{B}})$; that is, μ_j is continuously differentiable on $\mathcal{U}_{\mathcal{B}}$.

Assumption 3.2 implies that the expected per-period reward of the static fluid policy equals the fluid relaxation benchmark, as stated in Lemma 3.2. We prove Lemma 3.2 in Appendix A.3.

Lemma 3.2. *For every $\boldsymbol{\rho} \in \mathcal{U}$, we have $\mathbb{E}[R_{a^{\text{F}}}] = \phi^{\text{F}}(\boldsymbol{\rho})$.*

Finally, we impose Assumption 3.3, which controls how often the fluid policy encounters actions that are nearly indistinguishable in terms of dual-price score but differ substantially in their binding-resource consumption.

Assumption 3.3 (Consumption-Weighted Near-Indifference). Define the dual-price score of an action a by

$$S_a(\boldsymbol{\rho}_{\mathcal{B}}) \triangleq R_a - \boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})^{\text{T}} B_{a,\mathcal{B}}$$

and define the binding-resource consumption difference of two actions a and b by $d_{ab}^{\mathcal{B}} \triangleq \|B_{a,\mathcal{B}} - B_{b,\mathcal{B}}\|_1$. For every compact set $\mathcal{K}_{\mathcal{B}} \subseteq \mathcal{U}_{\mathcal{B}}$, there exist constants $\beta_{\mathcal{K}} > 0$, $t_{\mathcal{K}} > 0$, and $C_{\mathcal{K}} < \infty$ such that

$$\sup_{\mathbf{x} \in \mathcal{K}_{\mathcal{B}}} \mathbb{E} \left[\sum_{b \neq a^{\text{F}}(\mathbf{x})} d_{a^{\text{F}}(\mathbf{x}),b}^{\mathcal{B}} \mathbb{1} \left\{ S_{a^{\text{F}}(\mathbf{x})}(\mathbf{x}) - S_b(\mathbf{x}) \leq t d_{a^{\text{F}}(\mathbf{x}),b}^{\mathcal{B}} \right\} \right] \leq C_{\mathcal{K}} t^{\beta_{\mathcal{K}}}, \quad 0 \leq t \leq t_{\mathcal{K}},$$

where $a^{\text{F}}(\mathbf{x})$ denotes the action with the highest dual-price score, as defined in (4).

Because the fluid policy selects the action a^{F} with the highest dual-price score, the difference $S_{a^{\text{F}}}(\mathbf{x}) - S_b(\mathbf{x})$ measures how much more attractive the selected action is than a competing action

b. Assumption 3.3 requires that, as the tolerance t decreases, the consumption-weighted frequency of such near-indifference events decays at a positive polynomial rate. The weight $d_{a^F, b}^{\mathcal{B}}$ measures the difference in binding-resource consumption between the two actions. Near-indifference is less consequential when the two actions consume similar binding-resource bundles, because switching between them has little effect on the future inventory of the binding resources.

To conclude this section, we note that Assumption 3.1, together with Assumption 3.2, requires that at least one resource constraint is binding in the fluid relaxation (i.e., $\mathcal{B} \neq \emptyset$). If, instead, all resource constraints are slack, then

$$\phi^F = \mathbb{E} \left[\max_a R_a \right],$$

and the optimal policy in the fluid relaxation simply selects the action with the highest reward in each period. Under mild conditions, the performance gap between this greedy policy in the original problem and the fluid relaxation upper bound converges to zero as the time horizon $n \rightarrow \infty$, as shown in Proposition 3.3. We prove Proposition 3.3 in Section A.4.

Proposition 3.3 (The All-Slack Case). *Suppose Assumption 2.1 holds. For a generic opportunity, let $a^0 \triangleq \arg \max_{a \in \mathcal{A}} R_a$ with a certain tie-breaking rule. Consider the greedy policy that selects a^0 when it is feasible and selects action 0 otherwise. Let $V_n^G(\mathbf{c})$ denote its expected reward. Define $\mu_j \triangleq \mathbb{E}[(B_{a^0})_j]$ for $j \in [m]$. Given a sequence of initial inventories $(\mathbf{c}_n)_{n \geq 1}$, let $\boldsymbol{\rho}_n = (\rho_{nj})_{j \in [m]} \triangleq \mathbf{c}_n/n$ and define $\delta_n \triangleq \min_{j \in [m]} \{\rho_{nj} - \mu_j\}$ as the minimum slackness across all resource constraints. If $\delta_n \geq \kappa \sqrt{\ln n/n}$, where $\kappa > 0$ is a constant depending only on the problem primitives, then*

$$n \mathbb{E} \left[\max_a R_a \right] - V_n^G(\mathbf{c}_n) = o(1),$$

where $o(1)$ denotes a term that converges to zero as $n \rightarrow \infty$, uniformly over all initial inventories satisfying the above slack condition.

Therefore, the all-slack regime is relatively straightforward. In this paper, we focus on the more interesting setting in which at least one resource constraint is binding.

4 The Main Result

In this section, we first define the logarithmic correction coefficient and then present the common expansion and constant optimality gap for the dynamic fluid policy in Theorem 4.1.

For any $\mathbf{x} \in \mathcal{U}_{\mathcal{B}}$, let a^F denote the static fluid action in (4) evaluated at $\boldsymbol{\rho}_{\mathcal{B}} = \mathbf{x}$, and define the

covariance matrix of its binding-resource consumption by

$$\Sigma(\mathbf{x}) \triangleq \mathbb{Cov}(B_{a^F, \mathcal{B}}),$$

where the expectation is taken over $\boldsymbol{\theta} \sim \Phi$. We then define the logarithmic correction coefficient by

$$\Gamma(\mathbf{x}) \triangleq -\frac{1}{2} \text{tr} \left(\nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) \Sigma(\mathbf{x}) \right). \quad (5)$$

Since $\phi_{\mathcal{B}}$ is concave on $\mathcal{U}_{\mathcal{B}}$ by the concavity of ϕ^F , its Hessian satisfies $\nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) \preceq 0$ for all $\mathbf{x} \in \mathcal{U}_{\mathcal{B}}$. Moreover, since $\Sigma(\mathbf{x}) \succeq 0$ by definition, (5) implies that $\Gamma(\mathbf{x}) \geq 0$ for every $\mathbf{x} \in \mathcal{U}_{\mathcal{B}}$.¹ We now state our main result.

Theorem 4.1 (Common Expansion and Constant Optimality Gap). *Suppose Assumption 2.1 and Assumptions 3.1–3.3 hold. For every compact set $\mathcal{K} \subseteq \mathcal{U}$, uniformly over states satisfying $\mathbf{c}/n \in \mathcal{K}$, we have*

$$\begin{aligned} V_n^F(\mathbf{c}) &= n \phi^F(\mathbf{c}/n) - \Gamma(\mathbf{c}_{\mathcal{B}}/n) \log n + O_{\mathcal{K}}(1), \\ V_n^{\text{OPT}}(\mathbf{c}) &= n \phi^F(\mathbf{c}/n) - \Gamma(\mathbf{c}_{\mathcal{B}}/n) \log n + O_{\mathcal{K}}(1). \end{aligned}$$

Here, $O_{\mathcal{K}}(1)$ denotes a remainder that is uniformly bounded over all $n \geq 1$ and all states with $\mathbf{c}/n \in \mathcal{K}$. Consequently, there exists a constant $C_{\mathcal{K}} < \infty$, depending only on the model primitives and the set $\mathcal{K}$, such that for every $n \geq 1$ and all states with $\mathbf{c}/n \in \mathcal{K}$,

$$0 \leq V_n^{\text{OPT}}(\mathbf{c}) - V_n^F(\mathbf{c}) \leq C_{\mathcal{K}}.$$

Theorem 4.1 compares the values of both the dynamic fluid policy and the optimal online policy with the deterministic fluid benchmark. Since they share the same leading fluid term and logarithmic correction, these terms cancel when the two values are compared, leaving a uniformly bounded optimality gap. Thus, the logarithmic term represents a common second-order correction to the fluid benchmark. Its coefficient, Γ , captures the interaction between the curvature of the fluid value function in the binding-resource directions and the covariance of the corresponding resource consumption under the static fluid policy in the fluid relaxation.

In Section 5, we apply Theorem 4.1 to multiple notable online resource allocation problems. We

¹ Indeed, let $H = \nabla^2 \phi_{\mathcal{B}}(\mathbf{x})$ and $Z = B_{a^F, \mathcal{B}} - \mathbb{E}[B_{a^F, \mathcal{B}}]$. Since $\Sigma(\mathbf{x}) = \mathbb{E}[ZZ^T]$ and $H \preceq 0$, we have $\text{tr}(H\Sigma(\mathbf{x})) = \mathbb{E}[\text{tr}(HZZ^T)] = \mathbb{E}[Z^T H Z] \leq 0$. Therefore, $\Gamma(\mathbf{x}) = -\frac{1}{2} \text{tr}(H\Sigma(\mathbf{x})) \geq 0$.

prove Theorem 4.1 in Section 6.

5 Applications

In this section, we apply Theorem 4.1 three online resource allocation problems and demonstrate the new results it yields in these settings.

5.1 The Uniform Multisecretary Model

In the uniform multisecretary problem (Section 3 of Bray 2025), the decision maker has a single resource with an integer capacity $c \in \mathbb{N}_+$. In each period t , the decision maker observes a value V_t and either accepts it, consuming one unit of capacity, or rejects it. The values $\{V_t\}$ are i.i.d. standard uniform random variables. This problem is a special case of our online resource allocation problem with a single resource ($m = 1$), an action set $\mathcal{A} = \{0, 1\}$, and reward-consumption pairs $(R_0, B_0) = (0, 0)$ and $(R_1, B_1) = (V, 1)$ with $V \sim \text{Unif}[0, 1]$. Propositions 1 and 2 of Bray (2025) establishes a tight $\Theta(\ln n)$ lower bound on the performance gap of any policy relative to the offline optimal benchmark when the initial capacity is proportional to the horizon.

We note that on the set $\mathcal{U} = (0, 1)$, Assumption 2.1 and Assumptions 3.1 to 3.3 are satisfied with binding-resource set $\mathcal{B} = \{1\}$, fluid value function $\phi^F(\rho) = \phi_{\mathcal{B}}(\rho) = \rho - \rho^2/2$, and optimal dual variable $\lambda^*(\rho) = 1 - \rho$. The dynamic fluid policy takes a simple threshold form: in each period with c remaining capacity and n periods remaining, accept the arrival whenever its value satisfies $V > 1 - c/n$.

Theorem 4.1 implies that, for every compact set $\mathcal{K} \subseteq (0, 1)$, uniformly over all states (c, n) satisfying $c/n \in \mathcal{K}$, both $V_n^{\text{OPT}}(c)$ and $V_n^F(c)$ admit the expansion

$$n \left(\rho - \frac{1}{2} \rho^2 \right) - \frac{1}{2} \rho(1 - \rho) \log n + O_{\mathcal{K}}(1), \quad \rho = c/n.$$

Therefore, the dynamic fluid policy has a uniformly bounded optimality gap relative to the optimal online policy. We provide further details in Section B.1.

5.2 The Stochastic Knapsack Problem of Lueker (1998)

The stochastic knapsack problem is a classical problem in operations research and computer science. In this problem, the decision maker sequentially observes n i.i.d. nonnegative weight-value pairs (W, V) and must decide whether to accept each item before observing the next. The decision maker

has a total capacity of $n\beta_0$. If the remaining capacity is c , an item can be accepted only if $W \leq c$. Acceptance yields reward V and consumes W units of capacity. This problem is a special case of our online resource allocation problem with one resource, action set $\mathcal{A} = \{0, 1\}$, and reward-consumption pairs $(R_0, B_0) = (0, 0)$ and $(R_1, B_1) = (V, W)$.

Lueker (1998) study an OnLineGreedy policy that coincides with our dynamic fluid policy for the stochastic knapsack problem. Theorem 2 of Lueker (1998) show that, under their Conditions 1–3, the value of the dynamic fluid policy admits the following expansion:^{2 3}

$$n \phi^F(\beta_0) - \Gamma(\beta_0) \log n + o(\log n).$$

Thus, the dynamic fluid policy differs from the offline optimal benchmark by $\Theta(\log n)$. Moreover, this order is tight: no non-anticipative policy can improve the gap by $o(\log n)$.

Lueker’s Conditions 1–2 imply our Assumption 2.1 and Assumptions 3.1–3.3. Thus, our model is more general and our assumptions are weaker. Moreover, Theorem 4.1 yields a stronger result: both $V_n^{\text{OPT}}(c)$ and $V_n^F(c)$ admit the expansion

$$n \phi^F(\beta_0) - \Gamma(\beta_0) \log n + O(1).$$

Consequently, the dynamic fluid policy has only a uniformly bounded performance gap relative to the optimal online policy. We provide further details in Section B.2.

5.3 The Online Linear Programming Problem of Bray (2025)

We next consider the online linear programming problem studied in Section 4 of Bray (2025). The decision maker begins with an inventory vector $n\beta \in \mathbb{R}_+^m$ and sequentially observes n i.i.d. requests $(u_t, \mathbf{a}_t)$. Accepting request t yields utility $u_t \geq 0$ and consumes the resource vector $\mathbf{a}_t \geq \mathbf{0}$, whereas rejecting it yields no utility and consumes no resources. This problem is a special case of our problem with action set $\mathcal{A} = \{0, 1\}$ and reward-consumption pairs $(R_0, B_0) = (0, \mathbf{0})$ and $(R_1, B_1) = (u, \mathbf{a})$.

In the known-distribution setting, Bray (2025) analyze the same policy (Algorithm 2 therein) as our dynamic fluid policy. Under mild non-degeneracy conditions, Theorems 1 and 3 of Bray (2025) together imply that this policy has a tight $\Theta(\log n)$ performance gap relative to the offline optimum

²As a notable example, in Lueker (1998), Conditions 1–3 are satisfied when the values V and weights W are independent and both follow the standard uniform distribution.

³We note that $\kappa(\beta_0)$ in Lueker’s Theorem 2 equals our $\phi^F(\beta_0)$ and $\sigma_0^2/(2\alpha_0)$ equals our $\Gamma(\beta_0)$. Thus, the first-order and logarithmic terms in Lueker’s expansion coincide with those in Theorem 4.1. We provide further details in Section B.2.

and that no non-anticipative policy can achieve a smaller-order gap.

Bray's non-degeneracy conditions, stated in their Assumptions 1–6, imply ours in Section 3, except for the local α -Hölder condition on $\nabla^2 \phi_{\mathcal{B}}$ required in part 2 of Assumption 3.1. We verify this implication in Section B.3. Under this additional local Hölder condition, Theorem 4.1 further establishes that the dynamic fluid policy has a uniformly bounded optimality gap relative to the optimal online policy.

6 Proof of Theorem 4.1

In this section we prove Theorem 4.1. The proof is based on comparing the value functions of both the dynamic fluid policy and the optimal online policy with a common corrected fluid reference $Q_n(\mathbf{c})$ satisfying

$$Q_n(\mathbf{c}) = n \phi^F(\mathbf{c}/n) - \Gamma(\mathbf{c}_{\mathcal{B}}/n) H_{n-1} + O_{\mathcal{K}}(1).$$

We motivate and construct this corrected fluid reference in Section 6.1. In Section 6.2, we demonstrate that, uniformly over all initial states $(\mathbf{c}, n)$ with $\mathbf{c}/n \in \mathcal{K}$, the value functions $V_n^F(\mathbf{c})$ and $V_n^{\text{OPT}}(\mathbf{c})$ deviate from $Q_n(\mathbf{c})$ by at most a fixed constant.

Throughout this section, given a continuation value function $u(\cdot)$, we define the Bellman operators associated with the optimal policy and the dynamic fluid policy by

$$\begin{aligned} (T^*u)(\mathbf{c}) &\triangleq \mathbb{E} \left[\max_{a \in \mathcal{A}(\mathbf{c}, \boldsymbol{\theta})} \left\{ R_a + u(\mathbf{c} - B_a) \right\} \right], \\ (T_n^F u)(\mathbf{c}) &\triangleq \mathbb{E} \left[R_{a_n^F(\mathbf{c})} + u(\mathbf{c} - B_{a_n^F(\mathbf{c})}) \right]. \end{aligned}$$

Thus, $V_n^{\text{OPT}} = T^* V_{n-1}^{\text{OPT}}$ and $V_n^F = T_n^F V_{n-1}^F$ for $n \geq 1$, with $V_0^{\text{OPT}} = V_0^F \equiv 0$. We also fix a compact target set $\mathcal{K} \subseteq \mathcal{U}$. Unless otherwise stated, all constants may depend on $\mathcal{K}$ and the model primitives, but not on the horizon n or the initial state $\mathbf{c}$, provided that $\mathbf{c}/n \in \mathcal{K}$.

6.1 The Corrected Fluid Reference

We define the fluid relaxation upper bound by $F_n(\mathbf{c}) \triangleq n \phi^F(\mathbf{c}/n)$ for $n \geq 1$ and let $F_0(\mathbf{c}) = 0$. Let $\mathcal{K}_{\mathcal{B}} \triangleq \{\boldsymbol{\rho}_{\mathcal{B}} : \boldsymbol{\rho} \in \mathcal{K}\}$ denote the projection of the target set $\mathcal{K}$ onto the binding coordinates. For any $\mathbf{x} \in \mathcal{U}_{\mathcal{B}}$, let $a^F(\mathbf{x})$ be the static dual-price action defined in (4) and let $Y_{\mathbf{x}} \triangleq B_{a^F(\mathbf{x}), \mathcal{B}}$ denote the corresponding consumption of binding resources.

We first provide some intuition for why the corrected fluid reference takes the form

$$n\phi_{\mathcal{B}}(\mathbf{x}) - \Gamma(\mathbf{x}) H_{n-1} + O(1).$$

Suppose that n periods remain and that the normalized inventory of the binding resources is $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}$. After taking action $a^F(\mathbf{x})$, the normalized binding-resource inventory in the next period is

$$\mathbf{x}' \triangleq \frac{n\mathbf{x} - Y_{\mathbf{x}}}{n-1} = \mathbf{x} + \frac{\mathbf{x} - Y_{\mathbf{x}}}{n-1}.$$

The one-period residual from the fluid relaxation upper bound is

$$\delta_n^F(\mathbf{x}) \triangleq n\phi_{\mathcal{B}}(\mathbf{x}) - \mathbb{E}\left[R_{a^F(\mathbf{x})} + (n-1)\phi_{\mathcal{B}}(\mathbf{x}')\right].$$

By Lemma 3.2, $\mathbb{E}[R_{a^F(\mathbf{x})}] = \phi_{\mathcal{B}}(\mathbf{x})$. Moreover, by Assumption 3.2, $\mathbb{E}[Y_{\mathbf{x}}] = \mathbf{x}$, and hence $\mathbb{E}[\mathbf{x}'] = \mathbf{x}$. Therefore,

$$\delta_n^F(\mathbf{x}) = (n-1)(\phi_{\mathcal{B}}(\mathbf{x}) - \mathbb{E}[\phi_{\mathcal{B}}(\mathbf{x}')]) \geq 0,$$

where the inequality follows from the concavity of $\phi_{\mathcal{B}}$ and Jensen's inequality. Hence, the residual $\delta_n^F(\mathbf{x})$ arises from the curvature of the deterministic fluid relaxation $\phi_{\mathcal{B}}$, rather than a one-period optimality loss of the dynamic fluid policy. Finally, a Taylor expansion of $\phi_{\mathcal{B}}(\mathbf{x}')$ around $\mathbf{x}$ yields

$$\begin{aligned} \delta_n^F(\mathbf{x}) &= -\nabla\phi_{\mathcal{B}}(\mathbf{x})^T \mathbb{E}[\mathbf{x} - Y_{\mathbf{x}}] - \frac{1}{2(n-1)} \mathbb{E}\left[(\mathbf{x} - Y_{\mathbf{x}})^T \nabla^2\phi_{\mathcal{B}}(\mathbf{x})(\mathbf{x} - Y_{\mathbf{x}})\right] + O\left(n^{-1-\alpha}\right) \\ &= \frac{\Gamma(\mathbf{x})}{n-1} + O\left(n^{-1-\alpha}\right). \end{aligned} \tag{6}$$

The first equality holds for sufficiently large n —so that the line segment between $\mathbf{x}$ and $\mathbf{x}'$ lies in a compact subset of $\mathcal{U}_{\mathcal{B}}$ —and uses the local α -Hölder continuity of $\nabla^2\phi_{\mathcal{B}}$ from part 2 of Assumption 3.1.⁴ The second equality follows from $\mathbb{E}[Y_{\mathbf{x}}] = \mathbf{x}$ and

$$\mathbb{E}\left[(\mathbf{x} - Y_{\mathbf{x}})^T \nabla^2\phi_{\mathcal{B}}(\mathbf{x})(\mathbf{x} - Y_{\mathbf{x}})\right] = \text{tr}\left(\nabla^2\phi_{\mathcal{B}}(\mathbf{x})\Sigma(\mathbf{x})\right) = -2\Gamma(\mathbf{x}).$$

Hence, $\Gamma(\mathbf{x})$ captures the curvature of the fluid relaxation $\phi_{\mathcal{B}}(\mathbf{x})$ along directions in which the binding-resource consumption varies. In particular, $\Gamma(\mathbf{x}) = 0$ if $\phi_{\mathcal{B}}(\mathbf{x})$ is locally affine along every direction in which $Y_{\mathbf{x}}$ varies.

Choose a sufficiently large n_0 so that the expansion (6) holds for every $n \geq n_0$. Keeping $\mathbf{x}$ fixed

⁴We provide further details in Lemma D.1 of the Appendix.

and summing the one-period residuals then gives

$$\sum_{j=n_0}^n \delta_j^F(\mathbf{x}) = \Gamma(\mathbf{x}) \mathbf{H}_{n-1} + O(1).$$

Note that this calculation provides only a frozen-state heuristic. Along the actual trajectory of the dynamic fluid policy, the normalized binding state $\mathbf{x}$ changes after each action, and hence so does $\Gamma(\mathbf{x})$. Therefore, although this calculation identifies the source and order of the logarithmic correction, it does not by itself yield a valid Bellman telescope. Such a telescope requires evaluating the state-dependent correction at the random successor state $\mathbf{x}'$, rather than repeatedly at the current state $\mathbf{x}$. Our subsequent analysis accounts for precisely this state dependence.

Finally, the Hessian $\nabla^2 \phi_{\mathcal{B}}(\mathbf{x})$ is only locally Hölder continuous by Assumption 3.1. Consequently, $\Gamma(\mathbf{x})$, and hence, $\delta_n^F(\mathbf{x})$, are also only locally Hölder continuous and need not be differentiable. To obtain the regularity needed for the subsequent analysis, we smooth $\Gamma(\mathbf{x})$ to construct a smooth proxy $\tilde{\delta}_n(\mathbf{x}) \in C^\infty(\mathcal{U}_{\mathcal{B}})$ for the residual $\delta_n^F(\mathbf{x})$. Using these smooth proxies, we define the cumulative correction and the corrected fluid reference by

$$A_n(\mathbf{x}) \triangleq \sum_{j=n_0}^n \tilde{\delta}_j(\mathbf{x}), \quad Q_n(\mathbf{c}) \triangleq F_n(\mathbf{c}) - A_n\left(\frac{\mathbf{c}_{\mathcal{B}}}{n}\right).$$

The smoothness of the proxies $\tilde{\delta}_j$ make $A_n(\mathbf{x})$ sufficiently smooth for second-order Taylor expansions used in the subsequent analysis in Section 6.2. Proposition 6.1 summarizes the key properties of the corrected fluid reference Q_n .

Proposition 6.1 (Corrected Fluid Reference). *There exist an integer n_0 , a sequence of smooth cumulative correction functions $(A_n)_{n \geq n_0}$, with $A_n \in C^\infty(\mathcal{U}_{\mathcal{B}})$, and a nonnegative summable sequence $(a_n)_{n \geq n_0}$ with $\sum_{n \geq n_0} a_n < \infty$, such that, for every $n \geq n_0$, the following statements hold.*

1. *Uniformly over $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$,*

$$A_n(\mathbf{x}) = \Gamma(\mathbf{x}) \mathbf{H}_{n-1} + O(1),$$

where $\mathcal{K}_{\mathcal{B}}^2$ is a compact set defined in Section C.1 satisfying $\mathcal{K}_{\mathcal{B}} \Subset \mathcal{K}_{\mathcal{B}}^2 \Subset \mathcal{U}_{\mathcal{B}}$. Consequently, defining the corrected reference by

$$Q_n(\mathbf{c}) \triangleq F_n(\mathbf{c}) - A_n(\mathbf{c}_{\mathcal{B}}/n),$$

we have, uniformly over $\mathbf{c}/n \in \mathcal{K}$,

$$Q_n(\mathbf{c}) = n\phi^F(\mathbf{c}/n) - \Gamma(\mathbf{c}_B/n) H_{n-1} + O(1).$$

2. Uniformly over all states with $\mathbf{c}/n \in \mathcal{D}_1$, we have

$$\left| Q_n(\mathbf{c}) - T_n^F Q_{n-1}(\mathbf{c}) \right| \leq a_n,$$

where $\mathcal{D}_1$ is a region defined in Section C.1 satisfying $\mathcal{K} \subseteq \mathcal{D}_1$.

In Appendix C, we develop the ingredients needed to construct the corrected reference $Q_n(\mathbf{c})$ and to support the subsequent analysis. In Appendix D, we construct the smoothed residual $\tilde{\delta}_n(\mathbf{x})$, use it to define $Q_n(\mathbf{c})$, and prove Proposition 6.1.

6.2 Comparison to the Corrected Fluid Reference

In this section, we compare the value functions $V_n^F(\mathbf{c})$ and $V_n^{\text{OPT}}(\mathbf{c})$ to the corrected fluid reference $Q_n(\mathbf{c})$. The two comparisons use different arguments. For the dynamic fluid policy, we follow its state trajectory until it leaves a local region and telescope the summable one-period residuals of the corrected reference. For the online-optimal policy, we instead construct a supersolution, because the optimal policy need not preserve the state dynamics used in the first comparison.

We begin with $V_n^F(\mathbf{c})$. By definition, $V_n^F = T_n^F V_{n-1}^F$ for $n \geq 1$. Although Q_n is not the value function of the dynamic fluid policy, part 2 of Proposition 6.1 demonstrates that it satisfies an approximate Bellman recursion:

$$Q_n(\mathbf{c}) \approx T_n^F Q_{n-1}(\mathbf{c}).$$

Moreover, the one-period residual is bounded by a_n , where $\sum_n a_n < \infty$. Hence, as long as the normalized binding state $\mathbf{x}$ remains in the local region containing $\mathcal{K}_B$ on which this residual bound holds, the difference between $V_n^F(\mathbf{c})$ and $Q_n(\mathbf{c})$ can accumulate through these summable residuals and therefore remains bounded by a constant.

Moreover, under the dynamic fluid policy, the normalized binding state $\mathbf{x}$ evolves as a martingale as long as $\mathbf{x} \in \mathcal{U}_B$. Let τ denote the first time at which the normalized inventory leaves the local region $\mathcal{D}_1$ specified in part 2 of Proposition 6.1, and let $h_\tau \triangleq n - \tau$ denote the number of remaining periods at the time of exit. Stopping at τ decomposes the difference between $V_n^F(\mathbf{c})$ with $Q_n(\mathbf{c})$ into

the accumulated residuals before stopping and a terminal error of order h_τ , yielding

$$|V_n^F(\mathbf{c}) - Q_n(\mathbf{c})| \leq O(\mathbb{E}[h_\tau]) + \sum_{j \leq n} a_j.$$

Finally, a maximal concentration inequality for martingales shows that the probability of leaving the local region with at least r periods remaining is at most Ce^{-cr} . Consequently, $\mathbb{E}[h_\tau] = O(1)$. Because the residual sequence $(a_j)_j$ is summable, the preceding inequality implies that the difference between V_n^F and Q_n is uniformly bounded. Proposition 6.2 formalizes this comparison, and is proved in Section E.

Proposition 6.2 (Comparison with the Value Function of the Dynamic Fluid Policy). *There exists a constant $C_F < \infty$ such that, uniformly over $\mathbf{c}/n \in \mathcal{K}$, $|V_n^F(\mathbf{c}) - Q_n(\mathbf{c})| \leq C_F$.*

We next compare $Q_n(\mathbf{c})$ with $V_n^{\text{OPT}}(\mathbf{c})$. Unlike the dynamic fluid policy, the normalized state $\mathbf{x}$ need not evolve as a martingale under the optimal online policy, hence the same state-concentration argument is no longer available. Recall that by definition, the optimal value function satisfies $V_n^{\text{OPT}} = T^*V_{n-1}^{\text{OPT}}$. We say that a sequence of continuation value functions $\{u_n\}_{n \geq 0}$ is a supersolution to the Bellman recursion if there exists $n_0 \in \mathbb{N}$ such that

$$T^*u_{n-1}(\mathbf{c}) \leq u_n(\mathbf{c}), \quad \forall \mathbf{c} \in \mathbb{R}_+^m, n \geq n_0. \quad (7)$$

By the monotone property of the Bellman operator T^* , if $\{u_n\}_{n \geq 0}$ is a supersolution to the Bellman recursion, then there exists a positive constant $C < \infty$ such that

$$V_n^{\text{OPT}}(\mathbf{c}) \leq u_n(\mathbf{c}) + C, \quad \forall \mathbf{c} \in \mathbb{R}_+^m, n \geq 0.$$

We construct a local Bellman supersolution $\{W_n\}$ based on the corrected fluid reference Q_n , for which (7) holds whenever $\mathbf{c}/n$ falls into a local region containing $\mathcal{K}$ and $n \geq n_0$. We then patch $\{W_n\}$ to obtain a global supersolution $\{\bar{W}_n\}$ satisfying (7) over the entire state space. Furthermore, we show that $\bar{W}_n$ exceeds Q_n by at most a fixed constant on the target set $\mathcal{K}$. This yields a constant upper bound on the optimal value function V_n^{OPT} relative to the reference Q_n , as formalized in Proposition 6.3. We prove Proposition 6.3 in Section F.

Proposition 6.3 (Comparison with the Optimal Value Function). *There exists a constant $C_{\text{OPT}} < \infty$ such that, uniformly over all $(\mathbf{c}, n)$ satisfying $\mathbf{c}/n \in \mathcal{K}$, $V_n^{\text{OPT}}(\mathbf{c}) \leq Q_n(\mathbf{c}) + C_{\text{OPT}}$.*

6.3 Completion of the Proof

Theorem 4.1 directly follows from Propositions 6.1–6.3. Indeed, Propositions 6.2 and 6.3, together with the ordering $V_n^F \leq V_n^{\text{OPT}}$, imply that, for suitable constants C_F and C_{OPT} ,

$$Q_n(\mathbf{c}) - C_F \leq V_n^F(\mathbf{c}) \leq V_n^{\text{OPT}}(\mathbf{c}) \leq Q_n(\mathbf{c}) + C_{\text{OPT}},$$

for any initial state $(\mathbf{c}, n)$ with $\mathbf{c}/n \in \mathcal{K}$. On the other hand, Proposition 6.1 gives, uniformly over all such initial states,

$$Q_n(\mathbf{c}) = n \phi^F(\mathbf{c}/n) - \Gamma(\mathbf{c}_B/n) H_{n-1} + O_{\mathcal{K}}(1).$$

Combining this expansion with the preceding bounds on $V_n^F(\mathbf{c})$ and $V_n^{\text{OPT}}(\mathbf{c})$ and the fact that $H_n - 1 \leq \ln n \leq H_{n-1}$ leads to Theorem 4.1.

7 Conclusion

In this paper, we studied a general online resource allocation problem. As our main result, we show that under standard non-degeneracy conditions, the classic dynamic fluid policy and the optimal online policy admit the same performance expansion: the fluid benchmark minus a logarithmic correction $\Gamma \log n$, up to uniformly bounded remainders. The coefficient Γ captures the interaction between random resource consumption and the curvature of the fluid value function in the binding-resource directions. Consequently, the dynamic fluid policy achieves a constant optimality gap relative to the optimal online policy, in contrast to the order-optimal $\Theta(\ln n)$ gap relative to the offline optimum established in the literature.

In the proof, we compare the value functions of the dynamic fluid policy and the optimal online policy with a common corrected fluid reference. For the dynamic fluid policy, we use a stopping-time and martingale-concentration argument. For the optimal online policy, we use the corrected fluid reference to construct a supersolution to its Bellman equation, to obtain a tight bound on its value function.

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A Proofs and Additional Details for Sections 2 and 3

A.1 Further Details of Lemma 2.1

A.1.1 Definition of the Offline Optimal Upper Bound $V_n^{\text{OFF}}(\mathbf{c})$

Let $\theta_1, \dots, \theta_n$ be independent copies of $\theta \sim \Phi$, where $\theta_t = (R_{a,t}, B_{a,t})_{a \in \mathcal{A}}$ records the action-specific rewards and resource consumptions in period t . The offline benchmark observes the entire sequence $\theta_1, \dots, \theta_n$ before selecting an action in each period and is defined by

$$V_n^{\text{OFF}}(\mathbf{c}) \triangleq \mathbb{E} \left[\max_{\substack{(a_1, \dots, a_n) \in \mathcal{A}^n \\ \sum_{t=1}^n B_{a_t, t} \leq \mathbf{c}}} \sum_{t=1}^n R_{a_t, t} \right].$$

A.1.2 Proof of Lemma 2.1

Since the offline benchmark $V_n^{\text{OFF}}(\mathbf{c})$ observes the entire sequence $\{\theta_t\}_{t=1}^n$ before making any decisions, the value of perfect information implies that $V_n^{\text{OPT}}(\mathbf{c}) \leq V_n^{\text{OFF}}(\mathbf{c})$. We next show that

$$V_n^{\text{OFF}}(\mathbf{c}) \leq n \cdot \phi^{\text{F}}(\rho),$$

that is, the fluid relaxation provides an upper bound on the offline optimal benchmark. This result is standard and can be established in several ways. For completeness, we provide a proof based on the dual formulation of the fluid relaxation. An alternative proof can be obtained from the primal formulation by following a similar argument as in the proof of Proposition 1 of Balseiro et al. (2023).

Fix a dual variable $\lambda \in \mathbb{R}_+^m$. For every realization of the arriving sequence $\{\theta_t\}$ and every feasible action sequence $(a_1, \dots, a_n) \in \mathcal{A}^n$,

$$\begin{aligned} \sum_{t=1}^n R_{a_t, t} &= \lambda^{\text{T}} \sum_{t=1}^n B_{a_t, t} + \sum_{t=1}^n (R_{a_t, t} - \lambda^{\text{T}} B_{a_t, t}) \\ &\leq \lambda^{\text{T}} \mathbf{c} + \sum_{t=1}^n (R_{a_t, t} - \lambda^{\text{T}} B_{a_t, t}) \\ &\leq \lambda^{\text{T}} \mathbf{c} + \sum_{t=1}^n \max_{a \in \mathcal{A}} \{R_{a, t} - \lambda^{\text{T}} B_{a, t}\}, \end{aligned}$$

where the first inequality follows from the capacity constraint $\sum_{t=1}^n B_{a_t, t} \leq \mathbf{c}$ and $\lambda \geq \mathbf{0}$. Maximizing this pathwise inequality over all feasible action sequences and taking expectations gives

$$\begin{aligned} V_n^{\text{OFF}}(\mathbf{c}) &\leq \lambda^{\text{T}} \mathbf{c} + \sum_{t=1}^n \mathbb{E} \left[\max_{a \in \mathcal{A}} \{R_{a, t} - \lambda^{\text{T}} B_{a, t}\} \right] \\ &= \lambda^{\text{T}} \mathbf{c} + n \mathcal{V}(\lambda). \end{aligned}$$

The equality follows from the fact that the opportunities are identically distributed across time and the definition of $\mathcal{V}(\lambda)$. Since the bound holds for every $\lambda \in \mathbb{R}_+^m$, we have:

$$V_n^{\text{OFF}}(\mathbf{c}) \leq \inf_{\lambda \in \mathbb{R}_+^m} \left\{ \lambda^{\text{T}} \mathbf{c} + n \mathcal{V}(\lambda) \right\}$$

$$\begin{aligned}
&= n \inf_{\boldsymbol{\lambda} \in \mathbb{R}_+^m} \left\{ \boldsymbol{\lambda}^\top \boldsymbol{\rho} + \mathcal{V}(\boldsymbol{\lambda}) \right\} \\
&= n \phi^F(\boldsymbol{\rho}),
\end{aligned}$$

where the first equality uses $\mathbf{c} = n\boldsymbol{\rho}$ and the second equality follows from the strong duality between (1) and (2).

A.2 Proof of Lemma 3.1

Let $\boldsymbol{\lambda}^*$ be an optimal solution to the dual problem (2) at some $\boldsymbol{\rho} \in \mathcal{U}$. From (2) we have

$$\phi^F(\boldsymbol{\rho}') \leq \boldsymbol{\lambda}^{*\top} \boldsymbol{\rho}' + \mathcal{V}(\boldsymbol{\lambda}^*) = \phi^F(\boldsymbol{\rho}) + \boldsymbol{\lambda}^{*\top} (\boldsymbol{\rho}' - \boldsymbol{\rho}), \quad \forall \boldsymbol{\rho}' \in \mathbb{R}_+^m,$$

where the equality follows from the optimality of $\boldsymbol{\lambda}^*$ at $\boldsymbol{\rho}$. Hence, $\boldsymbol{\lambda}^*$ is a supergradient of the concave function ϕ^F at $\boldsymbol{\rho}$.

On the other hand, since $\phi^F(\boldsymbol{\rho}) = \phi_{\mathcal{B}}(\boldsymbol{\rho}_{\mathcal{B}})$ and $\phi_{\mathcal{B}} \in C^{2,\alpha}(\mathcal{U}_{\mathcal{B}})$ by Assumption 3.1, the fluid value function ϕ^F is differentiable on $\mathcal{U}$, with gradient

$$\nabla \phi^F(\boldsymbol{\rho}) = (\boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}}), \mathbf{0}_{\mathcal{S}}).$$

Because a differentiable concave function admits a unique supergradient at every interior point of its domain, it follows that

$$\boldsymbol{\lambda}^* = \nabla \phi^F(\boldsymbol{\rho}) = (\boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}}), \mathbf{0}_{\mathcal{S}}).$$

Therefore, every optimal dual solution coincides with this vector, implying that the optimal dual solution is unique.

A.3 Proof of Lemma 3.2

For any $\boldsymbol{\rho} \in \mathcal{U}$, we have $\boldsymbol{\lambda}^*(\boldsymbol{\rho}) = (\boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}}), \mathbf{0}_{\mathcal{S}})$ by Lemma 3.1. Therefore, from (2),

$$\begin{aligned}
\phi^F(\boldsymbol{\rho}) &= \boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})^\top \boldsymbol{\rho}_{\mathcal{B}} + \mathbb{E} \left[\max_{a \in \mathcal{A}} \left\{ R_a - \boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})^\top B_{a,\mathcal{B}} \right\} \right] \\
&= \boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})^\top \boldsymbol{\rho}_{\mathcal{B}} + \mathbb{E} \left[R_{a^F(\boldsymbol{\rho}_{\mathcal{B}})} - \boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}})^\top B_{a^F(\boldsymbol{\rho}_{\mathcal{B}}),\mathcal{B}} \right] \\
&= \mathbb{E} \left[R_{a^F(\boldsymbol{\rho}_{\mathcal{B}})} \right],
\end{aligned}$$

where the first equality follows from the optimality of $\boldsymbol{\lambda}^*(\boldsymbol{\rho})$, the second equality follows from the definition of a^F in (4), and the third equality uses $\mathbb{E}[B_{a^F(\boldsymbol{\rho}_{\mathcal{B}}),\mathcal{B}}] = \boldsymbol{\rho}_{\mathcal{B}}$ from Assumption 3.2.

A.4 Proof of Proposition 3.3

In the proof, we set $\kappa = 2\bar{b}$, where $\bar{b}$ is the constant from Assumption 2.1. For each period t , let $Y_t = (Y_{tj})_{j \in [m]} \triangleq B_{a_t^0,t} \in \mathbb{R}_+^m$ denote the resource-consumption vector associated with action a^0 in period t . Then, $\mathbb{E}[Y_t] = \boldsymbol{\mu}$ for every t .

To prove the performance gap, define the safety threshold by

$$h_n \triangleq n\delta_n - \bar{b}.$$

For each $j \in [m]$, define the cumulative consumption deviation by

$$S_{kj} \triangleq \sum_{t=1}^k \{Y_{tj} - \mu_j\}.$$

Finally, define the corresponding stopping time by

$$\tau_n \triangleq \min \left\{ k \in \{1, \dots, n-1\} : S_{kj} > h_n \text{ for some } j \in [m] \right\},$$

with the convention $\tau_n = n$ if the set is empty.

Note that the constrained and unconstrained greedy policies take the same action and receive the same reward up to period τ_n . To see this, suppose that the two policies have agreed during the first $k < \tau_n$ periods. For each $j \in [m]$, the remaining capacity satisfies

$$c_{nj} - \sum_{t=1}^k Y_{tj} = (n-k)\mu_j + n(\rho_{nj} - \mu_j) - S_{kj} \geq (n-k)\mu_j + n\delta_n - S_{kj} \geq \bar{b},$$

where the first inequality uses the definition of δ_n and the second uses $S_{kj} \leq h_n$. By Assumption 2.1, every action consumes at most $\bar{b}$ units of each resource. Thus, every action is feasible in period $k+1$, so the constrained greedy policy also selects a_{k+1}^0 .

Since the constrained and unconstrained greedy policies agree up to period τ_n , and the one-period loss is at most $(\max_{a \in \mathcal{A}} R_a)^+$, whose expectation is at most $\bar{r}$ under Assumption 2.1, we have

$$0 \leq n \mathbb{E} \left[\max_a R_a \right] - V_n^G(\mathbf{c}_n) \leq \bar{r} \cdot \mathbb{E}(n - \tau_n)^+ \leq n\bar{r} \cdot \mathbb{P}\{\tau_n < n\}.$$

Moreover, the tail probability can be bounded as:

$$\begin{aligned} \mathbb{P}\{\tau_n < n\} &= \mathbb{P} \left\{ \bigcup_{j \in [m]} \left\{ \max_{k \leq n-1} S_{kj} > h_n \right\} \right\} \\ &\leq \sum_{j \in [m]} \mathbb{P} \left\{ \max_{k \leq n-1} S_{kj} > h_n \right\} \\ &\leq m \exp \left(-\frac{2h_n^2}{n\bar{b}^2} \right) \\ &\leq \frac{m}{n^2}. \end{aligned}$$

The first inequality follows from the union bound. The second inequality follows from the maximal Azuma-Hoeffding inequality (see Lemma A.1) because, for each resource $j \in [m]$, the deviations $Y_{tj} - \mu_j$ are independent and have mean zero, with $Y_{tj} - \mu_j \in [-\mu_j, \bar{b} - \mu_j]$. The third inequality follows from the definition of h_n , the choice $\kappa = 2\bar{b}$, and the bound $(2\sqrt{n \ln n} - 1)^2 \geq n \ln n$ for every $n \in \mathbb{N}_+$. Combining this tail estimate with the performance bound gives

$$0 \leq n \mathbb{E} \left[\max_{a \in \mathcal{A}} R_a \right] - V_n^G(\mathbf{c}_n) \leq \frac{m\bar{r}}{n} = o(1).$$

This proves the proposition.

Lemma A.1 (Maximal Azuma-Hoeffding Inequality; Theorem 3.2.1 of Roch 2024). *Let $(Z_t)_{t \in \mathbb{Z}_+}$ be a martingale with respect to the filtration $(\mathcal{F}_t)_{t \in \mathbb{Z}_+}$. Suppose there exist predictable processes $(A_t)_{t \geq 1}$*

and $(B_t)_{t \geq 1}$, with $A_t, B_t \in \mathcal{F}_{t-1}$, and constants $0 < c_t < \infty$ such that, for all $t \geq 1$, almost surely,

$$A_t \leq Z_t - Z_{t-1} \leq B_t \quad \text{and} \quad B_t - A_t \leq c_t.$$

Then, for every $\beta > 0$ and $t \geq 1$,

$$\mathbb{P} \left(\max_{0 \leq i \leq t} (Z_i - Z_0) \geq \beta \right) \leq \exp \left(- \frac{2\beta^2}{\sum_{i=1}^t c_i^2} \right).$$

B Proofs and Additional Details for Section 5

B.1 Further Details on the Uniform Multisecretary Model in Section 5.1

We restrict attention to $\rho \in \mathcal{U} \triangleq (0, 1)$. We first derive the fluid value function $\phi^F(\rho)$. For any dual variable $\lambda \geq 0$, we have:

$$\mathcal{V}(\lambda) = \mathbb{E}[(V - \lambda)^+] = \begin{cases} \frac{1}{2}(1 - \lambda)^2 & 0 \leq \lambda \leq 1 \\ 0 & \lambda \geq 1. \end{cases}$$

Thus,

$$\phi^F(\rho) = \min_{\lambda \geq 0} \{\lambda\rho + \mathcal{V}(\lambda)\} = \min_{\lambda \in [0, 1]} \left\{ \lambda\rho + \frac{1}{2}(1 - \lambda)^2 \right\} = \rho - \frac{1}{2}\rho^2,$$

where the optimal dual variable is $\lambda^*(\rho) = 1 - \rho$.

We next verify that Assumption 2.1 and Assumptions 3.1 to 3.3 hold. Assumption 2.1 holds with $\bar{r} = \bar{b} = 1$. Because ϕ^F is C^∞ , Assumption 3.1 holds with binding-resource set $\mathcal{B} = \{1\}$ and $\phi_{\mathcal{B}}(\rho) = \phi^F(\rho)$. It remains to verify Assumptions 3.2 and 3.3.

For Assumption 3.2, the relevant fluid dual-price action in (4) is

$$a^F(\rho) = \mathbb{1}\{V > 1 - \rho\}.$$

Direct integration yields

$$\mathbb{E} \left[B_{a^F(\rho)} \right] = \mathbb{P}(V > 1 - \rho) = \rho.$$

Therefore, Assumption 3.2 holds.

For Assumption 3.3, the only top-versus-competitor gap is $S_{a^F(\rho)} - S_{1-a^F(\rho)} = |V - (1 - \rho)|$, and its weight is $d_{01} = 1$. Thus, uniformly over $\rho \in \mathcal{K}$,

$$\mathbb{E} \left[d_{a^F(\rho), 1-a^F(\rho)} \mathbb{1} \left\{ S_{a^F(\rho)}(\rho) - S_{1-a^F(\rho)}(\rho) \leq t d_{a^F(\rho), 1-a^F(\rho)} \right\} \right] = \mathbb{P}(|V - (1 - \rho)| \leq t) \leq 2t.$$

Therefore, Assumption 3.3 holds with $\beta_{\mathcal{K}} = 1$ and $C_{\mathcal{K}} = 2$.

Finally, we derive the logarithmic coefficient in Theorem 4.1. Note that $B_{a^F(\rho)} = a^F(\rho)$, which is Bernoulli with parameter ρ . Therefore,

$$\Sigma(\rho) = \mathbb{V}\text{ar} \left[B_{a^F(\rho)} \right] = \rho(1 - \rho), \quad \Gamma(\rho) = \frac{1}{2}\rho(1 - \rho).$$

B.2 Further Details on the Stochastic Knapsack Problem in Section 5.2

Let W and V denote the weight and profit of an item, respectively. For a threshold $m \geq 0$, define

$$F_a(m) \triangleq \mathbb{E}[W \mathbf{1}\{V \geq mW\}], \quad F_c(m) \triangleq \mathbb{E}[V \mathbf{1}\{V \geq mW\}].$$

Therefore, $F_a(m)$ is the expected capacity consumed by accepting all items with a profit-to-weight ratio of at least m , and $F_c(m)$ is the corresponding expected reward.

Condition 1 of Lueker (1998) assumes that, for some finite d , $0 \leq W, V \leq d$ with probability one. Condition 2 assumes that there is an $m_0 > 0$ satisfying $\beta_0 = F_a(m_0)$ and that, on a neighborhood N_m of m_0 , the derivative $f_a \triangleq F'_a$ exists, is strictly negative, and is Lipschitz continuous.

Since $f_a(m_0) < 0$ and f_a is Lipschitz continuous, we may shrink N_m so that there exist constants $0 < \underline{\alpha} \leq \bar{\alpha} < \infty$ such that

$$\underline{\alpha} \leq -f_a(m) \leq \bar{\alpha}, \quad m \in N_m.$$

Thus, function F_a is strictly decreasing on N_m . Choose an open neighborhood $\mathcal{U}$ of β_0 such that $\bar{\mathcal{U}} \subseteq F_a(N_m)$, and define

$$m_\rho \triangleq F_a^{-1}(\rho), \quad \alpha_\rho \triangleq -f_a(m_\rho) > 0, \quad \rho \in \mathcal{U}.$$

After shrinking $\mathcal{U}$ if necessary, the compact set $\{m_\rho : \rho \in \bar{\mathcal{U}}\}$ lies in the interior of N_m .

In Lemma B.1, we show that the boundary events are immaterial under Lueker's Conditions 1 and 2; specifically, changing the acceptance rule from $V \geq mW$ to $V > mW$ changes neither expected consumption nor expected reward.

Lemma B.1. *Under Conditions 1 and 2 of Lueker (1998), for any $m \in N_m$,*

$$W \mathbf{1}\{V = mW\} = 0, \quad V \mathbf{1}\{V = mW\} = 0, \quad W^2 \mathbf{1}\{V = mW\} = 0, \quad a.s.$$

Lemma B.1 implies that $\frac{d}{d\lambda} \mathbb{E}[(V - \lambda W)^+] = -F_a(\lambda)$, as stated in Lemma B.2.

Lemma B.2. *For any $\lambda \in N_m$,*

$$\frac{d}{d\lambda} \mathbb{E}[(V - \lambda W)^+] = -\mathbb{E}[W \mathbf{1}\{V \geq \lambda W\}] = -F_a(\lambda).$$

The rest of this section is organized as follows. In Section B.2.1, we derive the fluid relaxation and characterize the optimal dual variable for the stochastic knapsack problem. Section B.2.2 shows that our dynamic fluid policy coincides with the OnLineGreedy policy in Lueker (1998). Section B.2.3 verifies that Conditions 1 and 2 of Lueker (1998) imply Assumption 2.1 and Assumptions 3.1 to 3.3. Section B.2.4 shows that the first-order and logarithmic terms in Lueker's expansion coincide with ours in Theorem 4.1. Finally, Section B.2.5 analyzes the special case with uniformly distributed weights and values.

Proof of Lemma B.1. For every $m \in N_m$ and every $h > 0$ such that $m + h \in N_m$, we have

$$F_a(m) - F_a(m + h) = \mathbb{E}[W \mathbf{1}\{mW \leq V < (m + h)W\}].$$

Letting $h \downarrow 0$ yields

$$\begin{aligned} \mathbb{E}[W \mathbf{1}\{V = mW\}] &= \lim_{h \downarrow 0} \mathbb{E}[W \mathbf{1}\{mW \leq V < (m + h)W\}] \\ &= \lim_{h \downarrow 0} \{F_a(m) - F_a(m + h)\} = 0, \end{aligned}$$

where the first equality follows from the dominated convergence theorem and the second equality from the continuity of F_a . Moreover, since $W \leq d$ almost surely by Condition 1, we have

$$0 \leq \mathbb{E}[W^2 \mathbf{1}\{V = mW\}] \leq d \cdot \mathbb{E}[W \mathbf{1}\{V = mW\}] = 0.$$

This proves $\mathbb{E}[W^2 \mathbf{1}\{V = mW\}] = 0$. Finally, because $V = mW$ on the boundary event, we have

$$\mathbb{E}[V \mathbf{1}\{V = mW\}] = m \cdot \mathbb{E}[W \mathbf{1}\{V = mW\}] = 0.$$

Since $W \mathbf{1}\{V = mW\}$, $V \mathbf{1}\{V = mW\}$, and $W^2 \mathbf{1}\{V = mW\}$ are nonnegative and have zero expectations, they must each be zero almost surely. $\square$

Proof of Lemma B.2. Fix $\lambda \in N_m$. For every nonzero h ,

$$\left| (V - (\lambda + h)W)^+ - (V - \lambda W)^+ \right| \leq |h|W.$$

Dividing by $|h|$ gives

$$\left| \frac{(V - (\lambda + h)W)^+ - (V - \lambda W)^+}{h} \right| \leq W.$$

Because $0 \leq W \leq d$ almost surely, the dominated convergence theorem gives

$$\begin{aligned} \frac{d}{d\lambda} \mathbb{E}[(V - \lambda W)^+] &= \lim_{h \rightarrow 0} \mathbb{E} \left\{ \frac{(V - (\lambda + h)W)^+ - (V - \lambda W)^+}{h} \right\} \\ &= \mathbb{E} \left\{ \lim_{h \rightarrow 0} \frac{(V - (\lambda + h)W)^+ - (V - \lambda W)^+}{h} \right\} \\ &= \mathbb{E} \left[\frac{d}{d\lambda} (V - \lambda W)^+ \right] = -\mathbb{E}[W \mathbf{1}\{V \geq \lambda W\}] = -F_a(\lambda). \end{aligned}$$

The third and fourth equalities follow from the identity

$$\frac{d}{d\lambda} (V - \lambda W)^+ = -W \mathbf{1}\{V \geq \lambda W\}, \quad a.s.$$

Indeed, the derivative exists almost surely because $W \mathbf{1}\{V = \lambda W\} = 0$ almost surely by Lemma B.1. $\square$

B.2.1 Fluid Relaxation and Optimal Dual Variable

Fix $\rho \in \mathcal{U}$. Because the action set is binary, an opportunity-dependent randomized control can be represented by a measurable function $x : \mathbb{R}_+^2 \rightarrow [0, 1]$, where $x(W, V)$ is the conditional probability of accepting an item with weight W and profit V . Specializing the fluid relaxation in (1) to this accept–reject problem gives

$$\begin{aligned} \phi^F(\rho) &= \sup_{x: \mathbb{R}_+^2 \rightarrow [0, 1]} \mathbb{E}[Vx(W, V)] \\ \text{subject to} \quad &\mathbb{E}[Wx(W, V)] \leq \rho. \end{aligned}$$

Let $\lambda \geq 0$ be a multiplier for the expected-capacity constraint. The dual problem in (2) becomes

$$\phi^F(\rho) = \inf_{\lambda \geq 0} \{ \lambda \rho + \mathbb{E}[(V - \lambda W)^+] \} = \inf_{\lambda \geq 0} g_\rho(\lambda),$$

where $g_\rho(\lambda) \triangleq \lambda \rho + \mathbb{E}[(V - \lambda W)^+]$. Note that the function g_ρ is convex. Moreover, for $\lambda \in N_m$, Lemma B.2 yields

$$g'_\rho(\lambda) = \rho - F_a(\lambda).$$

By the first-order condition, the optimal dual variable $\lambda^*(\rho)$ is uniquely given by

$$\lambda^*(\rho) = F_a^{-1}(\rho) = m_\rho > 0. \quad (8)$$

Substituting this minimizer into the dual objective yields

$$\begin{aligned} \phi^F(\rho) &= g_\rho(m_\rho) = m_\rho \rho + \mathbb{E}[(V - m_\rho W)^+] \\ &= m_\rho F_a(m_\rho) + F_c(m_\rho) - m_\rho F_a(m_\rho) \\ &= F_c(m_\rho), \end{aligned} \quad (9)$$

where the second equality uses $F_a(m_\rho) = \rho$, the fact that $\mathbb{E}[(V - m_\rho W)^+] = \mathbb{E}[(V - m_\rho W) \mathbb{1}\{V > m_\rho W\}]$, and the definitions of F_a and F_c .

In addition,

$$x_\rho^*(W, V) \triangleq \mathbb{1}\{V > m_\rho W\}.$$

is an optimal primal solution to the fluid relaxation. This threshold control exhausts the expected capacity

$$\mathbb{E}[W x_\rho^*(W, V)] = \mathbb{E}[W \mathbb{1}\{V > m_\rho W\}] = F_a(m_\rho) = \rho, \quad (10)$$

and its expected reward is

$$\mathbb{E}[V x_\rho^*(W, V)] = F_c(m_\rho) = \phi^F(\rho).$$

Finally, from the envelope theorem we have

$$\phi^{F'}(\rho) = m_\rho = F_a^{-1}(\rho). \quad (11)$$

B.2.2 Policy Correspondence

Consider a state with k periods remaining and capacity c , where $c/k \in \mathcal{U}$. The dynamic fluid policy re-solves the fluid problem at the normalized remaining capacity c/k and uses the optimal dual price

$$\lambda^*(c/k) = m_{c/k} = F_a^{-1}(c/k)$$

according to (8). Therefore, our dynamic fluid policy and the OnLineGreedy policy of Lueker (1998) use the same optimal fluid dual price and hence the same acceptance threshold.

B.2.3 Verification of Assumptions

In this section, we verify that Conditions 1 and 2 of Lueker (1998) imply Assumption 2.1 and Assumptions 3.1 to 3.3. First, Condition 1 implies Assumption 2.1 with $\bar{r} = \bar{b} = d$. We then set $\mathcal{B} = \{1\}$ and $\phi_{\mathcal{B}}(\rho) = \phi^F(\rho)$ and verify that Assumptions 3.1 to 3.3 are satisfied.

To verify Assumption 3.1, from (11) and the inverse function theorem we have

$$\phi''_{\mathcal{B}}(\rho) = \frac{1}{f_a(m_\rho)} = -\frac{1}{\alpha_\rho}. \quad (12)$$

Since f_a is Lipschitz and bounded away from zero on N_m , the map $\rho \mapsto 1/f_a(m_\rho)$ is also locally Lipschitz. Thus, $\phi_{\mathcal{B}} \in C^{2,1}(\mathcal{U})$, which verifies Assumption 3.1 with Hölder exponent $\alpha = 1$.

We now verify Assumption 3.2. The static fluid action in (4) is $a^F(\rho) = x_\rho^*(W, V) = \mathbb{1}\{V > m_\rho W\}$. Equation (10) gives $\mathbb{E}[B_{a^F(\rho)}] = \rho$. Since there is only one resource and it binds throughout $\mathcal{U}$, the strict-slackness and smooth slack-consumption requirements are vacuous. Hence, Assumption 3.2 holds.

Finally, we verify Assumption 3.3. With a binary action set, the consumption difference between the two actions is W , and the top-versus-competitor score gap is $|V - m_\rho W|$, as m_ρ is the optimal dual variable by (8). Fix a compact set $\mathcal{K} \subseteq \mathcal{U}$ and choose $t_{\mathcal{K}} > 0$ so that $m_\rho \pm t \in N_m$ for every $\rho \in \mathcal{K}$ and $0 \leq t \leq t_{\mathcal{K}}$. We have

$$\mathbb{E}[W \mathbb{1}\{|V - m_\rho W| \leq tW\}] = F_a(m_\rho - t) - F_a(m_\rho + t) = - \int_{m_\rho - t}^{m_\rho + t} f_a(s) ds \leq 2\bar{\alpha}t.$$

Hence, Assumption 3.3 holds with $\beta_{\mathcal{K}} = 1$ and $C_{\mathcal{K}} = 2\bar{\alpha}$.

B.2.4 Coefficient Correspondence

Define

$$\kappa(\rho) \triangleq F_c(m_\rho) \quad \text{and} \quad \sigma_\rho^2 \triangleq \text{Var}(W \mathbb{1}\{V \geq m_\rho W\})$$

following Lueker (1998). By (9), we have $\phi^F(\rho) = F_c(m_\rho) = \kappa(\rho)$. Moreover, by definition,

$$\Sigma(\rho) = \text{Var}(W \mathbb{1}\{V \geq m_\rho W\}) = \sigma_\rho^2.$$

Substituting this equation and (12) into (5) yields

$$\Gamma(\rho) = -\frac{1}{2}\phi''_{\mathcal{B}}(\rho)\Sigma(\rho) = \frac{\sigma_\rho^2}{2\alpha_\rho}.$$

In Lueker's notation, $\alpha_0 = \alpha_{\beta_0}$ and $\sigma_0^2 = \sigma_{\beta_0}^2$. Hence, at $\rho = \beta_0$, $\kappa(\beta_0) = \phi^F(\beta_0)$ and $\sigma_0^2/(2\alpha_0) = \Gamma(\beta_0)$. Therefore, the first-order and logarithmic coefficients in Lueker's Theorem 2 coincide with ours in Theorem 4.1.

B.2.5 The Case with Uniformly Distributed Weights and Values

In this section, we consider the special case in which V and W are independent standard uniform random variables. We restrict attention to $\rho = c/n \in \mathcal{U} \triangleq (0, 1/2)$, on which the capacity constraint is binding in the fluid relaxation.

The Fluid Value Function and Dynamic Fluid Policy For any dual variable $\lambda \geq 0$, we have:

$$\mathcal{V}(\lambda) = \mathbb{E}[(V - \lambda W)^+] = \begin{cases} \frac{1}{2} - \frac{\lambda}{2} + \frac{\lambda^2}{6}, & 0 \leq \lambda \leq 1, \\ \frac{1}{6\lambda}, & \lambda \geq 1. \end{cases}$$

Therefore,

$$\phi^F(\rho) = \min_{\lambda \geq 0} \{\lambda\rho + \mathcal{V}(\lambda)\} = \begin{cases} \sqrt{\frac{2}{3}}\rho, & 0 < \rho \leq 1/6, \\ \frac{1}{2} - \frac{3}{2}(1/2 - \rho)^2, & 1/6 \leq \rho < 1/2, \end{cases} \quad (13)$$

and the optimal dual variable is

$$\lambda^*(\rho) = \begin{cases} \frac{1}{\sqrt{6}\rho}, & 0 < \rho \leq 1/6, \\ 3(1/2 - \rho), & 1/6 \leq \rho < 1/2. \end{cases} \quad (14)$$

To obtain the above, note that the objective $\lambda\rho + \mathcal{V}(\lambda)$ is convex in λ and has an interior minimizer. As a result, the optimal solution $\lambda^*(\rho)$ is characterized by the first-order condition; substituting it into the objective gives $\phi^F(\rho)$.

At a state with c units of remaining capacity and n periods remaining, the dynamic fluid policy accepts the item if and only if $W \leq c$ and $V > \lambda^*(c/n)W$, with $\lambda^*(\rho)$ given above.

Verification of Assumptions We next verify that Assumption 2.1 and Assumptions 3.1 – 3.3 hold for the uniform case, with binding-resource set $\mathcal{B} = \{1\}$ and fluid function $\phi_B(\rho) = \phi^F(\rho)$.

Assumption 2.1 holds with $\bar{r} = \bar{b} = 1$.

To verify Assumption 3.1, we differentiate (13) to obtain the following derivatives:

$$\begin{aligned} \phi^{F'}(\rho) &= \begin{cases} \frac{1}{\sqrt{6}\rho}, & 0 < \rho < 1/6, \\ 3(1/2 - \rho), & 1/6 < \rho < 1/2, \end{cases} \\ \phi^{F''}(\rho) &= \begin{cases} -\frac{1}{2\sqrt{6}\rho^{3/2}}, & 0 < \rho < 1/6, \\ -3, & 1/6 < \rho < 1/2. \end{cases} \end{aligned}$$

At $\rho = 1/6$, the two branches of the fluid function and their first two derivatives agree:

$$\phi^F(1/6-) = \phi^F(1/6+) = 1/3, \quad \phi^{F'}(1/6-) = \phi^{F'}(1/6+) = 1, \quad \phi^{F''}(1/6-) = \phi^{F''}(1/6+) = -3.$$

Consequently, $\phi^F(\rho)$ is twice continuously differentiable on $(0, 1/2)$. In addition, the second-order derivative $\phi^{F''}(\rho)$ is Lipschitz (i.e., $\alpha = 1$) on every compact subset of $(0, 1/2)$. To see this, note that on a compact interval to the left of the junction $\rho = 1/6$, the derivative of $\phi^{F''}$ is bounded; on the right-hand branch, $\phi^{F''}$ is constant. If $x < 1/6 < y$, continuity at the junction and a left-branch derivative bound L give $|\phi^{F''}(x) - \phi^{F''}(y)| \leq L(1/6 - x) \leq L|y - x|$. Therefore, $\phi_B = \phi^F \in C^{2,1}(0, 1/2)$, and this verifies Assumption 3.1.

We now verify that Assumption 3.2 holds. For every dual variable $\lambda \geq 0$,

$$\mathbb{E}[W \mathbb{1}\{V \geq \lambda W\}] = \begin{cases} \frac{1}{2} - \frac{\lambda}{3}, & 0 \leq \lambda \leq 1, \\ \frac{1}{6\lambda^2}, & \lambda \geq 1. \end{cases} \quad (15)$$

The fluid dual-price action in (4) is

$$a^F(\rho) \triangleq \mathbb{1}\{V > \lambda^*(\rho)W\}.$$

Therefore,

$$\mathbb{E} \left[B_{a^F(\rho)} \right] = \mathbb{E} [W a^F(\rho)] = \mathbb{E} [W \mathbb{1}\{V > \lambda^*(\rho) W\}] = \rho,$$

where the last equality follows from (14) and (15). Consequently, Assumption 3.2 holds. Because $\mathcal{B}^c = \emptyset$, the slack-resource conditions are vacuous.

Finally, we verify Assumption 3.3. The only top-versus-competitor gap is $S_{a^F}(\rho) - S_{1-a^F}(\rho) = |V - \lambda^*(\rho)W|$, and its weight is $d_{01} = W$. Therefore, uniformly over $\rho \in \mathcal{K}$,

$$\begin{aligned} \mathbb{E} \left[d_{a^F(\rho), 1-a^F(\rho)} \mathbb{1} \left\{ S_{a^F}(\rho) - S_{1-a^F}(\rho) \leq t d_{a^F(\rho), 1-a^F(\rho)} \right\} \right] &= \mathbb{E} \left[W \cdot \mathbb{1} \left\{ |V - \lambda^*(\rho)W| \leq tW \right\} \right] \\ &\leq 2t \cdot \mathbb{E} [W^2] = \frac{2t}{3}. \end{aligned}$$

The inequality follows because, conditional on W , the interval of V -values satisfying the event has length at most $2tW$. Therefore, Assumption 3.3 holds with $\beta_{\mathcal{K}} = 1$ and $C_{\mathcal{K}} = \frac{2}{3}$.

Implication of Theorem 4.1 and Derivation of the Logarithmic Coefficient By Theorem 4.1, for every compact set $\mathcal{K} \Subset (0, 1/2)$, uniformly over all states satisfying $\rho = c/n \in \mathcal{K}$, both $V_n^{\text{OPT}}(c)$ and $V_n^F(c)$ admit the expansion

$$n\phi^F(\rho) - \Gamma(\rho) \log n + O_{\mathcal{K}}(1).$$

Consequently, the dynamic fluid policy has a uniformly bounded optimality gap relative to the optimal online policy.

We now derive the logarithmic coefficient. By direct integration, we have

$$\mathbb{E} \left[(W a^F(\rho))^2 \right] = \mathbb{E} [W^2 \mathbb{1}\{V > \lambda^*(\rho) W\}] = \begin{cases} (12\lambda^*(\rho)^3)^{-1}, & 0 < \rho \leq 1/6, \\ \frac{1}{3} - \frac{1}{4}\lambda^*(\rho), & 1/6 \leq \rho < 1/2. \end{cases}$$

Consequently, $\Sigma(\rho) = \mathbb{E}[(W a^F(\rho))^2] - \rho^2$, and

$$\Gamma(\rho) = -\frac{1}{2}\Sigma(\rho)\phi^{F''}(\rho) = \begin{cases} \frac{3 - \sqrt{6\rho}}{24}, & 0 < \rho \leq 1/6, \\ \frac{-24\rho^2 + 18\rho - 1}{16}, & 1/6 \leq \rho < 1/2. \end{cases}$$

B.3 Further Details on the Online Linear Programming Problem in Section 5.3

In this section, we show that Assumptions 1–6 of Bray (2025), with Assumption 6 equivalent to their Lemma 1, imply our Assumption 2.1 and Assumptions 3.1–3.3, except for the local α -Hölder condition on $\nabla^2\phi_{\mathcal{B}}$ required in part 2 of Assumption 3.1.

Bray's Assumptions 1–4 imply our Assumption 2.1. In the remainder of this section, we take $\mathcal{B} = [m]$ and $\phi_{\mathcal{B}} = \phi^F$ and verify the other assumptions. We first characterize the fluid value function ϕ^F and its derivatives in Section B.3.1. We then verify Assumptions 3.1–3.3 in the remaining subsections.

B.3.1 Characterization of the Fluid Value Function ϕ^F and Its Derivatives

We follow Bray's notation, so b , β , and y denote vectors. Under the binary-action representation $(R_0, B_0) = (0, \mathbf{0})$ and $(R_1, B_1) = (u, \mathbf{a})$ and for a normalized inventory vector b , Bray's limiting dual

objective $\Lambda_\infty^b(y) = b^\top y + \mathbb{E}[(u - \mathbf{a}^\top y)^+]$ is exactly the objective of our fluid dual problem (2). For notational convenience, define $g(y) \triangleq \mathbb{E}[(u - \mathbf{a}^\top y)^+]$, so that $\Lambda_\infty^b(y) = b^\top y + g(y)$.

Given a dual price y , define the expected resource consumption by

$$q(y) \triangleq \mathbb{E}[\mathbf{a} \mathbf{1}\{u > \mathbf{a}^\top y\}] \in \mathbb{R}_+^m.$$

For y in the neighborhood of y_β specified by Bray's Assumption 6, we have

$$\nabla_y \Lambda_\infty^b(y) = b - q(y) \quad \text{and} \quad \nabla_y^2 \Lambda_\infty^b(y) = H(y) \triangleq -Dq(y), \quad (16)$$

where $Dq(y)$ denotes the Jacobian matrix of $q(y)$. We prove (16) at the end of this subsection.

Bray's Assumption 5 guarantees a minimizer $y_\beta > \mathbf{0}$. Because y_β lies in the interior of $\mathbb{R}_+^m$, the gradient of $\Lambda_\infty^b(y)$ at y_β must be zero. Substituting $b = \beta$ and $y = y_\beta$ into (16) yields

$$\mathbf{0} = \nabla_y \Lambda_\infty^\beta(y_\beta) = \beta - q(y_\beta), \quad \text{so} \quad q(y_\beta) = \beta. \quad (17)$$

We remark that β is componentwise positive (i.e., $\beta > \mathbf{0}$). Suppose, to the contrary, that $\beta_j = 0$ for some j . The identity $q(y_\beta) = \beta$ gives $q_j(y_\beta) = \beta_j = 0$. Since $q_j(y) \geq 0$ for every y , the differentiable function q_j attains its global minimum at the interior point y_β . Its first-order condition is therefore $\frac{\partial}{\partial y_k} q_j(y_\beta) = 0$ for every $k \in [m]$. Equivalently, row j of $Dq(y_\beta)$ is zero, which contradicts the nonsingularity of $Dq(y_\beta)$. Hence, $\beta > \mathbf{0}$.

By Lemma 1 of Bray (2025), the matrix $H(y)$ is continuous and full rank in a neighborhood of y_β . The implicit function theorem then yields open neighborhoods $\mathcal{U}$ of β and $\mathcal{V}$ of y_β and a continuously differentiable map $b \mapsto y_b \in \mathcal{V}$ such that, for every $b \in \mathcal{U}$, $q(y_b) = b$ and y_b is the unique $y \in \mathcal{V}$ satisfying $q(y) = b$. We state this result in Proposition B.3 and defer its proof to the end of the subsection.

Proposition B.3. *There exist open, convex neighborhoods $\mathcal{U} \subseteq (0, \infty)^m$ of β and $\mathcal{V} \subseteq (0, \infty)^m$ of y_β and a continuously differentiable map $y : \mathcal{U} \rightarrow \mathcal{V}$ such that, for every $b \in \mathcal{U}$, $q(y(b)) = b$ and $y(b)$ is the unique $y \in \mathcal{V}$ satisfying $q(y) = b$. For ease of notation, let $y_b \triangleq y(b)$. Then $D_b y_b = -H(y_b)^{-1}$.*

We note that y_b is the unique global minimizer of Λ_∞^b . The identity $q(y_b) = b$ implies $\nabla_y \Lambda_\infty^b(y_b) = \mathbf{0}$ by (16). Moreover, the map $y \mapsto (u - \mathbf{a}^\top y)^+$ is convex for every realization of $(u, \mathbf{a})$, so Λ_∞^b is convex. Hence, its zero-gradient point y_b is a global minimizer. Finally, since H is positive definite on $\mathcal{V}$, Λ_∞^b is strictly convex on $\mathcal{V}$; therefore, y_b is the unique global minimizer.

We now characterize the fluid value function ϕ^F and its derivatives. Consider the threshold control

$$x_b(u, \mathbf{a}) \triangleq \mathbf{1}\{u > \mathbf{a}^\top y_b\}.$$

Its expected resource consumption is $\mathbb{E}[\mathbf{a} x_b] = q(y_b) = b$, so it is feasible for the fluid relaxation at b and exhausts every resource. Its expected reward satisfies

$$\begin{aligned} \mathbb{E}[u x_b] &= \mathbb{E}[(u - \mathbf{a}^\top y_b)^+] + \mathbb{E}[\mathbf{a}^\top y_b \mathbf{1}\{u > \mathbf{a}^\top y_b\}] \\ &= g(y_b) + y_b^\top q(y_b) \\ &= g(y_b) + y_b^\top b = \Lambda_\infty^b(y_b). \end{aligned}$$

Therefore, weak duality implies that the threshold control x_b and the dual price y_b are optimal for the primal and dual problems of the fluid relaxation, respectively, and

$$\phi^F(b) = \Lambda_\infty^b(y_b) = b^\top y_b + g(y_b).$$

For each coordinate $i \in [m]$, since y_b is the unique minimizer of $\phi^F(b)$, the envelope theorem gives $\frac{\partial \phi^F(b)}{\partial b_i} = (y_b)_i$. Consequently,

$$\nabla \phi^F(b) = y_b, \quad \nabla^2 \phi^F(b) = D y_b = -H(y_b)^{-1}, \quad (18)$$

where the last equality follows from Proposition B.3.

Proof of Equation (16). Fix y in the neighborhood on which Bray's Assumption 6 holds, and let e_j be the j th standard basis vector. For every $h > 0$,

$$0 \leq \mathbb{E}[a_j \mathbf{1}\{u = \mathbf{a}^T y\}] \leq q_j(y - h e_j) - q_j(y + h e_j).$$

Since $q(y)$ is continuous at y by Bray's Assumption 6, letting $h \downarrow 0$ gives

$$\mathbb{E}[a_j \mathbf{1}\{u = \mathbf{a}^T y\}] = 0.$$

Because $a_j \mathbf{1}\{u = \mathbf{a}^T y\}$ is nonnegative and has expectation zero, $a_j \mathbf{1}\{u = \mathbf{a}^T y\} = 0$ almost surely.

Next, for any $h \neq 0$, we have

$$\left| [u - \mathbf{a}^T (y + h e_j)]^+ - (u - \mathbf{a}^T y)^+ \right| \leq |h| a_j.$$

Dividing by $|h|$ gives

$$\left| \frac{[u - \mathbf{a}^T (y + h e_j)]^+ - (u - \mathbf{a}^T y)^+}{h} \right| \leq a_j.$$

Since $0 \leq a_j \leq \alpha_j$ by Bray's Assumption 4, the dominated convergence theorem yields

$$\begin{aligned} \frac{\partial g(y)}{\partial y_j} &= \lim_{h \rightarrow 0} \mathbb{E} \left\{ \frac{[u - \mathbf{a}^T (y + h e_j)]^+ - (u - \mathbf{a}^T y)^+}{h} \right\} \\ &= \mathbb{E} \left\{ \lim_{h \rightarrow 0} \frac{[u - \mathbf{a}^T (y + h e_j)]^+ - (u - \mathbf{a}^T y)^+}{h} \right\} \\ &= \mathbb{E} \left[\frac{\partial}{\partial y_j} (u - \mathbf{a}^T y)^+ \right] = -\mathbb{E}[a_j \mathbf{1}\{u > \mathbf{a}^T y\}] = -q_j(y), \end{aligned}$$

The third and fourth equalities use

$$\frac{\partial}{\partial y_j} (u - \mathbf{a}^T y)^+ = -a_j \mathbf{1}\{u > \mathbf{a}^T y\}, \quad a.s.$$

which holds because $a_j \mathbf{1}\{u = \mathbf{a}^T y\} = 0$ almost surely.

Computing this derivative coordinatewise gives the first equality in (16), and differentiating once more gives the second equality in (16). $\square$

Proof of Proposition B.3. Let $\mathcal{V}_0$ denote the neighborhood of y_β on which Bray's Lemma 1 holds. Define function $F : \mathbb{R}^m \times \mathcal{V}_0 \rightarrow \mathbb{R}^m$ by $F(b, y) \triangleq b - q(y)$. (17) gives $F(\beta, y_\beta) = \mathbf{0}$. The Jacobians of F with respect to b and y are

$$D_b F(b, y) = I_m, \quad D_y F(b, y) = -Dq(y) = H(y).$$

By Bray's Lemma 1, F is continuously differentiable on $\mathbb{R}^m \times \mathcal{V}_0$. Moreover, $D_y F(\beta, y_\beta) = H(y_\beta)$ is invertible. The implicit function theorem therefore yields open neighborhoods $\mathcal{U}$ of β and $\mathcal{V} \subseteq \mathcal{V}_0$ of y_β , together with a continuously differentiable map $b \mapsto y_b \in \mathcal{V}$ such that, for each $b \in \mathcal{U}$, $F(b, y_b) = \mathbf{0}$ and y_b is the unique $y \in \mathcal{V}$ satisfying $F(b, y) = \mathbf{0}$. Equivalently,

$$F(b, y_b) = \mathbf{0} \iff q(y_b) = b, \quad b \in \mathcal{U}.$$

By shrinking both neighborhoods if necessary, we may take $\mathcal{U}$ and $\mathcal{V}$ to be convex. In addition, since $\beta > \mathbf{0}$ and $y_\beta > \mathbf{0}$, we may shrink them further so that $\mathcal{U}, \mathcal{V} \subseteq (0, \infty)^m$.

To compute the derivative of y_b , differentiate $F(b, y_b) = \mathbf{0}$ with respect to b . By the chain rule,

$$\mathbf{0} = D_b F(b, y_b) + D_y F(b, y_b) D_b y_b = I_m + H(y_b) D_b y_b.$$

Hence, $D_b y_b = -H(y_b)^{-1}$. □

B.3.2 Verification of Assumption 3.1

Define the set of binding resources $\mathcal{B} = [m]$ and the fluid function $\phi_{\mathcal{B}} = \phi^F$. For every $b \in \mathcal{U}$, we have $\nabla^2 \phi_{\mathcal{B}}(b) = -H(y_b)^{-1}$ by (18). Since (i) the map $y(b) = y_b$ is continuous on $\mathcal{U}$ by Proposition B.3, (ii) $H(y)$ is continuous and invertible on $\mathcal{V}$ by Bray's Lemma 1, and (iii) matrix inversion is continuous on the set of nonsingular matrices, $\nabla^2 \phi_{\mathcal{B}}$ is continuous on $\mathcal{U}$, and hence $\phi_{\mathcal{B}} \in C^2(\mathcal{U})$.

Assumption 3.1 part 2 additionally requires that $\nabla^2 \phi_{\mathcal{B}}$ is locally α -Hölder continuous for some $\alpha \in (0, 1]$. This is the only requirement in Assumption 3.1 not implied by Bray's assumptions.

B.3.3 Verification of Assumption 3.2

The static fluid policy in (4) is

$$a^F(b) = \mathbf{1}\{u > \mathbf{a}^T y_b\} = x_b(u, \mathbf{a}).$$

By Proposition B.3, $q(y_b) = b$ for all $b \in \mathcal{U}$. Hence, for any resource $j \in [m]$,

$$\mu_j(b) = \mathbb{E}[a_j \mathbf{1}\{u > \mathbf{a}^T y_b\}] = q_j(y_b) = b_j.$$

As a result, all resource constraints bind for every $b \in \mathcal{U}$. This verifies Assumption 3.2.

B.3.4 Verification of Assumption 3.3

Because the action set is binary, there is only one competitor to the selected action. For either selected action, the consumption difference is $d_{01}^{\mathcal{B}} = \|\mathbf{a}\|_1$, and the score gap between the selected action and its competitor is $|u - \mathbf{a}^T y_b|$. Hence, the expectation inside the supremum in Assumption 3.3 equals

$$\mathbb{E}[\|\mathbf{a}\|_1 \mathbf{1}\{|u - \mathbf{a}^T y_b| \leq t \|\mathbf{a}\|_1\}].$$

Fix a compact set $\mathcal{K}_{\mathcal{B}} \subseteq \mathcal{U}$. Since the map $b \mapsto y_b \in \mathcal{V}$ is continuous on $\mathcal{U}$, $A \triangleq \{y_b : b \in \mathcal{K}_{\mathcal{B}}\} \subseteq \mathcal{V}$ is compact. Since $\mathcal{V}$ is an open and convex set, we can choose $t_{\mathcal{K}} > 0$ sufficiently small so that the compact set

$$B \triangleq \{y_b + s\mathbf{1} : b \in \mathcal{K}_{\mathcal{B}}, |s| \leq 2t_{\mathcal{K}}\} \subseteq \mathcal{V}.$$

Define

$$M_{\mathcal{K}} \triangleq \sup \left\{ \|Dq(y)\| : y \in B \right\} < \infty.$$

The upper bound $M_{\mathcal{K}}$ is finite because the Jacobian $Dq(y)$ is continuous by Bray's Assumption 6 and B is compact.

Fix a point $b \in \mathcal{K}_{\mathcal{B}}$ and set $y = y_b$. Take $0 \leq t \leq t_{\mathcal{K}}$, and let $\delta = t + \eta$ with $0 < \eta \leq t_{\mathcal{K}}$. Note that $\delta \leq 2t_{\mathcal{K}}$, hence $y + s\mathbf{1} \in B$ whenever $|s| \leq \delta$. When $\|\mathbf{a}\|_1 > 0$, the event $|u - y^T \mathbf{a}| \leq t \|\mathbf{a}\|_1$ implies

$$u - (y - \delta \mathbf{1})^T \mathbf{a} \geq (\delta - t) \|\mathbf{a}\|_1 = \eta \|\mathbf{a}\|_1 > 0$$

and

$$u - (y + \delta \mathbf{1})^T \mathbf{a} \leq (t - \delta) \|\mathbf{a}\|_1 = -\eta \|\mathbf{a}\|_1 < 0.$$

Therefore, on $\{\|\mathbf{a}\|_1 > 0\}$, we have

$$\mathbf{1}\{|u - y^T \mathbf{a}| \leq t \|\mathbf{a}\|_1\} \leq \mathbf{1}\{u > (y - \delta \mathbf{1})^T \mathbf{a}\} - \mathbf{1}\{u > (y + \delta \mathbf{1})^T \mathbf{a}\}.$$

Multiplying the preceding inequality by $\mathbf{a}$ and taking expectations yields:

$$\begin{aligned} \mathbb{E}[\|\mathbf{a}\|_1 \mathbf{1}\{|u - y^T \mathbf{a}| \leq t \|\mathbf{a}\|_1\}] &\leq \mathbf{1}^T \left\{ q(y - (t + \eta)\mathbf{1}) - q(y + (t + \eta)\mathbf{1}) \right\} \\ &= - \int_{-(t+\eta)}^{t+\eta} \mathbf{1}^T Dq(y + s\mathbf{1}) \mathbf{1} ds \leq 2mM_{\mathcal{K}}(t + \eta). \end{aligned}$$

The equality follows from the fundamental theorem of calculus along the line $s \mapsto y + s\mathbf{1}$, and the second inequality follows because $|\mathbf{1}^T Dq(y + s\mathbf{1}) \mathbf{1}| \leq \|\mathbf{1}\|_2^2 \|Dq(y + s\mathbf{1})\|_2 \leq mM_{\mathcal{K}}$. Letting $\eta \rightarrow 0$ implies that, for any $b \in \mathcal{K}_{\mathcal{B}}$ and $0 \leq t \leq t_{\mathcal{K}}$,

$$\mathbb{E}[\|\mathbf{a}\|_1 \mathbf{1}\{|u - \mathbf{a}^T y_b| \leq t \|\mathbf{a}\|_1\}] \leq 2mM_{\mathcal{K}}t.$$

Consequently, Assumption 3.3 holds with $\beta_{\mathcal{K}} = 1$ and $C_{\mathcal{K}} = 2mM_{\mathcal{K}}$.

C Preliminaries for the Proof of Theorem 4.1

In this section, we develop the ingredients needed to prove Propositions 6.1–6.3 in Appendices D–F. Specifically, Section C.1 constructs a sequence of nested local regions used throughout the subsequent analysis, and Section C.2 establishes several auxiliary lemmas used in the proofs.

C.1 Notation and Nested Local Regions

Fix the compact target set $\mathcal{K} \subseteq \mathcal{U}$ from Section 6 and let

$$\mathcal{K}_{\mathcal{B}} \triangleq \{\boldsymbol{\rho}_{\mathcal{B}} : \boldsymbol{\rho} \in \mathcal{K}\} \subseteq \mathcal{U}_{\mathcal{B}}$$

denote its projection onto the binding coordinates.

Recall that $\mathcal{S} \triangleq \mathcal{B}^c$ denote the set of slack resources. If $\mathcal{S} \neq \emptyset$, then, by the compactness of $\mathcal{K}$, the continuity of $\mu_j(\cdot)$ on $\mathcal{U}_{\mathcal{B}}$ for each $j \in \mathcal{S}$, and the strict slackness in Assumption 3.2, we have

$$\underline{\delta} \triangleq \inf_{\boldsymbol{\rho} \in \mathcal{K}} \min_{j \in \mathcal{S}} \left\{ \rho_j - \mu_j(\boldsymbol{\rho}_{\mathcal{B}}) \right\} > 0.$$

Choose $\eta > 0$ such that $3\eta \leq \underline{\delta}$. Otherwise, if $\mathcal{S} = \emptyset$, we set $\eta = 0$.

Starting from $\mathcal{K}_{\mathcal{B}}$, we choose compact sets $\mathcal{K}_{\mathcal{B}}^1$ and $\mathcal{K}_{\mathcal{B}}^2$ such that

$$\mathcal{K}_{\mathcal{B}} \subseteq \mathcal{K}_{\mathcal{B}}^1 \subseteq \mathcal{K}_{\mathcal{B}}^2 \subseteq \mathcal{U}_{\mathcal{B}}.$$

Recall that $A \subseteq B$ means that A is contained in B and has positive distance from the boundary of B . Consequently, $\mathcal{K}_{\mathcal{B}}^1$ and $\mathcal{K}_{\mathcal{B}}^2$ provide successive buffers around $\mathcal{K}_{\mathcal{B}}$. In the subsequent analysis, $\mathcal{K}_{\mathcal{B}}^1$ will serve as the stopping region for analyzing the dynamic fluid policy, and $\mathcal{K}_{\mathcal{B}}^2$ will contain the one-period successor of any state in $\mathcal{K}_{\mathcal{B}}^1$ when the remaining horizon is sufficiently large.

In addition, we define two open sets:

$$\begin{aligned} \mathcal{D}_1 &\triangleq \left\{ \boldsymbol{\rho} : \boldsymbol{\rho}_{\mathcal{B}} \in \text{int } \mathcal{K}_{\mathcal{B}}^1, \rho_j - \mu_j(\boldsymbol{\rho}_{\mathcal{B}}) > 2\eta \quad \forall j \in \mathcal{S} \right\}, \\ \mathcal{D}_2 &\triangleq \left\{ \boldsymbol{\rho} : \boldsymbol{\rho}_{\mathcal{B}} \in \text{int } \mathcal{K}_{\mathcal{B}}^2, \rho_j - \mu_j(\boldsymbol{\rho}_{\mathcal{B}}) > \eta \quad \forall j \in \mathcal{S} \right\}. \end{aligned}$$

Set $\mathcal{D}_1$ is associated with $\mathcal{K}_{\mathcal{B}}^1$ and set $\mathcal{D}_2$ is associated with $\mathcal{K}_{\mathcal{B}}^2$. By the preceding nesting and our choice of η , we have:

$$\mathcal{K} \subset \mathcal{D}_1 \subset \mathcal{D}_2.$$

Finally, let $\alpha \in (0, 1]$ denote the Hölder exponent in part 2 of Assumption 3.1, and let $\beta_{\star}, C_{\star}$, and $t_{\star}$ be the constants specified in Assumption 3.3 for $\mathcal{K}_{\mathcal{B}}^2$. Since replacing $t_{\star}$ and $\beta_{\star}$ by their respective minima with 1 preserves the bound in Assumption 3.3, we may assume without loss of generality that

$$t_{\star} \in (0, 1] \quad \text{and} \quad \beta_{\star} \in (0, 1].$$

Define

$$\xi_{\star} \triangleq \frac{\beta_{\star}}{1 + \beta_{\star}} \in (0, 1/2], \quad \gamma \triangleq \min \{ \alpha, \beta_{\star} \} \in (0, 1].$$

C.2 Useful Lemmas

In this section, we establish multiple auxiliary lemmas used in Appendices D–F.

C.2.1 Extension Across Slack Coordinates

Although Assumptions 3.1 and 3.2 are stated on the original neighborhood $\mathcal{U}$, the normalized inventories of slack resources may increase as the number of remaining periods decreases. Therefore, the normalized state need not remain in $\mathcal{U}$ as it evolves. Lemma C.1 shows that, as long as the slack resources retain strictly slack, the fluid value, optimal dual prices, and dual-price actions continue to depend only on the binding coordinates.

Recall that $a^{\text{F}}(\mathbf{x})$ denotes the static dual-price action defined in (4) and $a_n^{\text{F}}(\mathbf{c})$ denotes the action taken by the dynamic fluid policy at state $(\mathbf{c}, n)$ as defined in (3). We now state Lemma C.1.

Lemma C.1 (Extension Across Slack Coordinates). *Define the extended region*

$$\tilde{\mathcal{U}} \triangleq \{ \boldsymbol{\rho} : \boldsymbol{\rho}_{\mathcal{B}} \in \mathcal{U}_{\mathcal{B}}, \rho_j > \mu_j(\boldsymbol{\rho}_{\mathcal{B}}) \text{ for every } j \in \mathcal{S} \}.$$

Then, the following statements hold.

1. *For every $\boldsymbol{\rho} \in \tilde{\mathcal{U}}$, the fluid relaxation (1) satisfies $\phi^{\text{F}}(\boldsymbol{\rho}) = \phi_{\mathcal{B}}(\boldsymbol{\rho}_{\mathcal{B}})$, and the static action $a^{\text{F}}(\boldsymbol{\rho}_{\mathcal{B}})$ is an optimal control.*

2. For every $\boldsymbol{\rho} \in \tilde{\mathcal{U}}$, the fluid relaxation has the unique optimal dual variable $(\boldsymbol{\lambda}(\boldsymbol{\rho}_{\mathcal{B}}), \mathbf{0}_{\mathcal{S}})$.
3. At any state $(\mathbf{c}, n)$ such that $\mathbf{c}/n \in \tilde{\mathcal{U}}$ and every action is feasible, $a_n^{\mathbf{F}}(\mathbf{c}) = a^{\mathbf{F}}(\mathbf{c}_{\mathcal{B}}/n)$ almost surely.

Proof. For each $\mathbf{x} \in \mathcal{U}_{\mathcal{B}}$, there exists a reference point $\boldsymbol{\rho}^0(\mathbf{x}) \in \mathcal{U}$ such that $\boldsymbol{\rho}^0(\mathbf{x})_{\mathcal{B}} = \mathbf{x}$ by the definition of the projection $\mathcal{U}_{\mathcal{B}}$. Since the static action depends only on $\mathbf{x}$, Assumption 3.2 and Lemma 3.2 yield:

$$\mathbb{E}[B_{a^{\mathbf{F}}(\mathbf{x}), \mathcal{B}}] = \mathbf{x}, \quad \mathbb{E}[B_{a^{\mathbf{F}}(\mathbf{x}), j}] = \mu_j(\mathbf{x}) \quad \forall j \in \mathcal{S}, \quad \mathbb{E}[R_{a^{\mathbf{F}}(\mathbf{x})}] = \phi_{\mathcal{B}}(\mathbf{x}). \quad (19)$$

Proof of Part 1. Fix $\boldsymbol{\rho} \in \tilde{\mathcal{U}}$ and let $\mathbf{x} = \boldsymbol{\rho}_{\mathcal{B}}$. Since $\mu_j(\mathbf{x}) < \rho_j$ for every $j \in \mathcal{S}$, (19) imply that $a^{\mathbf{F}}(\mathbf{x})$ is feasible to (1) at $\boldsymbol{\rho}$ and attains objective value $\phi_{\mathcal{B}}(\mathbf{x})$. By Lemma 3.1, $(\boldsymbol{\lambda}(\mathbf{x}), \mathbf{0}_{\mathcal{S}})$ is dual optimal at $\boldsymbol{\rho}^0(\mathbf{x})$. Because $\boldsymbol{\rho}^0(\mathbf{x})_{\mathcal{B}} = \mathbf{x}$ and $\phi^{\mathbf{F}}(\boldsymbol{\rho}^0(\mathbf{x})) = \phi_{\mathcal{B}}(\mathbf{x})$, its dual objective value is

$$\boldsymbol{\lambda}(\mathbf{x})^{\mathbf{T}} \mathbf{x} + \mathcal{V}((\boldsymbol{\lambda}(\mathbf{x}), \mathbf{0}_{\mathcal{S}})) = \phi_{\mathcal{B}}(\mathbf{x}).$$

Thus, $(\boldsymbol{\lambda}(\mathbf{x}), \mathbf{0}_{\mathcal{S}})$ is dual feasible at $\boldsymbol{\rho}$ and has a dual objective value $\phi_{\mathcal{B}}(\mathbf{x})$. Since $a^{\mathbf{F}}(\mathbf{x})$ is primal feasible at $\boldsymbol{\rho}$ and achieves the same value, strong duality implies that $a^{\mathbf{F}}(\mathbf{x})$ is optimal for (1), $(\boldsymbol{\lambda}(\mathbf{x}), \mathbf{0}_{\mathcal{S}})$ is dual optimal, and $\phi^{\mathbf{F}}(\boldsymbol{\rho}) = \phi_{\mathcal{B}}(\mathbf{x}) = \phi_{\mathcal{B}}(\boldsymbol{\rho}_{\mathcal{B}})$.

Proof of Part 2. It remains to show that $(\boldsymbol{\lambda}(\mathbf{x}), \mathbf{0}_{\mathcal{S}})$ is the unique optimal dual solution. Let $\tilde{\boldsymbol{\lambda}}$ be any dual optimum at $\boldsymbol{\rho}$. The strong duality and the optimality of $a^{\mathbf{F}}(\mathbf{x})$ imply:

$$\begin{aligned} 0 &= \tilde{\boldsymbol{\lambda}}^{\mathbf{T}} \boldsymbol{\rho} + \mathcal{V}(\tilde{\boldsymbol{\lambda}}) - \mathbb{E}[R_{a^{\mathbf{F}}(\mathbf{x})}] \\ &= \sum_{i \in [m]} \tilde{\lambda}_i \left(\rho_i - \mathbb{E}[B_{a^{\mathbf{F}}(\mathbf{x}), i}] \right) + \mathbb{E} \left[\max_{a \in \mathcal{A}} \{ R_a - \tilde{\boldsymbol{\lambda}}^{\mathbf{T}} B_a \} - (R_{a^{\mathbf{F}}(\mathbf{x})} - \tilde{\boldsymbol{\lambda}}^{\mathbf{T}} B_{a^{\mathbf{F}}(\mathbf{x})}) \right]. \end{aligned}$$

The first term on the right-hand side is nonnegative because $\tilde{\boldsymbol{\lambda}} \geq \mathbf{0}$ and $a^{\mathbf{F}}(\mathbf{x})$ is feasible at $\boldsymbol{\rho}$. The second term is also nonnegative because it is the difference between a pointwise maximum and the score of one feasible candidate. Since their sum is zero, both terms must vanish. In particular, since $\rho_s - \mu_s(\mathbf{x}) > 0$ for every $s \in \mathcal{S}$, we must have $\tilde{\lambda}_s = 0$ for every slack coordinate. Thus, $\tilde{\boldsymbol{\lambda}} = (\tilde{\boldsymbol{\lambda}}_{\mathcal{B}}, \mathbf{0}_{\mathcal{S}})$. Because $\boldsymbol{\rho}_{\mathcal{B}} = \boldsymbol{\rho}^0(\mathbf{x})_{\mathcal{B}} = \mathbf{x}$, this vector has the same dual objective value at $\boldsymbol{\rho}$ and $\boldsymbol{\rho}^0(\mathbf{x})$ and is therefore dual optimal at $\boldsymbol{\rho}^0(\mathbf{x})$. Lemma 3.1 then yields $\tilde{\boldsymbol{\lambda}} = (\boldsymbol{\lambda}(\mathbf{x}), \mathbf{0}_{\mathcal{S}})$.

Proof of Part 3. When every action is feasible, (3) and (4) maximize the same objective using the same tie-breaking rule. Hence $a_n^{\mathbf{F}}(\mathbf{c}) = a^{\mathbf{F}}(\mathbf{c}_{\mathcal{B}}/n)$ almost surely. $\square$

C.2.2 Illustration of the Buffer Region $\mathcal{D}_2$

We illustrate the role of the buffer region $\mathcal{D}_2$ in Lemma C.2 and Corollary C.3.

Lemma C.2 (One-Period Containment). *There exists an integer n_0 such that, for any $n \geq n_0$ and $\mathbf{c}/n \in \mathcal{D}_1$,*

1. Every action $a \in \mathcal{A}$ is feasible at $\mathbf{c}$ almost surely;

2. For every action $a \in \mathcal{A}$, the corresponding one-step successor ratio satisfies

$$\boldsymbol{\rho}^a \triangleq \frac{\mathbf{c} - B_a}{n-1} \in \mathcal{D}_2 \quad \text{almost surely.}$$

Proof. Proof of Part 1. Let $\boldsymbol{\rho} = \mathbf{c}/n$ and define $g_s(\boldsymbol{\rho}) \triangleq \rho_s - \mu_s(\boldsymbol{\rho}_{\mathcal{B}})$ for all $s \in \mathcal{S}$. Since $\mathcal{K}_{\mathcal{B}}^1 \in \mathbb{R}_+^{|\mathcal{B}|}$ is compact, the binding coordinates have a strictly positive lower bound

$$\underline{\rho}_{\mathcal{B}} \triangleq \min_{\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^1} \min_{i \in \mathcal{B}} x_i > 0.$$

If $\mathcal{S} \neq \emptyset$, then $\boldsymbol{\rho} \in \mathcal{D}_1$ implies $g_s(\boldsymbol{\rho}) > 2\eta$ for every $s \in \mathcal{S}$. Since $\mu_s \geq 0$, it follows that $\rho_s > 2\eta$ for every $s \in \mathcal{S}$. Accordingly, define

$$\underline{\rho} \triangleq \begin{cases} \min \{ \underline{\rho}_{\mathcal{B}}, 2\eta \}, & \mathcal{S} \neq \emptyset, \\ \underline{\rho}_{\mathcal{B}}, & \mathcal{S} = \emptyset. \end{cases}$$

In either case, $\underline{\rho} > 0$, and every coordinate of $\mathbf{c} = n\boldsymbol{\rho}$ is at least $n\underline{\rho}$. Since every coordinate of B_a is at most $\bar{b}$ almost surely, part 1 holds whenever $n\underline{\rho} \geq \bar{b}$.

Proof of Part 2. For a one-period transition, the preceding definitions give

$$\boldsymbol{\rho}^a = \frac{\mathbf{c} - B_a}{n-1} = \boldsymbol{\rho} + \frac{\boldsymbol{\rho} - B_a}{n-1}.$$

Since consumption is uniformly bounded by Assumption 2.1, the binding coordinates satisfy

$$\|\boldsymbol{\rho}_{\mathcal{B}}^a - \boldsymbol{\rho}_{\mathcal{B}}\| = O(n^{-1})$$

uniformly over actions. Therefore, for all sufficiently large n , $\boldsymbol{\rho}_{\mathcal{B}}^a \in \text{int } \mathcal{K}_{\mathcal{B}}^2$. Meanwhile, for each $s \in \mathcal{S}$,

$$\rho_s^a - \mu_s(\boldsymbol{\rho}_{\mathcal{B}}^a) = g_s(\boldsymbol{\rho}) + \frac{\rho_s - B_{a,s}}{n-1} - \left\{ \mu_s(\boldsymbol{\rho}_{\mathcal{B}}^a) - \mu_s(\boldsymbol{\rho}_{\mathcal{B}}) \right\} \geq g_s(\boldsymbol{\rho}) - \frac{C}{n}$$

for some absolute constant $C < \infty$, because the resource consumption is uniformly bounded and function μ_j is locally Lipschitz continuous by Assumption 3.2. Since $g_s(\boldsymbol{\rho}) > 2\eta$, it follows that $\rho_s^a - \mu_s(\boldsymbol{\rho}_{\mathcal{B}}^a) > \eta$ for all $s \in \mathcal{S}$ and sufficiently large n . Therefore, $\boldsymbol{\rho}^a \in \mathcal{D}_2$ uniformly over actions. $\square$

For a continuation value function u and a feasible action a , define the one-step continuation difference by

$$D_a u(\mathbf{c}) \triangleq u(\mathbf{c}) - u(\mathbf{c} - B_a).$$

Recall that the fluid-relaxation upper bound is denoted by $F_n(\mathbf{c}) \triangleq n\phi^F(\mathbf{c}/n)$. For $n \geq 2$, a normalized inventory $\mathbf{x}$ of the binding resources, and a feasible action a , define the one-step fluid continuation cost by

$$m_{n,a}(\mathbf{x}) \triangleq (n-1) \left[\phi_{\mathcal{B}} \left(\frac{n\mathbf{x}}{n-1} \right) - \phi_{\mathcal{B}} \left(\frac{n\mathbf{x} - B_{a,\mathcal{B}}}{n-1} \right) \right]. \quad (20)$$

Hence, $m_{n,a}(\mathbf{x})$ represents the loss in fluid continuation value from consuming $B_{a,\mathcal{B}}$ units of the inventory $n\mathbf{x}$ when n periods remain.

Corollary C.3. *Under the conditions of Lemma C.2, for every action a , the fluid continuation cost satisfies*

$$D_a F_{n-1}(\mathbf{c}) = m_{n,a}(\mathbf{c}_B/n) \quad \text{almost surely,}$$

and the actions defined in (3) and (4) agree:

$$a_n^F(\mathbf{c}) = a^F(\mathbf{c}_B/n) \quad \text{almost surely.}$$

Proof. Set $\boldsymbol{\rho} = \mathbf{c}/n$. By Lemma C.2, both $\boldsymbol{\rho}^{a=0}$ and $\boldsymbol{\rho}^a$ belong to $\mathcal{D}_2 \subset \tilde{\mathcal{U}}$. Hence, Lemma C.1 yields

$$\begin{aligned} F_{n-1}(\mathbf{c}) &= (n-1) \phi_B \left(\frac{\mathbf{c}_B}{n-1} \right), \\ F_{n-1}(\mathbf{c} - B_a) &= (n-1) \phi_B \left(\frac{\mathbf{c}_B - B_{a,B}}{n-1} \right). \end{aligned}$$

Their difference is exactly $m_{n,a}(\boldsymbol{\rho}_B)$, which proves the first claim.

In addition, since $\boldsymbol{\rho} \in \mathcal{D}_1 \subset \mathcal{U}$ and every action is feasible by Lemma C.2, part 3 of Lemma C.1 proves the second claim. $\square$

C.2.3 Marginal Value Approximation

Lemma C.4. *There exist constants $C < \infty$ and $n_0 \in \mathbb{N}$ such that, for every $n \geq n_0$, $\mathbf{x} \in \mathcal{K}_B^2$, and action a ,*

$$|m_{n,a}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{a,B}| \leq \frac{C}{n} \|B_{a,B}\|.$$

Moreover, for any two actions $a, b \in \mathcal{A}$,

$$\left| \left(m_{n,a}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{a,B} \right) - \left(m_{n,b}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{b,B} \right) \right| \leq \frac{C}{n} \|B_{a,B} - B_{b,B}\| \leq \frac{C}{n} d_{ab}^B.$$

Proof. Since $\mathcal{K}_B^2 \Subset \mathcal{U}_B$, we can choose an integer $n_{\text{loc}} \geq 2$ large enough such that, for every $n \geq n_{\text{loc}}$, every $\mathbf{x} \in \mathcal{K}_B^2$, the points

$$\frac{n\mathbf{x} - B_{a,B}}{n-1}, \quad \forall a \in \mathcal{A}$$

and the line segments joining them lie in a fixed compact subset $\hat{\mathcal{K}}_B \Subset \mathcal{U}_B$.

Proof of Part 1. For notational convenience, let $b \triangleq B_{a,B}$. By the fundamental theorem of calculus,

$$m_{n,a}(\mathbf{x}) = \int_0^1 \nabla \phi_B \left(\frac{n\mathbf{x} - tb}{n-1} \right)^\top b \, dt.$$

To see this, Define $p(t) = (n\mathbf{x} - tb)/(n-1)$. The chain rule gives

$$\frac{d}{dt} \phi_B(p(t)) = -\frac{1}{n-1} \nabla \phi_B(p(t))^\top b.$$

Integrating over $t \in [0, 1]$ and multiplying by $n-1$ yields the stated representation of $m_{n,a}(\mathbf{x})$.

Since $\boldsymbol{\lambda}(\mathbf{x}) = \nabla \phi_{\mathcal{B}}(\mathbf{x})$, we have

$$\begin{aligned} |m_{n,a}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{a,\mathcal{B}}| &= \left| \int_0^1 \left(\nabla \phi_{\mathcal{B}} \left(\frac{n\mathbf{x} - tb}{n-1} \right) - \nabla \phi_{\mathcal{B}}(\mathbf{x}) \right)^\top b \, dt \right| \\ &\leq \int_0^1 \left\| \nabla \phi_{\mathcal{B}} \left(\mathbf{x} + \frac{\mathbf{x} - tb}{n-1} \right) - \nabla \phi_{\mathcal{B}}(\mathbf{x}) \right\| \|b\| \, dt. \end{aligned}$$

By Assumption 3.1, the Hessian $\nabla^2 \phi_{\mathcal{B}}$ is uniformly bounded on the fixed compact set $\hat{\mathcal{K}}_{\mathcal{B}}$. Hence, the mean-value theorem, together with the boundedness of $\mathbf{x}$ and b , implies that, uniformly over $t \in [0, 1]$,

$$\left\| \nabla \phi_{\mathcal{B}} \left(\mathbf{x} + \frac{\mathbf{x} - tb}{n-1} \right) - \nabla \phi_{\mathcal{B}}(\mathbf{x}) \right\| \leq \frac{C}{n}. \quad (21)$$

Substituting (21) into the preceding bound yields the first claim.

Proof of Part 2. Define function

$$g_{n,\mathbf{x}}(b) \triangleq (n-1) \left[\phi_{\mathcal{B}} \left(\frac{n\mathbf{x}}{n-1} \right) - \phi_{\mathcal{B}} \left(\frac{n\mathbf{x} - b}{n-1} \right) \right] - \boldsymbol{\lambda}(\mathbf{x})^\top b.$$

We have

$$\nabla_b g_{n,\mathbf{x}}(b) = \nabla \phi_{\mathcal{B}} \left(\frac{n\mathbf{x} - b}{n-1} \right) - \boldsymbol{\lambda}(\mathbf{x}) \quad \text{and} \quad \|\nabla_b g_{n,\mathbf{x}}(b)\| \leq \frac{C}{n},$$

where the bound follows from the same argument for (21). Therefore,

$$\begin{aligned} \left| \left(m_{n,a}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{a,\mathcal{B}} \right) - \left(m_{n,b}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{b,\mathcal{B}} \right) \right| &= \left| g_{n,\mathbf{x}}(B_{a,\mathcal{B}}) - g_{n,\mathbf{x}}(B_{b,\mathcal{B}}) \right| \\ &\leq \frac{C}{n} \|B_{a,\mathcal{B}} - B_{b,\mathcal{B}}\| \\ &\leq \frac{C}{n} \|B_{a,\mathcal{B}} - B_{b,\mathcal{B}}\|_1 = \frac{C}{n} d_{ab}^{\mathcal{B}}, \end{aligned}$$

where the first inequality follows from the mean-value theorem and the gradient bound. $\square$

C.2.4 Action Selection Stability

Lemma C.5 (Action Selection Stability under Near-Indifference). *Fix a compact set $\mathcal{K}_{\mathcal{B}} \in \mathcal{U}_{\mathcal{B}}$ and a point $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}$. Let $(\beta_{\mathcal{K}}, C_{\mathcal{K}}, t_{\mathcal{K}})$ be the constants specified in Assumption 3.3 for $\mathcal{K}_{\mathcal{B}}$. Let $i = a^{\mathcal{F}}(\mathbf{x})$ maximize the scores $S_a(\mathbf{x})$ defined in Assumption 3.3, and let j maximize the perturbed scores $S_a(\mathbf{x}) - e_a$. Suppose there exists $t \in [0, t_{\mathcal{K}}]$ such that, for every $a \in \mathcal{A}$,*

$$|e_i - e_a| \leq t d_{ia}^{\mathcal{B}} \quad \text{almost surely.}$$

Then

$$\begin{aligned} 0 &\leq \mathbb{E}[S_i(\mathbf{x}) - S_j(\mathbf{x})] \leq C_{\mathcal{K}} t^{1+\beta_{\mathcal{K}}}, \\ 0 &\leq \mathbb{E}[(S_j(\mathbf{x}) - e_j) - (S_i(\mathbf{x}) - e_i)] \leq C_{\mathcal{K}} t^{1+\beta_{\mathcal{K}}}. \end{aligned}$$

Proof. By the definitions of actions i and j , for any action $\ell \in \mathcal{A}$,

$$\begin{aligned} S_{\ell}(\mathbf{x}) &\leq S_i(\mathbf{x}), \\ S_{\ell}(\mathbf{x}) - e_{\ell} &\leq S_j(\mathbf{x}) - e_j. \end{aligned} \quad (22)$$

If $i = j$, both differences in the statement of the lemma vanish. Now consider the switching event $\{j = a \neq i\}$. (22) yields:

$$\begin{aligned} 0 &\leq S_i(\mathbf{x}) - S_a(\mathbf{x}) \leq e_i - e_a \leq |e_i - e_a| \leq t d_{ia}^{\mathcal{B}}, \\ 0 &\leq (S_a(\mathbf{x}) - e_a) - (S_i(\mathbf{x}) - e_i) \leq e_i - e_a \leq |e_i - e_a| \leq t d_{ia}^{\mathcal{B}}. \end{aligned}$$

Here, the second inequality in line one follows from the second inequality in (22) with $j = a$ and $\ell = i$, and the second inequality in line two follows from the first inequality in (22) with $\ell = a$.

Thus, on $\{j = a \neq i\}$, both the baseline loss $S_i(\mathbf{x}) - S_a(\mathbf{x})$ and the perturbed gain $(S_a(\mathbf{x}) - e_a) - (S_i(\mathbf{x}) - e_i)$ are bounded by $t d_{ia}^{\mathcal{B}}$. Moreover, the preceding inequalities imply that

$$\{j = a \neq i\} \subseteq \left\{0 \leq S_i(\mathbf{x}) - S_a(\mathbf{x}) \leq t d_{ia}^{\mathcal{B}}\right\}.$$

Consequently, each of the two differences in the statement is bounded pointwise by

$$\sum_{a \neq i} t d_{ia}^{\mathcal{B}} \cdot \mathbf{1}\left\{0 \leq S_i(\mathbf{x}) - S_a(\mathbf{x}) \leq t d_{ia}^{\mathcal{B}}\right\}.$$

Taking expectations and applying Assumption 3.3 gives $t \cdot C_{\mathcal{K}} t^{\beta_{\mathcal{K}}} = C_{\mathcal{K}} t^{1+\beta_{\mathcal{K}}}$, which proves both claims. $\square$

C.2.5 Hölder Continuity of the Logarithmic Coefficient $\Gamma(\mathbf{x})$

In this section, we prove that the logarithmic coefficient $\Gamma(\mathbf{x})$ is γ -Hölder continuous on the compact set $\mathcal{K}_{\mathcal{B}}^2$, where $\gamma \triangleq \min\{\alpha, \beta_{\star}\}$.

Lemma C.6 (Hölder Continuity of $\Gamma(\mathbf{x})$). *The logarithmic coefficient $\Gamma(\mathbf{x})$ defined in (5) is γ -Hölder continuous on $\mathcal{K}_{\mathcal{B}}^2$.*

Proof. Recall from (5) that

$$\Gamma(\mathbf{x}) \triangleq -\frac{1}{2} \operatorname{tr} \left(\nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) \Sigma(\mathbf{x}) \right).$$

Fix $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{K}_{\mathcal{B}}^2$. Adding and subtracting $\nabla^2 \phi_{\mathcal{B}}(\mathbf{x}_2) \Sigma(\mathbf{x}_1)$ inside the trace yields

$$\Gamma(\mathbf{x}_1) - \Gamma(\mathbf{x}_2) = -\frac{1}{2} \operatorname{tr} \left(\left\{ \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}_1) - \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}_2) \right\} \Sigma(\mathbf{x}_1) + \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}_2) \left\{ \Sigma(\mathbf{x}_1) - \Sigma(\mathbf{x}_2) \right\} \right).$$

Since $|\operatorname{tr}(AB)| \leq d \|A\| \|B\|$ for any $A, B \in \mathbb{R}^{d \times d}$, it follows that

$$|\Gamma(\mathbf{x}_1) - \Gamma(\mathbf{x}_2)| \leq \frac{|\mathcal{B}|}{2} \left\{ \left\| \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}_1) - \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}_2) \right\| \left\| \Sigma(\mathbf{x}_1) \right\| + \left\| \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}_2) \right\| \left\| \Sigma(\mathbf{x}_1) - \Sigma(\mathbf{x}_2) \right\| \right\}.$$

By Assumption 3.1, $\nabla^2 \phi_{\mathcal{B}}$ is bounded and α -Hölder continuous on $\mathcal{K}_{\mathcal{B}}^2$. Moreover, by Lemma C.7, Σ is bounded and $\beta_{\star}$ -Hölder continuous on $\mathcal{K}_{\mathcal{B}}^2$. Hence, there exists a constant $C < \infty$ such that

$$|\Gamma(\mathbf{x}_1) - \Gamma(\mathbf{x}_2)| \leq C \left(\|\mathbf{x}_1 - \mathbf{x}_2\|^{\alpha} + \|\mathbf{x}_1 - \mathbf{x}_2\|^{\beta_{\star}} \right).$$

which implies that $\Gamma(\mathbf{x})$ is γ -Hölder continuous on $\mathcal{K}_{\mathcal{B}}^2$ with $\gamma \triangleq \min\{\alpha, \beta_{\star}\}$. $\square$

Lemma C.7 (Hölder Continuity of the Covariance Matrix). *The covariance matrix $\Sigma(\mathbf{x})$ is bounded and $\beta_{\star}$ -Hölder continuous on $\mathcal{K}_{\mathcal{B}}^2$.*

Proof. Define the second-moment matrix

$$M_2(\mathbf{x}) \triangleq \mathbb{E} \left[B_{a^F(\mathbf{x}), \mathcal{B}} B_{a^F(\mathbf{x}), \mathcal{B}}^\top \right].$$

By Assumption 3.2 part 1,

$$\Sigma(\mathbf{x}) = M_2(\mathbf{x}) - \mathbf{x}\mathbf{x}^\top.$$

First, Σ is bounded on $\mathcal{K}_{\mathcal{B}}^2$. This is because bounded consumption in Assumption 2.1 implies that M_2 is bounded, and compactness of $\mathcal{K}_{\mathcal{B}}^2$ implies that $\mathbf{x}\mathbf{x}^\top$ is bounded.

Second, the map $\mathbf{x} \mapsto \mathbf{x}\mathbf{x}^\top$ is Lipschitz on $\mathcal{K}_{\mathcal{B}}^2$ and hence $\beta_\star$ -Hölder continuous as $\beta_\star \leq 1$. Thus, to show that Σ is $\beta_\star$ -Hölder continuous on $\mathcal{K}_{\mathcal{B}}^2$, it suffices to show that M_2 is $\beta_\star$ -Hölder continuous, which we do next.

Step 1. Preparation Fix $\mathbf{x}, \mathbf{x}' \in \mathcal{K}_{\mathcal{B}}^2$ and let $i = a^F(\mathbf{x})$ and $j = a^F(\mathbf{x}')$. Optimality of i at $\mathbf{x}$ and of j at $\mathbf{x}'$ gives

$$S_i(\mathbf{x}) - S_j(\mathbf{x}) \geq 0, \quad S_j(\mathbf{x}') - S_i(\mathbf{x}') \geq 0.$$

Therefore,

$$\begin{aligned} 0 &\leq (S_i(\mathbf{x}) - S_j(\mathbf{x})) + (S_j(\mathbf{x}') - S_i(\mathbf{x}')) = \left(\boldsymbol{\lambda}(\mathbf{x}') - \boldsymbol{\lambda}(\mathbf{x}) \right)^\top (B_{i, \mathcal{B}} - B_{j, \mathcal{B}}) \\ &\leq \|\boldsymbol{\lambda}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x}')\| \|B_{i, \mathcal{B}} - B_{j, \mathcal{B}}\| \leq C_1 \|\mathbf{x} - \mathbf{x}'\| d_{ij}^{\mathcal{B}}, \end{aligned}$$

for some constant $C_1 < \infty$. The second inequality is because $\boldsymbol{\lambda} = \nabla \phi_{\mathcal{B}}$ is continuously differentiable on $\mathcal{K}_{\mathcal{B}}^2 \subseteq \mathcal{U}_{\mathcal{B}}$ by Assumption 3.1, and hence is Lipschitz on $\mathcal{K}_{\mathcal{B}}^2$.

Note that on the event $\{j = a \neq i\}$, the preceding inequality yields

$$0 \leq S_i(\mathbf{x}) - S_a(\mathbf{x}) \leq C_1 \|\mathbf{x} - \mathbf{x}'\| d_{ia}^{\mathcal{B}}.$$

Therefore,

$$d_{ij}^{\mathcal{B}} \mathbb{1}\{i \neq j\} = \sum_{a \neq i} d_{ia}^{\mathcal{B}} \mathbb{1}\{j = a\} \leq \sum_{a \neq i} d_{ia}^{\mathcal{B}} \cdot \mathbb{1}\{0 \leq S_i(\mathbf{x}) - S_a(\mathbf{x}) \leq C \|\mathbf{x} - \mathbf{x}'\| d_{ia}^{\mathcal{B}}\}.$$

Hence, whenever $C_1 \|\mathbf{x} - \mathbf{x}'\| \leq t_\star$, Assumption 3.3 yields

$$\mathbb{E} \left[d_{a^F(\mathbf{x}), a^F(\mathbf{x}')}^{\mathcal{B}} \mathbb{1}\{a^F(\mathbf{x}) \neq a^F(\mathbf{x}')\} \right] \leq C_2 \|\mathbf{x} - \mathbf{x}'\|^{\beta_\star}. \quad (23)$$

for some constant $C_2 < \infty$.

Step 2. Hölder Continuity of the Second-Moment Matrix. For any two vectors $u, v \in \mathbb{R}^d$,

$$\|uu^\top - vv^\top\| = \|u(u-v)^\top + (u-v)v^\top\| \leq (\|u\| + \|v\|) \|u - v\|.$$

Hence, by bounded consumption, for some constant $C_3 < \infty$,

$$\left\| B_{a^F(\mathbf{x}), \mathcal{B}} B_{a^F(\mathbf{x}), \mathcal{B}}^\top - B_{a^F(\mathbf{x}'), \mathcal{B}} B_{a^F(\mathbf{x}'), \mathcal{B}}^\top \right\| \leq C_3 \cdot d_{a^F(\mathbf{x}), a^F(\mathbf{x}')}^{\mathcal{B}}. \quad (24)$$

The random matrices inside the expectations defining $M_2(\mathbf{x})$ and $M_2(\mathbf{x}')$ coincide when $a^F(\mathbf{x}) = a^F(\mathbf{x}')$. Therefore, by (23) and (24), whenever $C_1 \|\mathbf{x} - \mathbf{x}'\| \leq t_\star$,

$$\|M_2(\mathbf{x}) - M_2(\mathbf{x}')\| \leq C_2 C_3 \cdot \|\mathbf{x} - \mathbf{x}'\|^{\beta_\star}.$$

It remains to extend this local bound to all of $\mathcal{K}_B^2$. Choose $r \triangleq t_\star/C_1$. The preceding inequality applies whenever $\|\mathbf{x} - \mathbf{x}'\| < r$. If instead $\|\mathbf{x} - \mathbf{x}'\| \geq r$, boundedness of M_2 gives

$$\|M_2(\mathbf{x}) - M_2(\mathbf{x}')\| \leq 2 \sup_{\mathbf{z} \in \mathcal{K}_B^2} \|M_2(\mathbf{z})\| \leq \left(2 \sup_{\mathbf{z} \in \mathcal{K}_B^2} \|M_2(\mathbf{z})\| \cdot r^{-\beta_\star} \right) \|\mathbf{x} - \mathbf{x}'\|^{\beta_\star}.$$

Hence, M_2 is $\beta_\star$ -Hölder continuous on $\mathcal{K}_B^2$. □

D Proof of Proposition 6.1

In this section, we construct the corrected fluid reference $Q_n(\mathbf{c})$ and prove Proposition 6.1.

Throughout this section, we set $\xi_\star \triangleq \beta_\star/(1 + \beta_\star)$, where $\beta_\star \in (0, 1]$ and $\gamma = \min\{\alpha, \beta_\star\} > 0$ are specified in Appendix C.1. Unless otherwise stated, all constants in this section may depend on $\mathcal{K}$ and the model primitives, but not on the horizon n or the initial state $\mathbf{c}$, provided that $\mathbf{c}/n \in \mathcal{K}$.

D.1 Construction of the Corrected Fluid Reference

In this section, we construct the corrected fluid reference $Q_n(\mathbf{c})$. The construction proceeds in three steps. First, we derive a second-order expansion of the one-period residual δ_n^F (Appendix D.1.1). Because its leading coefficient Γ is Hölder continuous (see Lemma C.6) but need not be differentiable, we then smooth Γ by convolution with a C^∞ mollifier to obtain a smooth approximation $\tilde{\delta}_n$ of the raw residual (Appendix D.1.2). Finally, we sum these smoothed one-period residuals to obtain the cumulative residual A_n , and subtract A_n from the fluid relaxation upper bound F_n to obtain the corrected fluid reference Q_n (Appendix 6.1). In Appendix D.1.4, we show that Q_n approximately solves the Bellman recursion $V_n^F = T_n^F V_{n-1}^F$ (Lemma D.6).

D.1.1 The One-Period Residual

Suppose that n periods remain and that the normalized inventory of the binding resources is $\mathbf{x} \in \mathcal{K}_B^2$. After taking action $a^F(\mathbf{x})$, the normalized binding inventory in the next period is

$$\mathbf{x}' \triangleq \frac{n\mathbf{x} - Y_{\mathbf{x}}}{n-1} = \mathbf{x} + \frac{\mathbf{x} - Y_{\mathbf{x}}}{n-1},$$

where $Y_{\mathbf{x}} \triangleq B_{a^F(\mathbf{x}), \mathcal{B}}$ denotes the consumption under action $a^F(\mathbf{x})$. We define the one-period residual relative to the fluid relaxation upper bound by

$$\delta_n^F(\mathbf{x}) \triangleq n\phi_{\mathcal{B}}(\mathbf{x}) - \mathbb{E}\left[R_{a^F(\mathbf{x})} + (n-1)\phi_{\mathcal{B}}(\mathbf{x}')\right]. \quad (25)$$

Through a second-order Taylor expansion of $\delta_n^F(\mathbf{x})$, Lemma D.1 shows that $\delta_n^F(\mathbf{x})$ equals $\frac{\Gamma(\mathbf{x})}{n-1}$ plus a summable remainder.

Lemma D.1 (Second-Order Expansion of the One-Period Residual). *Uniformly over $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$,*

$$\delta_n^{\text{F}}(\mathbf{x}) = \frac{\Gamma(\mathbf{x})}{n-1} + O(n^{-1-\alpha}).$$

In particular, there exists a constant $C < \infty$ such that $|\delta_n^{\text{F}}(\mathbf{x})| \leq C/n$ uniformly over $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$ for all sufficiently large n .

Proof. Fix $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$ and define $h \triangleq \frac{\mathbf{x} - Y_{\mathbf{x}}}{n-1}$. By bounded consumption and compactness of $\mathcal{K}_{\mathcal{B}}^2$, there exists a constant $C_1 < \infty$ such that $\|h\| \leq \frac{C_1}{n}$ uniformly over $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$ for all sufficiently large n .

By Lemma D.2,

$$\phi_{\mathcal{B}}(\mathbf{x} + h) = \phi_{\mathcal{B}}(\mathbf{x}) + \boldsymbol{\lambda}(\mathbf{x})^{\text{T}} h + \frac{1}{2} h^{\text{T}} \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) h + r_n(\mathbf{x}, h), \quad (26)$$

where

$$r_n(\mathbf{x}, h) = \int_0^1 (1-t) h^{\text{T}} \left[\nabla^2 \phi_{\mathcal{B}}(\mathbf{x} + th) - \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) \right] h dt.$$

Since $\nabla^2 \phi_{\mathcal{B}}$ is α -Hölder continuous by Assumption 3.1, there exists a constant $L < \infty$ such that

$$\left\| \nabla^2 \phi(\mathbf{x} + th) - \nabla^2 \phi(\mathbf{x}) \right\| \leq L \cdot t^{\alpha} \|h\|^{\alpha}.$$

Therefore,

$$\begin{aligned} |r_n(\mathbf{x}, h)| &\leq \int_0^1 (1-t) \|h\|^2 \left\| \nabla^2 \phi(\mathbf{x} + th) - \nabla^2 \phi(\mathbf{x}) \right\| dt \\ &\leq L \|h\|^{2+\alpha} \int_0^1 (1-t) t^{\alpha} dt \leq C_2 n^{-(2+\alpha)} \end{aligned} \quad (27)$$

for some constant $C_2 < \infty$ uniformly over $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$.

We now substitute (26) into the definition of $\delta_n^{\text{F}}(\mathbf{x})$. Note that $\mathbb{E}[R_{a^{\text{F}}(\mathbf{x})}] = \phi_{\mathcal{B}}(\mathbf{x})$ by Lemma 3.2. Moreover, Assumption 3.2 gives $\mathbb{E}[Y_{\mathbf{x}}] = \mathbf{x}$ and hence $\mathbb{E}[h] = 0$. Finally,

$$(n-1)^2 \cdot \mathbb{E} \left[h^{\text{T}} \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) h \right] = \mathbb{E} \left[(\mathbf{x} - Y_{\mathbf{x}})^{\text{T}} \nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) (\mathbf{x} - Y_{\mathbf{x}}) \right] = \text{tr} \left(\nabla^2 \phi_{\mathcal{B}}(\mathbf{x}) \Sigma(\mathbf{x}) \right) = -2\Gamma(\mathbf{x}).$$

Combining these identities with (26) and (27) yields

$$\delta_n^{\text{F}}(\mathbf{x}) = n\phi_{\mathcal{B}}(\mathbf{x}) - \mathbb{E} \left[R_{a^{\text{F}}(\mathbf{x})} + (n-1) \phi_{\mathcal{B}}(\mathbf{x} + h) \right] = \frac{\Gamma(\mathbf{x})}{n-1} + O(n^{-1-\alpha}).$$

Finally, since Γ is continuous by Lemma C.6 and therefore uniformly bounded on the compact set $\mathcal{K}_{\mathcal{B}}^2$, there exists a constant C such that $|\delta_n^{\text{F}}(\mathbf{x})| \leq C/n$ for all $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$ and sufficiently large n . $\square$

Lemma D.2 (Taylor Expansion). *Suppose that function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is twice continuously differentiable (i.e., $\phi \in C^2$). Then, for any $x, h \in \mathbb{R}^d$,*

$$\phi(x + h) = \phi(x) + \nabla \phi(x)^{\text{T}} h + \frac{1}{2} h^{\text{T}} \nabla^2 \phi(x) h + \int_0^1 (1-t) h^{\text{T}} \left[\nabla^2 \phi(x + th) - \nabla^2 \phi(x) \right] h dt.$$

Proof. Define the one-dimensional function $g(t) = \phi(x + th)$ for $t \in [0, 1]$. By the chain rule,

$$g'(t) = \nabla \phi(x + th)^T h, \quad g''(t) = h^T \nabla^2 \phi(x + th) h.$$

Applying the fundamental theorem of calculus twice gives

$$\begin{aligned} g(1) - g(0) &= \int_0^1 g'(s) ds = g'(0) + \int_0^1 (g'(s) - g'(0)) ds \\ &= g'(0) + \int_0^1 \int_0^s g''(t) dt ds \\ &= g'(0) + \int_0^1 (1-t) g''(t) dt, \end{aligned}$$

where the last equality follows by changing the integration order. Therefore,

$$\phi(x + h) = \phi(x) + \nabla \phi(x)^T h + \int_0^1 (1-t) h^T \nabla^2 \phi(x + th) h dt. \quad (28)$$

Adding and subtracting $\nabla^2 \phi(x)$ inside the integral and using $\int_0^1 (1-t) dt = \frac{1}{2}$ yields the desired result. $\square$

D.1.2 Smoothing the One-Period Residual

Lemma D.1 shows that the leading term of the one-period residual $\delta_n^F(\mathbf{x})$ is $\Gamma(\mathbf{x})/(n-1)$, which is Hölder continuous by Lemma C.6 but need not be differentiable.

A direct correction of the form $A_n(\mathbf{x}) = \Gamma(\mathbf{x}) H_{n-1}$ would reproduce the desired logarithmic term but would not provide the regularity required for the subsequent analysis. We therefore smooth the one-period residuals before accumulating them. This yields controlled first- and second-order derivatives while preserving the leading-order value of the correction.

Recall from Lemma C.6 that Γ is bounded and γ -Hölder continuous on $\mathcal{K}_B^2$. Let L_Γ be the Hölder constant for Γ ,⁵ and let $C_\Gamma \triangleq \sup_{\mathbf{x} \in \mathcal{K}_B^2} \Gamma(\mathbf{x})$. Define

$$\Gamma^M(\mathbf{x}) \triangleq \inf_{\mathbf{y} \in \mathcal{K}_B^2} \left\{ \Gamma(\mathbf{y}) + L_\Gamma \|\mathbf{x} - \mathbf{y}\|^\gamma \right\}, \quad \Gamma^{\text{ext}}(\mathbf{x}) \triangleq \min \left\{ C_\Gamma, \Gamma^M(\mathbf{x}) \right\}.$$

Lemma D.3 shows that $\Gamma^{\text{ext}}(\mathbf{x})$ is a global extension of $\Gamma(\mathbf{x})$ and is bounded, nonnegative, and γ -Hölder continuous on the entire space.

Lemma D.3. *The function Γ^{ext} satisfies $\Gamma^{\text{ext}}(\mathbf{x}) = \Gamma(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{K}_B^2$. Moreover, $\Gamma^{\text{ext}}(\mathbf{x}) \in [0, C_\Gamma]$ for all $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$ and is globally γ -Hölder continuous with Hölder constant L_Γ .*

Proof. By footnote 1, $\Gamma(\mathbf{x})$ is nonnegative. Thus, by definition, $\Gamma^M(\mathbf{x})$ and $\Gamma^{\text{ext}}(\mathbf{x})$ are nonnegative as well. Moreover, the definition of $\Gamma^{\text{ext}}(\mathbf{x})$ immediately gives $\Gamma^{\text{ext}}(\mathbf{x}) \in [0, C_\Gamma]$.

We next verify that Γ^{ext} agrees with Γ on $\mathcal{K}_B^2$. Fix $\mathbf{x} \in \mathcal{K}_B^2$. Since $\Gamma(\mathbf{x})$ is γ -Hölder continuous on $\mathcal{K}_B^2$ with Hölder constant L_Γ ,

$$\Gamma(\mathbf{y}) + L_\Gamma \|\mathbf{x} - \mathbf{y}\|^\gamma \geq \Gamma(\mathbf{x}) \quad \text{for every } \mathbf{y} \in \mathcal{K}_B^2.$$

⁵That is, for any $\mathbf{x}, \mathbf{x}' \in \mathcal{K}_B^2$, $\|\Gamma(\mathbf{x}) - \Gamma(\mathbf{x}')\| \leq L_\Gamma \|\mathbf{x} - \mathbf{x}'\|^\gamma$.

Taking the infimum over $\mathbf{y}$ gives $\Gamma^M(\mathbf{x}) \geq \Gamma(\mathbf{x})$. On the other hand, choosing $\mathbf{y} = \mathbf{x}$ in the definition of Γ^M gives $\Gamma^M(\mathbf{x}) \leq \Gamma(\mathbf{x})$. Thus $\Gamma^M(\mathbf{x}) = \Gamma(\mathbf{x})$. Because $0 \leq \Gamma(\mathbf{x}) \leq C_\Gamma$, the upper truncation leaves this value unchanged, and hence $\Gamma^{\text{ext}}(\mathbf{x}) = \Gamma(\mathbf{x})$ for all $\mathbf{x} \in \mathcal{K}_B^2$.

It remains to show that Γ^{ext} is globally γ -Hölder continuous. We first establish this property for Γ^M . For any $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^{|\mathcal{B}|}$ and $\mathbf{y} \in \mathcal{K}_B^2$,

$$\Gamma(\mathbf{y}) + L_\Gamma \|\mathbf{x} - \mathbf{y}\|^\gamma \leq \Gamma(\mathbf{y}) + L_\Gamma \|\mathbf{x}' - \mathbf{y}\|^\gamma + L_\Gamma \|\mathbf{x} - \mathbf{x}'\|^\gamma,$$

where the inequality follows from the triangle inequality $(a+b)^\gamma \leq a^\gamma + b^\gamma$ for any $a, b \geq 0$ and $0 < \gamma \leq 1$. Taking the infimum over $\mathbf{y} \in \mathcal{K}_B^2$ gives

$$\Gamma^M(\mathbf{x}) \leq \Gamma^M(\mathbf{x}') + L_\Gamma \|\mathbf{x} - \mathbf{x}'\|^\gamma.$$

Exchanging $\mathbf{x}$ and $\mathbf{x}'$ yields

$$|\Gamma^M(\mathbf{x}) - \Gamma^M(\mathbf{x}')| \leq L_\Gamma \|\mathbf{x} - \mathbf{x}'\|^\gamma.$$

Thus, Γ^M is globally γ -Hölder continuous with Hölder constant L_Γ . Since the truncation map $u \mapsto \min\{C_\Gamma, u\}$ is 1-Lipschitz on $[0, \infty)$, it follows that Γ^{ext} is also globally γ -Hölder continuous with Hölder constant L_Γ . $\square$

We next smooth the global extension Γ^{ext} of Γ by averaging it over neighborhoods of radius ε_n . We do so by convolving Γ^{ext} with a C^∞ mollifier kernel. Convolution is a standard operation for smoothing functions. In general, for a bounded measurable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and an integrable function $g : \mathbb{R}^d \rightarrow \mathbb{R}$, their convolution is defined by

$$(f * g)(\mathbf{x}) \triangleq \int_{\mathbb{R}^d} f(\mathbf{x} - z) g(z) dz.$$

Thus, if g is nonnegative, integrates to one, and is supported near the origin, $(f * g)(\mathbf{x})$ is a weighted local average of f in a neighborhood of $\mathbf{x}$. We review the basic properties of convolution in Appendix D.3.

Let φ be a nonnegative, infinitely continuously differentiable function (i.e., $\varphi \in C^\infty$) supported on the unit ball with integral one, and let $\varphi_\varepsilon(z) = \varepsilon^{-|\mathcal{B}|} \varphi(z/\varepsilon)$. Let

$$\varepsilon_n = n^{-\xi_\star}, \quad \Gamma_n^{\text{sm}} \triangleq \Gamma^{\text{ext}} * \varphi_{\varepsilon_n}$$

and for $n \geq 2$, define

$$\tilde{\delta}_n \triangleq \frac{\Gamma_n^{\text{sm}}}{n-1} = \frac{\Gamma^{\text{ext}} * \varphi_{\varepsilon_n}}{n-1} \quad (29)$$

with $\tilde{\delta}_1 = 0$. $\tilde{\delta}_n$ is a smooth approximation to the raw residual δ_n^F . Because $\tilde{\delta}_n$ is obtained by convolution with the C^∞ function φ_{ε_n} , it is itself C^∞ (see Proposition D.8). Lemma D.4 bounds the approximation error from smoothing and the first two derivatives of the smoothed residual $\tilde{\delta}_n$.

Lemma D.4 (Bounds from Smoothing). *Recall that $\alpha, \beta_\star \in (0, 1]$ are specified in Appendix C.1, and*

$$\gamma \triangleq \min\{\alpha, \beta_\star\} \in (0, 1], \quad \xi_\star \triangleq \frac{\beta_\star}{1 + \beta_\star} \in (0, 1/2].$$

There exists an integer $n_0 \geq 1$ and a constant $C < \infty$ such that, for every $n \geq n_0$ and every $\mathbf{x} \in \mathcal{K}_B^2$,

$$|\tilde{\delta}_n(\mathbf{x}) - \delta_n^F(\mathbf{x})| \leq C n^{-1-\xi_\star \gamma}.$$

Moreover, for every $n \geq n_0$ and every $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$,

$$\|\nabla \tilde{\delta}_n(\mathbf{x})\| \leq C n^{-1+\xi_\star(1-\gamma)} \quad \text{and} \quad \|\nabla^2 \tilde{\delta}_n(\mathbf{x})\| \leq C n^{-1+\xi_\star(2-\gamma)}.$$

Proof. Since $\delta_n^F(\mathbf{x}) = \frac{\Gamma(\mathbf{x})}{n-1} + O(n^{-1-\alpha})$ by Lemma D.1, $\tilde{\delta}_n \triangleq \frac{\Gamma_n^{\text{sm}}}{n-1}$ by (29), and $-1 - \xi_\star \gamma > -1 - \alpha$, it suffices to show that, for every $n \geq n_0$ and every $\mathbf{x} \in \mathcal{K}_B^2$,

$$|\Gamma_n^{\text{sm}}(\mathbf{x}) - \Gamma(\mathbf{x})| \leq C n^{-\xi_\star \gamma},$$

and for every $n \geq n_0$ and every $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$,

$$\|\nabla \Gamma_n^{\text{sm}}(\mathbf{x})\| \leq C n^{\xi_\star(1-\gamma)} \quad \text{and} \quad \|\nabla^2 \Gamma_n^{\text{sm}}(\mathbf{x})\| \leq C n^{\xi_\star(2-\gamma)}.$$

Since $\Gamma^{\text{ext}} = \Gamma$ on $\mathcal{K}_B^2$, $\Gamma^{\text{ext}} \in [0, C_\Gamma]$, and Γ^{ext} is globally γ -Hölder with constant L_Γ , all three bounds follow directly from Proposition D.9 with $f = \Gamma^{\text{ext}}$ and $\varepsilon = \varepsilon_n = n^{-\xi_\star}$. $\square$

D.1.3 Cumulative Residual and Corrected Fluid Reference

Lemma D.4 shows that the smoothed residual $\tilde{\delta}_n$ preserves the leading term $\Gamma(\mathbf{x})/(n-1)$ of the one-period residual δ_n^F up to a summable error, while providing the first- and second-order derivative bounds required for the subsequent analysis. Summing $1/(j-1)$ yields the logarithmic correction, whereas any term of order $j^{-1-\varepsilon}$, with $\varepsilon > 0$, has a uniformly bounded cumulative sum.

We define the cumulative residual A_n and the corrected fluid reference Q_n by

$$A_n(\mathbf{x}) \triangleq \sum_{j=n_0}^n \tilde{\delta}_j(\mathbf{x}), \quad Q_n(\mathbf{c}) \triangleq F_n(\mathbf{c}) - A_n(\mathbf{c}_B/n).$$

Here, n_0 is a fixed integer that exceeds all finite minimum-horizon thresholds introduced thus far, and we may increase n_0 finitely many times in the subsequent analysis. For $n < n_0$, we set $A_n = 0$ and $Q_n = F_n$.

Lemma D.5 provides uniform L_∞ bounds on A_n and its first- and second-order derivatives, together with a logarithmic expansion of A_n on $\mathcal{K}_B^2$.

Lemma D.5 (Cumulative Residual Bounds and Logarithmic Expansion). *There exists a constant $C < \infty$ such that, for every $n \geq 1$ and uniformly over $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$,*

$$\begin{aligned} |A_n(\mathbf{x})| &\leq C \ln n, \\ \|\nabla A_n(\mathbf{x})\|_\infty &\leq C n^{\xi_\star(1-\gamma)} \ln n, \\ \|\nabla^2 A_n(\mathbf{x})\|_\infty &\leq C n^{\xi_\star(2-\gamma)}. \end{aligned}$$

Moreover, uniformly over $\mathbf{x} \in \mathcal{K}_B^2$, $A_n(\mathbf{x}) = \Gamma(\mathbf{x}) H_{n-1} + O(1)$.

Proof. We bound A_n and its first two derivatives by summing the corresponding bounds on $\tilde{\delta}_j$. Since the lower index n_0 is fixed, all omitted initial terms affect the bounds only through constants.

Since $\Gamma^{\text{ext}} \in [0, C_\Gamma]$ and the mollifier φ_{ε_j} is nonnegative with unit mass, Proposition D.8 implies that $|\tilde{\delta}_j(\mathbf{x})| \leq C j^{-1}$ uniformly over $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$. Together with the derivative bounds in Lemma D.4,

we obtain,

$$\begin{aligned} |\tilde{\delta}_j(\mathbf{x})| &\leq C j^{-1}, \\ \|\nabla \tilde{\delta}_j(\mathbf{x})\|_\infty &\leq C j^{-1+\xi_\star(1-\gamma)}, \\ \|\nabla^2 \tilde{\delta}_j(\mathbf{x})\|_\infty &\leq C j^{-1+\xi_\star(2-\gamma)}, \end{aligned}$$

uniformly over $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$. Since $A_n(\mathbf{x}) \triangleq \sum_{j=n_0}^n \tilde{\delta}_j(\mathbf{x})$, uniformly over $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$,

$$|A_n(\mathbf{x})| \leq \sum_{j=n_0}^n |\tilde{\delta}_j(\mathbf{x})| \leq C \ln n.$$

Similarly,

$$\|\nabla A_n(\mathbf{x})\|_\infty \leq \sum_{j=n_0}^n \|\nabla \tilde{\delta}_j(\mathbf{x})\|_\infty \leq \begin{cases} C n^{\xi_\star(1-\gamma)}, & \gamma < 1, \\ C \ln n, & \gamma = 1, \end{cases}$$

and hence $\|\nabla A_n(\mathbf{x})\|_\infty \leq C n^{\xi_\star(1-\gamma)} \ln n$. Finally,

$$\|\nabla^2 A_n(\mathbf{x})\|_\infty \leq \sum_{j=n_0}^n \|\nabla^2 \tilde{\delta}_j(\mathbf{x})\|_\infty \leq C n^{\xi_\star(2-\gamma)}.$$

For the logarithmic expansion, Lemmas D.1 and D.4 yield, uniformly over $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$,

$$\tilde{\delta}_j(\mathbf{x}) = \frac{\Gamma(\mathbf{x})}{j-1} + O(j^{-1-\alpha}) + O(j^{-1-\xi_\star\gamma}).$$

Since $\alpha > 0$ and $\xi_\star\gamma > 0$, both error terms are summable. Consequently, uniformly in n and $\mathbf{x} \in \mathcal{K}_{\mathcal{B}}^2$,

$$A_n(\mathbf{x}) \triangleq \sum_{j=n_0}^n \tilde{\delta}_j(\mathbf{x}) = \Gamma(\mathbf{x}) H_{n-1} + O(1). \quad \square$$

D.1.4 Approximate Bellman Recursion

In this section, we show that the corrected fluid reference Q_n approximately solves the Bellman recursion $V_n^F = T_n^F V_{n-1}^F$, as formalized in Lemma D.6.

Lemma D.6 (Approximate Bellman Recursion). *There exist an integer $n_0 \geq 1$ and a summable deterministic sequence $(a_n)_{n \geq n_0}$ such that, for all $n \geq n_0$ and all states with $\mathbf{c}/n \in \mathcal{D}_1$,*

$$|Q_n(\mathbf{c}) - T_n^F Q_{n-1}(\mathbf{c})| \leq a_n.$$

In particular, one may take

$$a_n = C \left(n^{-1-\xi_\star\gamma} + n^{-2+\xi_\star(2-\gamma)} \right)$$

for some constant $C < \infty$.

Proof. Recall that, given a continuation value function $u(\cdot)$, we define the Bellman operator for the dynamic fluid policy by

$$(T_n^F u)(\mathbf{c}) \triangleq \mathbb{E} \left[R_{a_n^F(\mathbf{c})} + u(\mathbf{c} - B_{a_n^F(\mathbf{c})}) \right]. \quad (30)$$

Fix a state with $\mathbf{c}/n \in \mathcal{D}_1$ and let $\mathbf{x} = \mathbf{c}_{\mathcal{B}}/n$ denote the normalized inventory of binding resources.

Corollary C.3 gives that $a_n^F(\mathbf{c}) = a^F(\mathbf{x})$. Let $Y_{\mathbf{x}} \triangleq B_{a_n^F(\mathbf{c}), \mathcal{B}} = B_{a^F(\mathbf{x}), \mathcal{B}}$ denote the consumption of binding resources, and let

$$\mathbf{x}' \triangleq \frac{n\mathbf{x} - Y_{\mathbf{x}}}{n-1} = \mathbf{x} + \frac{\mathbf{x} - Y_{\mathbf{x}}}{n-1}$$

denote the normalized binding-resource inventory in the next period.

By definition,

$$\begin{aligned} Q_n(\mathbf{c}) &= F_n(\mathbf{c}) - A_n(\mathbf{x}), \\ T_n^F Q_{n-1}(\mathbf{c}) &= T_n^F F_{n-1}(\mathbf{c}) - \mathbb{E}[A_{n-1}(\mathbf{x}')]. \end{aligned}$$

Therefore,

$$\begin{aligned} Q_n(\mathbf{c}) - T_n^F Q_{n-1}(\mathbf{c}) &= F_n(\mathbf{c}) - T_n^F F_{n-1}(\mathbf{c}) + \mathbb{E}[A_{n-1}(\mathbf{x}')] - A_n(\mathbf{x}) \\ &= \underbrace{[\delta_n^F(\mathbf{x}) - \tilde{\delta}_n(\mathbf{x})]}_{(a)} + \underbrace{[\mathbb{E}[A_{n-1}(\mathbf{x}')] - A_{n-1}(\mathbf{x})]}_{(b)}, \end{aligned}$$

where the second equality follows from the fact that $F_n(\mathbf{c}) - T_n^F F_{n-1}(\mathbf{c}) = \delta_n^F(\mathbf{x})$ by (25) and (30) and $A_n(\mathbf{x}) = A_{n-1}(\mathbf{x}) + \delta_n(\mathbf{x})$ by the definition of $A_n(\mathbf{x})$.

Note that $|(a)| \leq C n^{-1-\xi_\star \gamma}$ uniformly by Lemma D.4. It remains to show that uniformly, $|(b)| \leq C n^{-2+\xi_\star(2-\gamma)}$, which we do now.

Let $\Delta \triangleq \mathbf{x}' - \mathbf{x} = (\mathbf{x} - Y_{\mathbf{x}})/(n-1)$. From (28) we have

$$A_{n-1}(\mathbf{x}') - A_{n-1}(\mathbf{x}) = \nabla A_{n-1}(\mathbf{x})^\top \Delta + \int_0^1 (1-t) \Delta^\top \nabla^2 A_{n-1}(\mathbf{x} + t\Delta) \Delta dt.$$

Since $\mathbb{E}[\Delta] = 0$ by Assumption 3.2, taking expectations gives

$$|\mathbb{E}[A_{n-1}(\mathbf{x}')] - A_{n-1}(\mathbf{x})| \leq \frac{1}{2} \|\nabla^2 A_{n-1}\|_\infty \mathbb{E}\|\Delta\|^2 \leq C n^{-2+\xi_\star(2-\gamma)},$$

where the second inequality follows from the uniform bound on $\|\nabla^2 A_{n-1}\|_\infty$ in Lemma D.5 and the bound $\|\Delta\|_\infty \leq C/n$ by bounded consumption and compactness of $\mathcal{K}_{\mathcal{B}}^2$.

Combining the bounds on (a) and (b) proves Lemma D.6. Finally, $(a_n)_{n \geq n_0}$ is summable because $\xi_\star \gamma > 0$ and $\xi_\star(2-\gamma) < 1$ as $\gamma > 0$ and $\xi_\star \leq 1/2$. $\square$

D.2 Proof of Proposition 6.1

Increase n_0 so that all bounds in this section hold simultaneously. Lemma D.5 proves part 1 of Proposition 6.1. To obtain the stated consequence for Q_n , fix $\mathbf{c}/n \in \mathcal{K}$. Then $\mathbf{c}_{\mathcal{B}}/n \in \mathcal{K}_{\mathcal{B}} \subseteq \mathcal{K}_{\mathcal{B}}^2$, and the definitions $F_n(\mathbf{c}) = n\phi^F(\mathbf{c}/n)$ and $Q_n(\mathbf{c}) = F_n(\mathbf{c}) - A_n(\mathbf{c}_{\mathcal{B}}/n)$, together with the expansion for A_n , give

$$Q_n(\mathbf{c}) = n\phi^F(\mathbf{c}/n) - \Gamma(\mathbf{c}_{\mathcal{B}}/n) H_{n-1} + O(1).$$

Lemma D.6 proves part 2 of Proposition 6.1.

D.3 Basic Properties of Convolution

In this section, we review the basic properties of convolution used in the smoothing in Appendix D.1.

For a bounded measurable function $f : \mathbb{R}^d \rightarrow \mathbb{R}$ and an integrable function $g : \mathbb{R}^d \rightarrow \mathbb{R}$, their convolution is defined by

$$(f * g)(\mathbf{x}) \triangleq \int_{\mathbb{R}^d} f(\mathbf{x} - z) g(z) dz.$$

By the change of variables $y = \mathbf{x} - z$,

$$(f * g)(\mathbf{x}) = \int_{\mathbb{R}^d} f(\mathbf{x} - z) g(z) dz = \int_{\mathbb{R}^d} g(\mathbf{x} - y) f(y) dy = (g * f)(\mathbf{x}). \quad (31)$$

Thus, convolution is commutative whenever the integrals are well defined.

For a multi-index $\nu = (\nu_1, \dots, \nu_d) \in \mathbb{N}_0^d$, let $|\nu| \triangleq \nu_1 + \dots + \nu_d$ and $D^\nu \triangleq \partial_1^{\nu_1} \dots \partial_d^{\nu_d}$. The next three propositions establish the properties used in the smoothing construction in Appendix D.1.

Proposition D.7 (Scaled Mollifier Properties). *Let $\varphi \in C^\infty(\mathbb{R}^d)$ be nonnegative, supported on the unit ball, and normalized so that $\int_{\mathbb{R}^d} \varphi(w) dw = 1$. For $\varepsilon > 0$, define*

$$\varphi_\varepsilon(z) \triangleq \varepsilon^{-d} \varphi(z/\varepsilon).$$

Then $\varphi_\varepsilon \in C^\infty(\mathbb{R}^d)$, $\varphi_\varepsilon \geq 0$,

$$\text{supp}(\varphi_\varepsilon) \subseteq \{z \in \mathbb{R}^d : \|z\| \leq \varepsilon\}, \quad \int_{\mathbb{R}^d} \varphi_\varepsilon(z) dz = 1.$$

Moreover, for every multi-index ν ,

$$D^\nu \varphi_\varepsilon(z) = \varepsilon^{-d-|\nu|} (D^\nu \varphi)(z/\varepsilon). \quad (32)$$

If $|\nu| \geq 1$, then

$$\int_{\mathbb{R}^d} D^\nu \varphi_\varepsilon(z) dz = 0.$$

Proof. Multiplication by the positive constant ε^{-d} and composition with the linear map $z \mapsto z/\varepsilon$ preserve smoothness and nonnegativity. If $\|z\| > \varepsilon$, then $\|z/\varepsilon\| > 1$, so the support assumption on φ gives $\varphi_\varepsilon(z) = 0$. This proves the support claim.

For the normalization, use the d -dimensional change of variables $z = \varepsilon w$. Its Jacobian gives $dz = \varepsilon^d dw$, and hence

$$\int_{\mathbb{R}^d} \varphi_\varepsilon(z) dz = \int_{\mathbb{R}^d} \varepsilon^{-d} \varphi(z/\varepsilon) dz = \int_{\mathbb{R}^d} \varphi(w) dw = 1.$$

Each derivative with respect to a coordinate of z contributes one factor ε^{-1} by the chain rule. Applying $|\nu|$ coordinate derivatives therefore gives

$$D^\nu \varphi_\varepsilon(z) = \varepsilon^{-d} \varepsilon^{-|\nu|} (D^\nu \varphi)(z/\varepsilon).$$

Note that if φ vanishes on an open neighborhood, all of its derivatives vanish there as well. Thus, for any $\nu \in \mathbb{N}_0^d$,

$$D^\nu \varphi_\varepsilon(z) = 0, \quad \forall z \text{ with } \|z\| > \varepsilon. \quad (33)$$

It remains to prove the zero-integral identity. Suppose $|\nu| \geq 1$, and choose a coordinate i for which $\nu_i \geq 1$. Let e_i denote the i th coordinate vector and write $z = (z_{-i}, z_i)$, where z_i is the i th coordinate. Fubini's theorem gives

$$\int_{\mathbb{R}^d} D^\nu \varphi_\varepsilon(z) dz = \int_{\mathbb{R}^{d-1}} \left[\int_{\mathbb{R}} \partial_i (D^{\nu-e_i} \varphi_\varepsilon)(z_{-i}, z_i) dz_i \right] dz_{-i}.$$

For each fixed z_{-i} , the function $t \mapsto (D^{\nu-e_i} \varphi_\varepsilon)(z_{-i}, t)$ vanishes for all sufficiently large $|t|$ by (33).

Thus, the one-dimensional fundamental theorem of calculus makes the inner integral equal to zero. The outer integral is consequently zero as well. $\square$

Proposition D.8 (Mollification as a Smooth Local Average). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and measurable, and let $\varepsilon > 0$. Then $f * \varphi_\varepsilon$ is well defined and, for every $\mathbf{x} \in \mathbb{R}^d$,*

$$(f * \varphi_\varepsilon)(\mathbf{x}) = \int_{\mathbb{R}^d} f(\mathbf{x} - z) \varphi_\varepsilon(z) dz = \int_{\mathbb{R}^d} f(y) \varphi_\varepsilon(\mathbf{x} - y) dy = \int_{\mathbb{R}^d} f(\mathbf{x} - \varepsilon w) \varphi(w) dw. \quad (34)$$

If $m \leq f \leq M$ on $\mathbb{R}^d$, then

$$m \leq f * \varphi_\varepsilon \leq M.$$

In particular, $\|f * \varphi_\varepsilon\|_\infty \leq \|f\|_\infty$, and $0 \leq f \leq M$ implies $0 \leq f * \varphi_\varepsilon \leq M$. Finally, $f * \varphi_\varepsilon \in C^\infty(\mathbb{R}^d)$ and, for every multi-index ν ,

$$D^\nu(f * \varphi_\varepsilon)(\mathbf{x}) = (f * D^\nu \varphi_\varepsilon)(\mathbf{x}) = \int_{\mathbb{R}^d} f(\mathbf{x} - z) (D^\nu \varphi_\varepsilon)(z) dz = \int_{\mathbb{R}^d} f(y) (D^\nu \varphi_\varepsilon)(\mathbf{x} - y) dy.$$

Proof. The first two equalities in (34) follow from the definition and the commutative property in (31), and the third equality from the change of variables $z = \varepsilon w$ and noting that $dz = \varepsilon^d dw$.

Since $\varphi_\varepsilon \geq 0$ and $\int \varphi_\varepsilon = 1$, for any $\mathbf{x}$,

$$|(f * \varphi_\varepsilon)(\mathbf{x})| \leq \int_{\mathbb{R}^d} |f(\mathbf{x} - z) \varphi_\varepsilon(z)| dz \leq \|f\|_\infty \int_{\mathbb{R}^d} \varphi_\varepsilon(z) dz = \|f\|_\infty.$$

If $m \leq f \leq M$, multiplying these inequalities by the nonnegative weight $\varphi_\varepsilon(z)$ and integrating gives

$$m = m \int_{\mathbb{R}^d} \varphi_\varepsilon(z) dz \leq (f * \varphi_\varepsilon)(\mathbf{x}) \leq M \int_{\mathbb{R}^d} \varphi_\varepsilon(z) dz = M.$$

Finally, we justify the differentiation formula. Fix $\mathbf{x} \in \mathbb{R}^d$ and a coordinate i . Using the second representation in (34), for $h \neq 0$,

$$\frac{(f * \varphi_\varepsilon)(\mathbf{x} + h e_i) - (f * \varphi_\varepsilon)(\mathbf{x})}{h} = \int_{\mathbb{R}^d} f(y) \frac{\varphi_\varepsilon(\mathbf{x} + h e_i - y) - \varphi_\varepsilon(\mathbf{x} - y)}{h} dy.$$

For $|h| \leq 1$, the one-dimensional mean-value theorem applied along the i th coordinate gives the pointwise bound

$$\left| f(y) \frac{\varphi_\varepsilon(\mathbf{x} + h e_i - y) - \varphi_\varepsilon(\mathbf{x} - y)}{h} \right| \leq \|f\|_\infty \|\partial_i \varphi_\varepsilon\|_\infty \mathbb{1}\{\|y - \mathbf{x}\| \leq \varepsilon + 1\}.$$

The right-hand side is integrable because it is bounded and supported on a ball of finite volume. The integrand on the left converges pointwise to $f(y) \partial_i \varphi_\varepsilon(\mathbf{x} - y)$ as $h \rightarrow 0$. The dominated convergence theorem therefore yields

$$\partial_i(f * \varphi_\varepsilon)(\mathbf{x}) = \int_{\mathbb{R}^d} f(y) \partial_i \varphi_\varepsilon(\mathbf{x} - y) dy.$$

Since $\varphi_\varepsilon \in C^\infty$, repeating this argument with $D^\mu \varphi_\varepsilon$ in place of φ_ε , for an arbitrary multi-index μ , proves inductively that derivatives of every order exist and satisfy the stated representation. $\square$

Proposition D.9 (Hölder Mollification Bounds). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be bounded and globally γ -Hölder*

continuous for some $\gamma \in (0, 1]$: that is, there is a constant $L < \infty$ such that

$$|f(\mathbf{x}) - f(\mathbf{y})| \leq L \|\mathbf{x} - \mathbf{y}\|^\gamma \quad \text{for all } \mathbf{x}, \mathbf{y} \in \mathbb{R}^d.$$

Then, for every $\varepsilon > 0$ and $\mathbf{x} \in \mathbb{R}^d$,

$$|(f * \varphi_\varepsilon)(\mathbf{x}) - f(\mathbf{x})| \leq L\varepsilon^\gamma \int_{\mathbb{R}^d} \|w\|^\gamma \varphi(w) dw \leq L\varepsilon^\gamma.$$

Moreover, for every multi-index ν with $|\nu| = k \in \{1, 2\}$,

$$|D^\nu(f * \varphi_\varepsilon)(\mathbf{x})| \leq L\varepsilon^{\gamma-k} \int_{\mathbb{R}^d} \|w\|^\gamma |D^\nu \varphi(w)| dw.$$

Consequently, for the fixed finite-dimensional norms used for the gradient and Hessian, there is a constant $C < \infty$, depending only on d , γ , and φ , such that

$$\|\nabla(f * \varphi_\varepsilon)(\mathbf{x})\| \leq CL\varepsilon^{\gamma-1}, \quad \|\nabla^2(f * \varphi_\varepsilon)(\mathbf{x})\| \leq CL\varepsilon^{\gamma-2}.$$

Proof. By the third representation in (34) and the Hölder bound, we have

$$\begin{aligned} |(f * \varphi_\varepsilon)(\mathbf{x}) - f(\mathbf{x})| &= \left| \int_{\mathbb{R}^d} [f(\mathbf{x} - \varepsilon w) - f(\mathbf{x})] \varphi(w) dw \right| \\ &\leq \int_{\mathbb{R}^d} |f(\mathbf{x} - \varepsilon w) - f(\mathbf{x})| \varphi(w) dw \\ &\leq L\varepsilon^\gamma \int_{\mathbb{R}^d} \|w\|^\gamma \varphi(w) dw. \end{aligned}$$

Because φ is supported on the unit ball, $\|w\|^\gamma \leq 1$ wherever $\varphi(w) \neq 0$. Its nonnegativity and unit integral therefore make the final integral at most one, which leads to the upper bound $L\varepsilon^\gamma$.

Now fix a multi-index ν with $|\nu| = k \in \{1, 2\}$. The derivative formula in Proposition D.8 and the identity $\int D^\nu \varphi_\varepsilon = 0$ from Proposition D.7 yield:

$$D^\nu(f * \varphi_\varepsilon)(\mathbf{x}) = \int_{\mathbb{R}^d} [f(\mathbf{x} - z) - f(\mathbf{x})] (D^\nu \varphi_\varepsilon)(z) dz.$$

Taking absolute values and applying the global Hölder bound yields

$$|D^\nu(f * \varphi_\varepsilon)(\mathbf{x})| \leq L \int_{\mathbb{R}^d} \|z\|^\gamma |D^\nu \varphi_\varepsilon(z)| dz.$$

By (32) and the change of variables $z = \varepsilon w$, we have:

$$\int_{\mathbb{R}^d} \|z\|^\gamma |D^\nu \varphi_\varepsilon(z)| dz = \int_{\mathbb{R}^d} \varepsilon^\gamma \|w\|^\gamma \varepsilon^{-d-k} |D^\nu \varphi(w)| \varepsilon^d dw = \varepsilon^{\gamma-k} \int_{\mathbb{R}^d} \|w\|^\gamma |D^\nu \varphi(w)| dw.$$

The last integral is finite because $D^\nu \varphi$ is continuous and compactly supported. This proves the componentwise derivative bound. For the Euclidean vector norm and its induced matrix norm,

$$\|\nabla(f * \varphi_\varepsilon)(\mathbf{x})\| \leq \sqrt{d} \max_{1 \leq i \leq d} |\partial_i(f * \varphi_\varepsilon)(\mathbf{x})|, \quad \|\nabla^2(f * \varphi_\varepsilon)(\mathbf{x})\| \leq d \max_{1 \leq i, j \leq d} |\partial_{ij}(f * \varphi_\varepsilon)(\mathbf{x})|.$$

Since the gradient and Hessian have finitely many components, the stated norm bounds follow after

enlarging the constant C . □

E Proof of Proposition 6.2

In this section, we prove Proposition 6.2. We fix a time horizon $n \geq n_0$ and a deterministic capacity vector $\mathbf{c}$ such that $\mathbf{c}/n \in \mathcal{K}$. We terminate the analysis once n_0 periods remain and let $\bar{N} \triangleq n - n_0$ denote the maximum number of decisions made before this cutoff.

Unlike elsewhere in the paper, in this section we index time forward: We let k denote the number of decisions made since the beginning of the problem, and $h_k \triangleq n - k$ denote the number of periods remaining after the k -th decision.

We now define the states, actions, and transitions for $1 \leq k \leq \bar{N}$. Let $\mathbf{c}_0 = \mathbf{c} \in \mathbb{R}_+^m$ denote the initial inventory and $\mathbf{c}_k$ the remaining inventory after the k -th decision. In period $k \geq 1$, the policy observes the independent opportunity $\boldsymbol{\theta}_k = (R_{a,k}, B_{a,k})_{a \in \mathcal{A}}$, chooses the action $a_k \triangleq a_{h_k+1}^F(\mathbf{c}_{k-1})$ defined in (3), collects a reward $\hat{R}_k \triangleq R_{a_k,k}$, and consumes a vector of resources $\mathbf{Y}_k \triangleq B_{a_k,k} \in \mathbb{R}_+^m$. The state variables satisfy

$$\mathbf{c}_k = \mathbf{c}_{k-1} - \mathbf{Y}_k, \quad \boldsymbol{\rho}_k = \mathbf{c}_k/h_k, \quad \mathbf{x}_k = (\boldsymbol{\rho}_k)_{\mathcal{B}}.$$

In addition, let $\{\mathcal{G}_k\}_{k \geq 0}$ denote the filtration of information, where $\mathcal{G}_k = \sigma(\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_k)$ for $k \geq 1$ and $\mathcal{G}_0 = \{\emptyset, \Omega\}$. Hence, $\mathbf{c}_k$, $\boldsymbol{\rho}_k$, and $\mathbf{x}_k$ are $\mathcal{G}_k$ -measurable, whereas $\boldsymbol{\theta}_{k+1}$ is independent of $\mathcal{G}_k$.

We define the stopping times σ and τ by

$$\sigma \triangleq \inf \{k : 0 \leq k < \bar{N}, \boldsymbol{\rho}_k \notin \mathcal{D}_1\}, \quad \tau \triangleq \sigma \wedge \bar{N},$$

with $\inf \emptyset = \infty$. Let $h_\tau \triangleq n - \tau$ denote the remaining horizon at stopping. If the trajectory exits the region $\mathcal{D}_1$ before the cutoff period $\bar{N}$, then $\tau = \sigma < \bar{N}$ and $h_\tau > n - \bar{N} = n_0$. If no such exit occurs before the cutoff, then $\tau = \bar{N}$ and $h_\tau = n_0$. Therefore, h_τ measures how early the analysis stops: $h_\tau > n_0$ can occur only if the trajectory exits $\mathcal{D}_1$ within the first $\bar{N} - 1$ periods.

On event $\{k < \tau\}$, we have $h_k \geq n_0 + 1$ and $\boldsymbol{\rho}_k \in \mathcal{D}_1$. Hence, Corollary C.3 gives $a_{k+1} = a^F(\mathbf{x}_k)$. Moreover, since $\boldsymbol{\theta}_{k+1}$ is independent from $\mathcal{G}_k$, we have

$$\begin{aligned} \mathbb{E}[\mathbf{Y}_{k+1, \mathcal{B}} \mid \mathcal{G}_k] &= \mathbf{x}_k, \\ \mathbb{E}[\mathbf{Y}_{k+1, s} \mid \mathcal{G}_k] &= \mu_s(\mathbf{x}_k), \quad s \in \mathcal{S} \triangleq \mathcal{B}^c. \end{aligned}$$

Finally, the capacity recursion yields $h_{k+1} = h_k - 1$ and

$$\boldsymbol{\rho}_{k+1} = \frac{h_k \boldsymbol{\rho}_k - \mathbf{Y}_{k+1}}{h_k - 1} = \boldsymbol{\rho}_k + \frac{\boldsymbol{\rho}_k - \mathbf{Y}_{k+1}}{h_k - 1}, \quad \mathbf{x}_{k+1} - \mathbf{x}_k = \frac{\mathbf{x}_k - \mathbf{Y}_{k+1, \mathcal{B}}}{h_k - 1}.$$

E.1 Trajectory Before Stopping

In this section, we analyze the inventory trajectory before stopping.

We first show in Lemma E.1 that the normalized binding resources is a martingale and thus have zero conditional drift, and that its one-step increment is of order $1/h_k$.

Lemma E.1 (Martingale for Binding Resources). *For $0 \leq k < \bar{N}$, define the one-step increment of the normalized binding resources before stopping by*

$$\Delta \mathbf{M}_{k+1}^{\mathcal{B}} \triangleq \mathbb{1}\{k < \tau\} \frac{\mathbf{x}_k - \mathbf{Y}_{k+1, \mathcal{B}}}{h_k - 1} = \mathbb{1}\{k < \tau\} (\mathbf{x}_{k+1} - \mathbf{x}_k) = \mathbf{x}_{(k+1) \wedge \tau} - \mathbf{x}_{k \wedge \tau}.$$

Then, $\mathbf{M}_k^\mathcal{B} \triangleq \sum_{j=1}^k \Delta \mathbf{M}_j^\mathcal{B} = \mathbf{x}_{k \wedge \tau} - \mathbf{x}_0$ is a martingale, and its one-step increment satisfies $\|\Delta \mathbf{M}_{k+1}^\mathcal{B}\| \leq C/h_k$ for some constant $C < \infty$.

Proof. On the event $\{k < \tau\}$, $\mathbf{x}_k \in \mathcal{K}_\mathcal{B}^1$ and hence $\mathbb{E}[\mathbf{Y}_{k+1,\mathcal{B}} \mid \mathcal{G}_k] = \mathbf{x}_k$ by Assumption 3.2. Since both $\{k < \tau\}$ and $\mathbf{x}_k$ are $\mathcal{G}_k$ -measurable,

$$\mathbb{E}[\Delta \mathbf{M}_{k+1}^\mathcal{B} \mid \mathcal{G}_k] = \frac{\mathbb{1}\{k < \tau\}}{h_k - 1} \left(\mathbf{x}_k - \mathbb{E}[\mathbf{Y}_{k+1,\mathcal{B}} \mid \mathcal{G}_k] \right) = \mathbf{0}.$$

Therefore,

$$\mathbb{E}[\mathbf{M}_{k+1}^\mathcal{B} \mid \mathcal{G}_k] = \mathbb{E}[\Delta \mathbf{M}_{k+1}^\mathcal{B} + \mathbf{M}_k^\mathcal{B} \mid \mathcal{G}_k] = \mathbf{M}_k^\mathcal{B},$$

so $\{\mathbf{M}_k^\mathcal{B}\}$ is a martingale.

Finally, before stopping, $\mathbf{x}_k \in \mathcal{K}_\mathcal{B}^1$ and is therefore uniformly bounded. Together with bounded consumption by Assumption 2.1, this yields $\|\Delta \mathbf{M}_{k+1}^\mathcal{B}\| \leq C/h_k$ for some constant $C < \infty$. $\square$

We next analyze the slack resources. For each slack resource $s \in \mathcal{S}$, define

$$z_{k,s} \triangleq \rho_{k \wedge \tau, s} - \mu_s(\mathbf{x}_{k \wedge \tau}).$$

On the event $\{k \leq \tau\}$, $\rho_{k,s}$ is the capacity of slack resource s per remaining period and $\mu_s(\mathbf{x}_k)$ is its expected per-period consumption. Hence, $z_{k,s}$ represents the per-period capacity in excess of expected consumption. By Assumption 3.2, $z_{k,s} > 0$ before stopping. This excess capacity induces a positive drift for the normalized inventory of slack resources.

We decompose the one-period increment of $(z_{k,s})_k$ into a predictable drift and a martingale-difference noise term. Specifically, write

$$z_{k+1,s} - z_{k,s} = b_{k,s} + \xi_{k+1,s},$$

where

$$b_{k,s} \triangleq \mathbb{E}[z_{k+1,s} - z_{k,s} \mid \mathcal{G}_k]$$

is the predictable drift, and

$$\xi_{k+1,s} \triangleq z_{k+1,s} - z_{k,s} - b_{k,s}$$

is a martingale difference satisfying $\mathbb{E}[\xi_{k+1,s} \mid \mathcal{G}_k] = 0$. In Lemma E.2 we show that,

$$b_{k,s} = \frac{z_{k,s}}{h_k - 1} + o\left(\frac{1}{h_k}\right), \quad |\xi_{k+1,s}| = O\left(\frac{1}{h_k}\right).$$

Therefore, because the slack state has a positive drift, a slack resource can exit the local region $\mathcal{D}_1$ only through a sufficiently large downward fluctuation of the cumulative noise $\sum_{j=1}^k \xi_{j,s}$, which is a martingale. We control such fluctuations using a maximal martingale concentration inequality in Appendix E.2.

Lemma E.2 (Positive Drift for Slack Resources). *Suppose $\mathcal{S} \neq \emptyset$. Fix $s \in \mathcal{S}$ and $0 \leq k < \bar{N}$. There is a deterministic nondecreasing function ω , with $\omega(u) \rightarrow 0$ as $u \downarrow 0$, such that, on $\{k < \tau\}$,*

$$\left| b_{k,s} - \frac{z_{k,s}}{h_k - 1} \right| \leq \frac{C}{h_k} \omega\left(\frac{C}{h_k}\right), \quad |\xi_{k+1,s}| \leq \frac{C}{h_k},$$

for some constant $C < \infty$. Moreover, by choosing the fixed cutoff n_0 sufficiently large,

$$b_{k,s} \geq \frac{\eta}{h_k} > 0$$

on $\{k < \tau\}$. Finally, we note that $b_{k,s} = \xi_{k+1,s} = 0$ on $\{k \geq \tau\}$.

Proof. Fix $s \in \mathcal{S}$ and work on the event $\{k < \tau\}$. Since $(k+1) \wedge \tau = k+1$ on this event,

$$z_{k,s} = \rho_{k,s} - \mu_s(\mathbf{x}_k), \quad z_{k+1,s} = \rho_{k+1,s} - \mu_s(\mathbf{x}_{k+1}).$$

Hence,

$$z_{k+1,s} - z_{k,s} = \left(\rho_{k+1,s} - \rho_{k,s} \right) - \left(\mu_s(\mathbf{x}_{k+1}) - \mu_s(\mathbf{x}_k) \right).$$

Moreover,

$$\rho_{k+1,s} - \rho_{k,s} = \frac{\rho_{k,s} - \mathbf{Y}_{k+1,s}}{h_k - 1} = \frac{z_{k,s}}{h_k - 1} - \frac{\mathbf{Y}_{k+1,s} - \mu_s(\mathbf{x}_k)}{h_k - 1},$$

where the second equality follows from $\rho_{k,s} = z_{k,s} + \mu_s(\mathbf{x}_k)$. Hence, the preceding identity yields:

$$z_{k+1,s} - z_{k,s} = \frac{z_{k,s}}{h_k - 1} - \frac{\mathbf{Y}_{k+1,s} - \mu_s(\mathbf{x}_k)}{h_k - 1} - (\mu_s(\mathbf{x}_{k+1}) - \mu_s(\mathbf{x}_k)). \quad (35)$$

Since $\mathbf{x}_k$ is $\mathcal{G}_k$ -measurable and $\mathbb{E}[\mathbf{Y}_{k+1,s} \mid \mathcal{G}_k] = \mu_s(\mathbf{x}_k)$ by Assumption 3.2, we have

$$\mathbb{E} \left[\frac{\mathbf{Y}_{k+1,s} - \mu_s(\mathbf{x}_k)}{h_k - 1} \mid \mathcal{G}_k \right] = 0.$$

Therefore, taking conditional expectations in (35) gives

$$b_{k,s} = \frac{z_{k,s}}{h_k - 1} - \mathbb{E} \left[\mu_s(\mathbf{x}_{k+1}) - \mu_s(\mathbf{x}_k) \mid \mathcal{G}_k \right]. \quad (36)$$

Predictable Drift Set $\mathbf{d}_{k+1} \triangleq \Delta \mathbf{M}_{k+1}^{\mathcal{B}}$. On $\{k < \tau\}$, Lemma E.1 yields

$$\mathbf{d}_{k+1} = \mathbf{x}_{k+1} - \mathbf{x}_k, \quad \mathbb{E}[\mathbf{d}_{k+1} \mid \mathcal{G}_k] = \mathbf{0}, \quad \|\mathbf{d}_{k+1}\| \leq \frac{C}{h_k}.$$

Since $\mathcal{K}_B^1 \in \mathcal{U}_B$, we can choose $\delta > 0$ such that the closed δ -neighborhood

$$\hat{\mathcal{K}} \triangleq \{\mathbf{y} : \text{dist}(\mathbf{y}, \mathcal{K}_B^1) \leq \delta\} \in \mathcal{U}_B.$$

Increase n_0 so that $C/(n_0 + 1) \leq \delta$. Since $\mathbf{x}_k \in \mathcal{K}_B^1$ and $h_k \geq n_0 + 1$ before stopping, for every $t \in [0, 1]$,

$$\text{dist}(\mathbf{x}_k + t\mathbf{d}_{k+1}, \mathcal{K}_B^1) \leq t\|\mathbf{d}_{k+1}\| \leq \frac{C}{h_k} \leq \delta.$$

Therefore, every segment from $\mathbf{x}_k$ to $\mathbf{x}_{k+1}$ lies in $\hat{\mathcal{K}}$. Because $\mathcal{S}$ is finite and each μ_s is continuously differentiable by Assumption 3.2, the gradients $\nabla \mu_s$ are uniformly continuous on $\hat{\mathcal{K}}$. Define

$$\omega(u) \triangleq \max_{s \in \mathcal{S}} \sup_{\substack{\mathbf{x}, \mathbf{y} \in \hat{\mathcal{K}} \\ \|\mathbf{x} - \mathbf{y}\| \leq u}} \|\nabla \mu_s(\mathbf{x}) - \nabla \mu_s(\mathbf{y})\|.$$

Then, ω is deterministic and nondecreasing, and uniform continuity implies that $\omega(u) \rightarrow 0$ as $u \downarrow 0$.

By the fundamental theorem of calculus along the line segment from $\mathbf{x}_k$ to $\mathbf{x}_{k+1}$, we have:

$$\mu_s(\mathbf{x}_{k+1}) - \mu_s(\mathbf{x}_k) = \int_0^1 \nabla \mu_s(\mathbf{x}_k + t\mathbf{d}_{k+1})^\top \mathbf{d}_{k+1} dt = \nabla \mu_s(\mathbf{x}_k)^\top \mathbf{d}_{k+1} + R_{k+1,s}^\mu, \quad (37)$$

where

$$R_{k+1,s}^\mu \triangleq \int_0^1 \left[\nabla \mu_s(\mathbf{x}_k + t\mathbf{d}_{k+1}) - \nabla \mu_s(\mathbf{x}_k) \right]^\top \mathbf{d}_{k+1} dt.$$

Since the entire line segment lies in $\hat{\mathcal{K}}$, the definition and monotonicity of ω imply

$$\begin{aligned} |R_{k+1,s}^\mu| &\leq \|\mathbf{d}_{k+1}\| \int_0^1 \|\nabla \mu_s(\mathbf{x}_k + t\mathbf{d}_{k+1}) - \nabla \mu_s(\mathbf{x}_k)\| dt \\ &\leq \|\mathbf{d}_{k+1}\| \int_0^1 \omega(t\|\mathbf{d}_{k+1}\|) dt \\ &\leq \|\mathbf{d}_{k+1}\| \omega(\|\mathbf{d}_{k+1}\|). \end{aligned}$$

Using $\|\mathbf{d}_{k+1}\| \leq C/h_k$ once more yields

$$|R_{k+1,s}^\mu| \leq \frac{C}{h_k} \omega\left(\frac{C}{h_k}\right). \quad (38)$$

Furthermore, since $\mathbb{E}[\mathbf{d}_{k+1} \mid \mathcal{G}_k] = \mathbf{0}$, (37) and (38) imply

$$\left| \mathbb{E}[\mu_s(\mathbf{x}_{k+1}) - \mu_s(\mathbf{x}_k) \mid \mathcal{G}_k] \right| \leq \frac{C}{h_k} \omega\left(\frac{C}{h_k}\right).$$

Combining this bound with (36) gives

$$\left| b_{k,s} - \frac{z_{k,s}}{h_k - 1} \right| \leq \frac{C}{h_k} \omega\left(\frac{C}{h_k}\right).$$

Finally, choose n_0 sufficiently large so that $C\omega(C/h) \leq \eta$ for $h \geq n_0$. By the definition of region $\mathcal{D}_1$ in Appendix C.1, $z_{k,s} = \rho_{k,s} - \mu_s(\mathbf{x}_k) > 2\eta$ before stopping. Thus,

$$b_{k,s} \geq \frac{2\eta}{h_k - 1} - \frac{\eta}{h_k} \geq \frac{\eta}{h_k} > 0.$$

Martingale-Difference Noise It remains to bound the martingale-difference noise $\xi_{k+1,s}$. Define $\Delta\mu_{k+1,s} \triangleq \mu_s(\mathbf{x}_{k+1}) - \mu_s(\mathbf{x}_k)$. By the definition of $\xi_{k+1,s}$ together with (35) and (36),

$$\xi_{k+1,s} = -\frac{\mathbf{Y}_{k+1,s} - \mu_s(\mathbf{x}_k)}{h_k - 1} - \left(\Delta\mu_{k+1,s} - \mathbb{E}[\Delta\mu_{k+1,s} \mid \mathcal{G}_k] \right).$$

Assumption 2.1 gives $0 \leq \mathbf{Y}_{k+1,s}, \mu_s(\mathbf{x}_k) \leq \bar{b}$. Therefore,

$$\left| \frac{\mathbf{Y}_{k+1,s} - \mu_s(\mathbf{x}_k)}{h_k - 1} \right| \leq \frac{\bar{b}}{h_k - 1} \leq \frac{2\bar{b}}{h_k}. \quad (39)$$

Moreover, by Assumption 3.2, the gradients $\nabla\mu_s(\mathbf{x})$ are uniformly bounded on the compact set $\hat{\mathcal{K}}$. Let

$$L \triangleq \max_{s \in \mathcal{S}} \sup_{\mathbf{x} \in \hat{\mathcal{K}}} \|\nabla\mu_s(\mathbf{x})\| < \infty.$$

Since $\mathbf{d}_{k+1} = \mathbf{x}_{k+1} - \mathbf{x}_k$ on $\{k < \tau\}$, Lemma E.1 gives (recall that we define $\mathbf{d}_{k+1} \triangleq \Delta\mathbf{M}_{k+1}^{\mathcal{B}}$):

$$\begin{aligned} |\Delta\mu_{k+1,s}| &\leq L\|\mathbf{d}_{k+1}\| \leq \frac{C}{h_k}, \\ |\Delta\mu_{k+1,s} - \mathbb{E}[\Delta\mu_{k+1,s} \mid \mathcal{G}_k]| &\leq |\Delta\mu_{k+1,s}| + \mathbb{E}[|\Delta\mu_{k+1,s}| \mid \mathcal{G}_k] \leq \frac{2C}{h_k}. \end{aligned} \quad (40)$$

Combining (39) and (40) yields

$$|\xi_{k+1,s}| \leq \frac{2\bar{b} + 2C}{h_k}. \quad \square$$

E.2 Bounded Expected Remaining Horizon at Stopping

In this section, we show that h_r , the remaining horizon at the stopping time, satisfies $\mathbb{E}[h_\tau] = O(1)$. To do so, define the event $\mathcal{E}_r \triangleq \{h_\tau \geq r\} = \{\tau \leq n - r\}$ for every integer $r \geq n_0$. If $r > n_0$, then, on $\mathcal{E}_r$, $\sigma = \tau < \bar{N}$ and $\boldsymbol{\rho}_\tau \notin \mathcal{D}_1$; hence, $\mathcal{E}_r$ is the event that the trajectory exits $\mathcal{D}_1$ with at least $r > n_0$ periods remaining. In Lemma E.3, we show that the probability of $\mathcal{E}_r$ decays exponentially in r , which in turn implies $\mathbb{E}[h_\tau] = O(1)$.

Lemma E.3 (Exponential Tail Bound for Early Exit). *There exist an integer n_0 and constants $C, c > 0$ such that, uniformly over $n \geq n_0$ and $\boldsymbol{\rho}_0 = \mathbf{c}/n \in \mathcal{K}$, for every $r \geq n_0$,*

$$\mathbb{P}(h_\tau \geq r) \leq Ce^{-cr}.$$

Consequently, $\sup_{n \geq n_0, \boldsymbol{\rho}_0 \in \mathcal{K}} \mathbb{E}[h_\tau] < \infty$.

To prove Lemma E.3, we first use a maximal concentration inequality for martingales to control the coordinatewise fluctuations of the binding and slack resources.

Lemma E.4 (Coordinatewise Maximal Bounds). *For $0 \leq k \leq \bar{N}$, define $M_k^{\mathcal{B},q}$ for each $q \in \mathcal{B}$ and, when $\mathcal{S} \neq \emptyset$, $M_k^{\mathcal{S},s}$ for each $s \in \mathcal{S}$, by*

$$M_k^{\mathcal{B},q} \triangleq (\mathbf{M}_k^{\mathcal{B}})_q = \sum_{\ell=1}^k (\Delta\mathbf{M}_\ell^{\mathcal{B}})_q, \quad M_k^{\mathcal{S},s} \triangleq \sum_{\ell=1}^k \xi_{\ell,s},$$

with $M_0^{\mathcal{B},q} = M_0^{\mathcal{S},s} = 0$. Then, there exists a constant $C < \infty$ such that, for every $d > 0$ and every integer r satisfying $n_0 < r \leq n$,

$$\begin{aligned} \mathbb{P}\left\{\max_{0 \leq j \leq n-r} |M_j^{\mathcal{B},q}| \geq d\right\} &\leq 2 \exp\left(-\frac{d^2 r}{2C^2}\right), \\ \mathbb{P}\left\{\max_{0 \leq j \leq n-r} |M_j^{\mathcal{S},s}| \geq d\right\} &\leq 2 \exp\left(-\frac{d^2 r}{2C^2}\right). \end{aligned}$$

The second inequality is vacuous when $\mathcal{S} = \emptyset$.

Proof. By Lemmas E.1 and E.2, the processes $\{M_k^{\mathcal{B},q}\}_{k \geq 0}$ and $\{M_k^{\mathcal{S},s}\}_{k \geq 0}$ are martingales whose increments satisfy

$$\left| M_{k+1}^{\mathcal{B},q} - M_k^{\mathcal{B},q} \right|, \left| M_{k+1}^{\mathcal{S},s} - M_k^{\mathcal{S},s} \right| \leq \frac{C}{h_k}$$

for some constant $C < \infty$.

We first prove the bound for $M_k^{\mathcal{B},q}$. For ease of notation, write $M_k = M_k^{\mathcal{B},q}$. The case $r = n$ is immediate because the maximum contains only $M_0 = 0$. Now suppose $r < n$. For $1 \leq \ell \leq n - r$, the increment $M_\ell - M_{\ell-1}$ lies in the interval

$$\left[-\frac{C}{h_{\ell-1}}, \frac{C}{h_{\ell-1}} \right],$$

whose width is $2C/h_{\ell-1}$. Moreover, since $h_{\ell-1} = n - \ell + 1$,

$$\sum_{\ell=1}^{n-r} \left(\frac{2C}{h_{\ell-1}} \right)^2 = 4C^2 \sum_{u=r+1}^n \frac{1}{u^2} \leq 4C^2 \sum_{u=r+1}^{\infty} \frac{1}{u^2} \leq 4C^2 \int_r^{\infty} \frac{1}{x^2} dx = \frac{4C^2}{r}.$$

Applying Lemma A.1 to M_k and $-M_k$, and then taking a union bound, yields

$$\mathbb{P} \left\{ \max_{0 \leq j \leq n-r} |M_j| \geq d \right\} \leq 2 \exp \left(-\frac{d^2 r}{2C^2} \right).$$

The bound for $M_k^{\mathcal{S},s}$ follows by the same argument. $\square$

We now apply the coordinatewise tail bound in Lemma E.4 to prove Lemma E.3.

Proof of Lemma E.3. Since $\boldsymbol{\rho}_0 \in \mathcal{K} \subset \mathcal{D}_1$ and $n > n_0$, we have $n_0 \leq h_\tau < n$. Therefore, $\mathbb{P}(h_\tau \geq r) = 0$ for $r \geq n$.

Now suppose $n_0 < r < n$. On $\mathcal{E}_r$, the trajectory $\boldsymbol{\rho}_k$ exits $\mathcal{D}_1$ before the cutoff $\bar{N}$. By the definition of $\mathcal{D}_1$, such an exit can occur only if either the binding projection leaves $\text{int } \mathcal{K}_B^1$ or some slack coordinate falls to 2η or below. We bound the probabilities of these two events separately.

Case 1. Exit Through a Binding Coordinate. Since $\mathcal{K}_B \Subset \mathcal{K}_B^1$, set $\mathcal{K}_B$ is separated from the complement of $\text{int } \mathcal{K}_B^1$ by a positive distance

$$\delta_B \triangleq \text{dist} \left(\mathcal{K}_B, (\text{int } \mathcal{K}_B^1)^c \right) > 0.$$

Consider the event

$$\mathcal{E}_{r,B} \triangleq \mathcal{E}_r \cap \left\{ \mathbf{x}_\tau \notin \text{int } \mathcal{K}_B^1 \right\}.$$

Since $\mathbf{x}_0 \in \mathcal{K}_B$ and $\mathbf{x}_\tau \notin \text{int } \mathcal{K}_B^1$ on $\mathcal{E}_{r,B}$, we have:

$$\delta_B \leq \|\mathbf{x}_\tau - \mathbf{x}_0\| = \|\mathbf{M}_\tau^{\mathcal{B}}\| \leq \sqrt{|\mathcal{B}|} \max_{q \in \mathcal{B}} |M_\tau^{\mathcal{B},q}|,$$

where the equality follows from $\mathbf{M}_\tau^{\mathcal{B}} = \mathbf{x}_\tau - \mathbf{x}_0$ by its definition in Lemma E.1. Thus, on $\mathcal{E}_{r,B}$, there exists a binding coordinate $q \in \mathcal{B}$ satisfying

$$|M_\tau^{\mathcal{B},q}| \geq \varsigma \triangleq \delta_B / \sqrt{|\mathcal{B}|}.$$

Since $\tau \leq n - r$ on $\mathcal{E}_r$, Lemma E.4, together with a union bound over the binding coordinates, gives

$$\mathbb{P}(\mathcal{E}_{r,\mathcal{B}}) \leq 2|\mathcal{B}| \exp\left(-\frac{\varsigma^2 r}{2C^2}\right).$$

Case 2. Exit Through a Slack Coordinate. Suppose $\mathcal{S} \neq \emptyset$. For every slack resource $s \in \mathcal{S}$, consider the event

$$\mathcal{E}_{r,s} \triangleq \mathcal{E}_r \cap \left\{z_{\tau,s} \leq 2\eta\right\}.$$

Telescoping the decomposition $z_{k+1,s} - z_{k,s} = b_{k,s} + \xi_{k+1,s}$ gives

$$z_{j,s} - z_{0,s} = \sum_{k=0}^{j-1} b_{k,s} + M_j^{S,s}.$$

By Lemma E.2, the predictable drift satisfies $b_{k,s} \geq 0$ before stopping and $b_{k,s} = 0$ after stopping. Therefore, $\sum_{k=0}^{j-1} b_{k,s} \geq 0$.

Since $\rho_0 \in \mathcal{K}$, the definitions of $\underline{\delta}$ and η in Appendix C.1 imply $z_{0,s} = \rho_{0,s} - \mu_s(\mathbf{x}_0) \geq \underline{\delta} \geq 3\eta$. On the other hand, $z_{\tau,s} \leq 2\eta$ on $\mathcal{E}_{r,s}$. Hence, taking $j = \tau$ in the preceding telescope gives

$$M_\tau^{S,s} = z_{\tau,s} - z_{0,s} - \sum_{k=0}^{\tau-1} b_{k,s} \leq 2\eta - 3\eta - 0 = -\eta.$$

Moreover, since $\tau \leq n - r$ on $\mathcal{E}_r$, Lemma E.4, applied with $d = \eta$, yields

$$\mathbb{P}(\mathcal{E}_{r,s}) \leq 2 \exp\left(-\frac{\eta^2 r}{2C^2}\right).$$

Step 3. Combining the Two Cases. Since ρ_k can leave $\mathcal{D}_1$ only through either a binding coordinate or a slack coordinate, we have $\mathcal{E}_r = \mathcal{E}_{r,\mathcal{B}} \cup \bigcup_{s \in \mathcal{S}} \mathcal{E}_{r,s}$. Applying a union bound gives

$$\mathbb{P}(\mathcal{E}_r) \leq 2me^{-cr},$$

where we may take

$$c \triangleq \begin{cases} \frac{\min\{\varsigma^2, \eta^2\}}{2C^2}, & \mathcal{S} \neq \emptyset, \\ \frac{\varsigma^2}{2C^2}, & \mathcal{S} = \emptyset. \end{cases}$$

After enlarging the constant C in the statement of Lemma E.3 to cover the case $r = n_0$, this proves the exponential tail bound.

Finally, since h_τ is integer-valued and $h_\tau \geq n_0$, the established tail bound yields

$$\mathbb{E}[h_\tau] = n_0 + \sum_{r=n_0+1}^{\infty} \mathbb{P}(h_\tau \geq r) \leq n_0 + C \sum_{r=n_0+1}^{\infty} e^{-cr} < \infty,$$

uniformly over $n \geq n_0$ and all initial states satisfying $\rho_0 = \mathbf{c}/n \in \mathcal{K}$. $\square$

E.3 Telescoping Analysis and Proof of Proposition 6.2

We are now ready to prove Proposition 6.2. Fix $n \geq n_0$ and $\mathbf{c}/n \in \mathcal{K}$. The proof has two steps. First, in Appendix E.3.1, using a telescoping analysis, we express the initial difference $V_n^F(\mathbf{c}) - Q_n(\mathbf{c})$ as the difference $V_{h_\tau}^F(\mathbf{c}_\tau) - Q_{h_\tau}(\mathbf{c}_\tau)$ at the stopping time, minus the residuals accumulated before stopping. We then bound these two terms separately in Appendix E.3.2.

E.3.1 The Telescoping Analysis

For $0 \leq k \leq \bar{N}$, define

$$e_k \triangleq V_{h_k \wedge \tau}^F(\mathbf{c}_{k \wedge \tau}) - Q_{h_k \wedge \tau}(\mathbf{c}_{k \wedge \tau}),$$

and

$$r_k \triangleq \begin{cases} Q_{h_k}(\mathbf{c}_k) - (T_{h_k}^F Q_{h_k-1})(\mathbf{c}_k), & k < \tau, \\ 0, & k \geq \tau. \end{cases}$$

Thus, after stopping, we freeze the difference e_k at e_τ and set $r_k = 0$. Since $\mathbf{c}_k \in \mathcal{G}_k$,

$$\begin{aligned} V_{h_k}^F(\mathbf{c}_k) &= \mathbb{E}[V_{h_k}^F(\mathbf{c}_k) \mid \mathcal{G}_k] = \mathbb{E}[\hat{R}_{k+1} + V_{h_k-1}^F(\mathbf{c}_{k+1}) \mid \mathcal{G}_k], \\ (T_{h_k}^F Q_{h_k-1})(\mathbf{c}_k) &= \mathbb{E}[(T_{h_k}^F Q_{h_k-1})(\mathbf{c}_k) \mid \mathcal{G}_k] = \mathbb{E}[\hat{R}_{k+1} + Q_{h_k-1}(\mathbf{c}_{k+1}) \mid \mathcal{G}_k]. \end{aligned} \quad (41)$$

On $\{k < \tau\}$, we have $k+1 \leq \tau$ and $h_{k+1} = h_k - 1$. Therefore, by (41),

$$\begin{aligned} e_k &= V_{h_k}^F(\mathbf{c}_k) - (T_{h_k}^F Q_{h_k-1})(\mathbf{c}_k) + (T_{h_k}^F Q_{h_k-1})(\mathbf{c}_k) - Q_{h_k}(\mathbf{c}_k) \\ &= \mathbb{E}[V_{h_k-1}^F(\mathbf{c}_{k+1}) - Q_{h_k-1}(\mathbf{c}_{k+1}) \mid \mathcal{G}_k] + (T_{h_k}^F Q_{h_k-1})(\mathbf{c}_k) - Q_{h_k}(\mathbf{c}_k) \\ &= \mathbb{E}[e_{k+1} \mid \mathcal{G}_k] - r_k. \end{aligned}$$

On $\{k \geq \tau\}$, $k \geq \tau$ implies $e_{k+1} = e_k = e_\tau$. Because e_τ is $\mathcal{G}_k$ -measurable conditional on $\{k \geq \tau\}$ and $r_k = 0$, the same recursion holds. As a result,

$$e_k = \mathbb{E}[e_{k+1} \mid \mathcal{G}_k] - r_k, \quad 0 \leq k < \bar{N}.$$

Taking expectations gives

$$\mathbb{E}[e_k] = \mathbb{E}[e_{k+1}] - \mathbb{E}[r_k], \quad 0 \leq k < \bar{N}.$$

Summing these identities over the range $k = 0, \dots, \bar{N} - 1$, we obtain

$$\mathbb{E}[e_0] = \mathbb{E}[e_{\bar{N}}] - \mathbb{E}\left[\sum_{k=0}^{\bar{N}-1} r_k\right].$$

Note that $e_0 = V_n^F(\mathbf{c}) - Q_n(\mathbf{c})$. Moreover, the definitions of e_k and r_k give $e_{\bar{N}} = e_\tau$, and $\sum_{k=0}^{\bar{N}-1} r_k = \sum_{k=0}^{\tau-1} r_k$, because $r_k = 0$ for every $k \geq \tau$. We therefore have

$$V_n^F(\mathbf{c}) - Q_n(\mathbf{c}) = \mathbb{E}\left[e_{\bar{N}} - \sum_{k=0}^{\bar{N}-1} r_k\right] = \mathbb{E}\left[e_\tau - \sum_{k=0}^{\tau-1} r_k\right]. \quad (42)$$

E.3.2 Bounding the Error Terms and Completing the Proof

The telescope in (42) contains two terms: the accumulated pre-stopping residual $\sum_{k=0}^{\tau-1} r_k$ and the terminal comparison error e_τ . We bound them in turn.

The accumulated residuals. For $k < \tau$, we have $\rho_k \in \mathcal{D}_1$ and $h_k \geq n_0 + 1$. Lemma D.6 therefore gives $|r_k| \leq a_{h_k}$. Since the pre-stopping horizons are $n, n-1, \dots, h_\tau + 1$,

$$\left| \sum_{k=0}^{\tau-1} r_k \right| \leq \sum_{k=0}^{\tau-1} |r_k| \leq \sum_{k=0}^{\tau-1} a_{h_k} = \sum_{j=h_\tau+1}^n a_j \leq \sum_{j=n_0+1}^{\infty} a_j < \infty.$$

The terminal error. Using $Q_h = F_h - A_h$ (see Section D.1.3), we can write the terminal difference as

$$e_\tau = V_{h_\tau}^F(\mathbf{c}_\tau) - F_{h_\tau}(\mathbf{c}_\tau) + A_{h_\tau}(\mathbf{x}_\tau).$$

Since $0 \leq V_h^F(\mathbf{c}'), F_h(\mathbf{c}') \leq h\bar{r}$, where $\bar{r}$ is defined in Assumption 2.1, we have

$$|V_h^F(\mathbf{c}') - F_h(\mathbf{c}')| \leq h\bar{r}.$$

Moreover, the uniform bound in Lemma D.5 gives

$$|A_{h_\tau}(\mathbf{x}_\tau)| \leq C \ln(h_\tau) \leq Ch_\tau,$$

for some constant $C < \infty$, where the second inequality uses $\ln h + 1 \leq h$ for every $h > 0$. Hence,

$$|e_\tau| \leq (\bar{r} + C) h_\tau.$$

Combining the bounds. Using the two preceding bounds, we have, uniformly over $\mathbf{c}/n \in \mathcal{K}$,

$$|V_n^F(\mathbf{c}) - Q_n(\mathbf{c})| = \left| \mathbb{E} \left[e_\tau - \sum_{k=0}^{\tau-1} r_k \right] \right| \leq \mathbb{E}[|e_\tau|] + \mathbb{E} \left[\left| \sum_{k=0}^{\tau-1} r_k \right| \right] \leq (\bar{r} + C) \cdot \mathbb{E}[h_\tau] + \sum_{j=n_0+1}^{\infty} a_j \leq C_F$$

for some constant $C_F < \infty$. The final inequality uses $\sup_{n \geq n_0, \rho_0 \in \mathcal{K}} \mathbb{E}[h_\tau] < \infty$ by Lemma E.3. This proves Proposition 6.2 for $n \geq n_0$. For the case of $n < n_0$, the convention $Q_n = F_n$ and the bounds $0 \leq V_n^F(\mathbf{c}), F_n(\mathbf{c}) \leq n\bar{r}$ give $|V_n^F(\mathbf{c}) - Q_n(\mathbf{c})| = |V_n^F(\mathbf{c}) - F_n(\mathbf{c})| \leq n_0\bar{r}$. Enlarging C_F once covers these finitely many horizons and completes the proof.

F Proof of Proposition 6.3

In this section, we prove Proposition 6.3. The proof has three steps. First, we relax the capacity constraints for the slack resources to obtain a reduced problem, whose optimal value satisfies $\bar{V}_n^{\text{OPT}}(\mathbf{c}_B) \geq V_n^{\text{OPT}}(\mathbf{c})$. We then construct a global upper bound for the reduced problem. Second, we add a smooth cutoff and a bounded correction to the corrected fluid reference to construct a local supersolution. Third, we take the minimum of the local supersolution and the global upper bound to obtain a global supersolution $\bar{W}_n(\mathbf{c}_B)$. The construction yields, for every state with $n \geq n_0$ and $\mathbf{c}/n \in \mathcal{K}$,

$$V_n^{\text{OPT}}(\mathbf{c}) \leq \bar{V}_n^{\text{OPT}}(\mathbf{c}_B) \leq \bar{W}_n(\mathbf{c}_B) + C_{\text{init}} \leq Q_n(\mathbf{c}) + C_{\text{OPT}}$$

for some constants $C_{\text{init}}, C_{\text{OPT}} < \infty$, which completes the proof.

F.1 A Reduced Problem and a Global Upper Bound

We consider a reduced problem in which we remove the capacity constraints on the slack resources and retain only those on the binding resources. Specifically, for an inventory of binding resources $\mathbf{z} \in \mathbb{R}_+^{|\mathcal{B}|}$, define the reduced feasible-action set and reduced Bellman operator by

$$\begin{aligned}\bar{\mathcal{A}}(\mathbf{z}, \boldsymbol{\theta}) &\triangleq \left\{ a \in \mathcal{A} : B_{a, \mathcal{B}} \leq \mathbf{z} \right\}, \\ (\bar{T}^*u)(\mathbf{z}) &\triangleq \mathbb{E} \left[\max_{a \in \bar{\mathcal{A}}(\mathbf{z}, \boldsymbol{\theta})} \left\{ R_a + u(\mathbf{z} - B_{a, \mathcal{B}}) \right\} \right].\end{aligned}$$

Let $\bar{V}_0^{\text{OPT}} = 0$ and $\bar{V}_n^{\text{OPT}} = \bar{T}^* \bar{V}_{n-1}^{\text{OPT}}$ for $n \geq 1$. Then, $(\bar{V}_n^{\text{OPT}})_{n \geq 0}$ is the optimal value function of the reduced problem. Since removing the slack-resource constraints enlarges the feasible-action set, the reduced problem provides an upper bound on the original problem, as formalized in Lemma F.1.

Lemma F.1. *For any state $(\mathbf{c}, n)$, we have $V_n^{\text{OPT}}(\mathbf{c}) \leq \bar{V}_n^{\text{OPT}}(\mathbf{c}_{\mathcal{B}})$.*

Proof. We proceed by induction on n . The claim is immediate for $n = 0$, since $\bar{V}_0^{\text{OPT}} = V_0^{\text{OPT}} = 0$. Suppose the claim holds at horizon $n - 1$. Then

$$\begin{aligned}V_n^{\text{OPT}}(\mathbf{c}) &= \mathbb{E} \left[\max_{a \in \mathcal{A}(\mathbf{c}, \boldsymbol{\theta})} \left\{ R_a + V_{n-1}^{\text{OPT}}(\mathbf{c} - B_a) \right\} \right] \\ &\leq \mathbb{E} \left[\max_{a \in \mathcal{A}(\mathbf{c}, \boldsymbol{\theta})} \left\{ R_a + \bar{V}_{n-1}^{\text{OPT}}(\mathbf{c}_{\mathcal{B}} - B_{a, \mathcal{B}}) \right\} \right] \\ &\leq \mathbb{E} \left[\max_{a \in \bar{\mathcal{A}}(\mathbf{c}_{\mathcal{B}}, \boldsymbol{\theta})} \left\{ R_a + \bar{V}_{n-1}^{\text{OPT}}(\mathbf{c}_{\mathcal{B}} - B_{a, \mathcal{B}}) \right\} \right] = \bar{V}_n^{\text{OPT}}(\mathbf{c}_{\mathcal{B}}).\end{aligned}$$

Here, the first inequality follows from the induction hypothesis, and the second inequality follows from $\mathcal{A}(\mathbf{c}, \boldsymbol{\theta}) \subseteq \bar{\mathcal{A}}(\mathbf{c}_{\mathcal{B}}, \boldsymbol{\theta})$. This completes the induction. $\square$

F.1.1 A Global Supersolution

We now construct a global supersolution of the Bellman operator $\bar{T}^*$ using the dual of the reduced problem. We first define supersolution in Definition F.1.

Definition F.1 (Supersolution). A sequence (W_n) is a supersolution of the Bellman operator $\bar{T}^*$ from horizon n_0 onward if

$$(\bar{T}^*W_{n-1})(\mathbf{z}) \leq W_n(\mathbf{z}) \tag{43}$$

for all $\mathbf{x} \in \mathbb{R}_+^{|\mathcal{B}|}$ and $n \geq n_0 + 1$.

To construct a supersolution, we consider the Lagrangian dual of the reduced problem obtained by dualizing the binding-resource capacity constraints. Specifically, for every $n \geq 0$, define

$$G_n(\mathbf{z}) \triangleq \inf_{\boldsymbol{\lambda} \geq 0} \left\{ \boldsymbol{\lambda}^T \mathbf{z} + n \bar{\phi}(\boldsymbol{\lambda}) \right\},$$

where, for $\boldsymbol{\lambda} \in \mathbb{R}_+^{|\mathcal{B}|}$,

$$\bar{\phi}(\boldsymbol{\lambda}) \triangleq \mathbb{E} \left[\max_{a \in \mathcal{A}} \left\{ R_a - \boldsymbol{\lambda}^T B_{a, \mathcal{B}} \right\} \right].$$

Lemma F.2 shows that $(G_n)_{n \geq 0}$ is a global supersolution of $\bar{T}^*$ and provides several additional properties of $(G_n)_{n \geq 0}$.

Lemma F.2. *The sequence $(G_n)_{n \geq 0}$ satisfies the following properties:*

1. $0 \leq G_n(\mathbf{z}) \leq n\bar{r}$ for any $n \geq 0$ and $\mathbf{z} \in \mathbb{R}_+^{|\mathcal{B}|}$.
2. (Supersolution) $\bar{T}^*G_{n-1}(\mathbf{z}) \leq G_n(\mathbf{z})$ for any $n \geq 1$ and $\mathbf{z} \in \mathbb{R}_+^{|\mathcal{B}|}$.
3. (Upper Bound on $\bar{V}_n^{\text{OPT}}$) $\bar{V}_n^{\text{OPT}}(\mathbf{z}) \leq G_n(\mathbf{z})$ for any $n \geq 0$ and $\mathbf{z} \in \mathbb{R}_+^{|\mathcal{B}|}$.
4. $G_n(\mathbf{z}) = n\phi_{\mathcal{B}}(\mathbf{z}/n)$ for any $n \geq 1$ and $\mathbf{z}/n \in \mathcal{U}_{\mathcal{B}}$.

Proof. Proof of Part 1. Since the rejection action $a = 0$ has zero reward and consumption, and all resource consumptions and prices $\boldsymbol{\lambda}$ are nonnegative, we have

$$0 \leq \max_{a \in \mathcal{A}} \left\{ R_a - \boldsymbol{\lambda}^T B_{a,\mathcal{B}} \right\} \leq \max_{a \in \mathcal{A}} (R_a)^+.$$

Consequently, $0 \leq \bar{\phi}(\boldsymbol{\lambda}) \leq \bar{r}$ by Assumption 2.1, and $0 \leq G_n(\mathbf{z}) \leq n\bar{\phi}(\mathbf{0}) \leq n\bar{r}$, where the second inequality follows by evaluating the dual objective of $G_n(\mathbf{z})$ at $\boldsymbol{\lambda} = \mathbf{0}$.

Proof of Part 2. Fix $n \geq 1$, $\mathbf{z} \in \mathbb{R}_+^{|\mathcal{B}|}$, and $\boldsymbol{\lambda} \geq \mathbf{0}$. By the definition of G_{n-1} ,

$$G_{n-1}(\mathbf{z} - B_{a,\mathcal{B}}) \leq \boldsymbol{\lambda}^T (\mathbf{z} - B_{a,\mathcal{B}}) + (n-1)\bar{\phi}(\boldsymbol{\lambda}).$$

It follows that

$$\bar{T}^*G_{n-1}(\mathbf{z}) \leq \boldsymbol{\lambda}^T \mathbf{z} + (n-1)\bar{\phi}(\boldsymbol{\lambda}) + \mathbb{E} \left[\max_{a \in \bar{\mathcal{A}}(\mathbf{z}, \boldsymbol{\theta})} \left\{ R_a - \boldsymbol{\lambda}^T B_{a,\mathcal{B}} \right\} \right] \leq \boldsymbol{\lambda}^T \mathbf{z} + n\bar{\phi}(\boldsymbol{\lambda}),$$

where the second inequality uses $\bar{\mathcal{A}}(\mathbf{z}, \boldsymbol{\theta}) \subseteq \mathcal{A}$. Taking the infimum over $\boldsymbol{\lambda} \geq \mathbf{0}$ yields $\bar{T}^*G_{n-1}(\mathbf{z}) \leq G_n(\mathbf{z})$.

Proof of Part 3. Suppose that $\bar{V}_{n-1}^{\text{OPT}} \leq G_{n-1}$, then

$$\bar{V}_n^{\text{OPT}} = \bar{T}^*\bar{V}_{n-1}^{\text{OPT}} \leq \bar{T}^*G_{n-1} \leq G_n,$$

where the first inequality follows from the monotone property of the Bellman operator $\bar{T}^*$ and the second inequality from part 2. Since $\bar{V}_0^{\text{OPT}} = G_0 = 0$, induction yields $\bar{V}_n^{\text{OPT}} \leq G_n$ for every $n \geq 0$.

Proof of Part 4. Fix a normalized state $\mathbf{x} \in \mathcal{U}_{\mathcal{B}}$. For every $\boldsymbol{\lambda} \geq \mathbf{0}$, we have

$$\begin{aligned} \boldsymbol{\lambda}^T \mathbf{x} + \bar{\phi}(\boldsymbol{\lambda}) &\geq \boldsymbol{\lambda}^T \mathbf{x} + \mathbb{E} \left[R_{a^F(\mathbf{x})} - \boldsymbol{\lambda}^T B_{a^F(\mathbf{x}),\mathcal{B}} \right] \\ &= \mathbb{E} \left[R_{a^F(\mathbf{x})} \right] + \boldsymbol{\lambda}^T \left(\mathbf{x} - \mathbb{E} \left[B_{a^F(\mathbf{x}),\mathcal{B}} \right] \right) \\ &= \mathbb{E} \left[R_{a^F(\mathbf{x})} \right] = \phi_{\mathcal{B}}(\mathbf{x}), \end{aligned} \tag{44}$$

where the inequality follows from the definition of $\bar{\phi}(\boldsymbol{\lambda})$, the second equality from $\mathbb{E} \left[B_{a^F(\mathbf{x}),\mathcal{B}} \right] = \mathbf{x}$ by Assumption 3.2, and the third equality from $\mathbb{E} \left[R_{a^F(\mathbf{x})} \right] = \phi_{\mathcal{B}}(\mathbf{x})$ by Lemma 3.2.

On the other hand, by the definition of the fluid policy in (4), $a^F(\mathbf{x})$ maximizes $R_a - \boldsymbol{\lambda}(\mathbf{x})^T B_{a,\mathcal{B}}$ over actions $a \in \mathcal{A}$. Hence, the inequality in (44) holds with equality at $\boldsymbol{\lambda} = \boldsymbol{\lambda}(\mathbf{x})$. Consequently,

$$\inf_{\boldsymbol{\lambda} \geq \mathbf{0}} \left\{ \boldsymbol{\lambda}^T \mathbf{x} + \bar{\phi}(\boldsymbol{\lambda}) \right\} = \phi_{\mathcal{B}}(\mathbf{x}).$$

Setting $\mathbf{x} = \mathbf{z}/n$ and factoring n out of the dual objective yields

$$G_n(\mathbf{z}) = n \inf_{\boldsymbol{\lambda} \geq \mathbf{0}} \left\{ \boldsymbol{\lambda}^\top \mathbf{x} + \bar{\phi}(\boldsymbol{\lambda}) \right\} = n\phi_{\mathcal{B}}(\mathbf{z}/n). \quad \square$$

Finally, we define the reduced corrected reference $(\bar{Q}_n)_{n \geq 0}$ globally by $\bar{Q}_0 = 0$ and, for every $n \geq 1$ and $\mathbf{z} \in \mathbb{R}_+^{|\mathcal{B}|}$,

$$\bar{Q}_n(\mathbf{z}) \triangleq G_n(\mathbf{z}) - A_n(\mathbf{z}/n).$$

If $\mathbf{c}/n \in \tilde{\mathcal{U}}$, where $\tilde{\mathcal{U}}$ is defined in Lemma C.1, then

$$\bar{Q}_n(\mathbf{c}_{\mathcal{B}}) = n\phi_{\mathcal{B}}(\mathbf{c}_{\mathcal{B}}/n) - A_n(\mathbf{c}_{\mathcal{B}}/n) = n\phi^{\mathcal{F}}(\mathbf{c}/n) - A_n(\mathbf{c}_{\mathcal{B}}/n) = Q_n(\mathbf{c}), \quad (45)$$

where the first equality follows from part 4 of Lemma F.2, the second equality from part 1 of Lemma C.1, and the third equality from the definition of Q_n in Appendix D.1.3.

F.2 Constructing a Local Supersolution

The global supersolution $(G_n)_{n \geq 0}$ constructed in Appendix F.1.1 may be loose. To obtain a tighter supersolution, we first construct a local supersolution (W_n^{loc}) on the region $\mathcal{K}_{\mathcal{B}}^1$; that is, the Bellman inequality (43) holds only for states satisfying $\mathbf{z}/n \in \text{int } \mathcal{K}_{\mathcal{B}}^1$. In Appendix F.3, we will take the minimum of the global supersolution (G_n) and the local supersolution (W_n^{loc}) to obtain a tighter global supersolution.

To ensure that (W_n^{loc}) is not selected outside $\mathcal{K}_{\mathcal{B}}^1$, where its Bellman inequality is not verified, we introduce a smooth cutoff term. This will ensure that taking the minimum of the global and local branches remains a global supersolution.

We first record a global bound on A_n . By Lemma D.5, there exists a constant $C_A < \infty$, such that for any $n \geq 0$ and $\mathbf{x} \in \mathbb{R}^{|\mathcal{B}|}$,

$$0 \leq A_n(\mathbf{x}) \leq C_A \ln n \leq C_A H_n.$$

To construct the smooth cutoff term, we choose a C^2 function $\chi : \mathbb{R}^{|\mathcal{B}|} \rightarrow [0, 1]$, with bounded first- and second-order derivatives, such that

$$\chi(\mathbf{x}) = \begin{cases} 0, & \mathbf{x} \in \mathcal{K}_{\mathcal{B}}, \\ \text{smooth transition}, & \mathbf{x} \in \mathcal{K}_{\mathcal{B}}^1 \setminus \mathcal{K}_{\mathcal{B}}, \\ 1, & \mathbf{x} \in (\text{int } \mathcal{K}_{\mathcal{B}}^1)^c. \end{cases}$$

Such a function exists because $\mathcal{K}_{\mathcal{B}} \Subset \mathcal{K}_{\mathcal{B}}^1$, hence the neighborhood on which χ vanishes can be chosen to have closure contained in $\text{int } \mathcal{K}_{\mathcal{B}}^1$. By construction, χ equals zero on the target set $\mathcal{K}_{\mathcal{B}}$, transitions smoothly within the local region $\mathcal{K}_{\mathcal{B}}^1$, and equals one outside $\text{int } \mathcal{K}_{\mathcal{B}}^1$. Fix $M \geq C_A$ and define the cutoff term as

$$L_n(\mathbf{z}) \triangleq M H_n \chi(\mathbf{z}/n), \quad n \geq 1,$$

with $L_0 \triangleq 0$.

Lemma F.3 establishes that the reduced corrected reference $\bar{Q}_n$, augmented by the smooth cutoff term L_n and a bounded correction term, forms a local supersolution on $\text{int } \mathcal{K}_{\mathcal{B}}^1$.

Lemma F.3 (Local Bellman Supersolution). *There exists an integer $n_0 \in \mathbb{N}$ and a bounded nonde-*

creasing sequence (ζ_n) such that

$$W_n^{\text{loc}}(\mathbf{z}) \triangleq \bar{Q}_n(\mathbf{z}) + L_n(\mathbf{z}) + \zeta_n$$

satisfies

$$\bar{T}^* W_{n-1}^{\text{loc}}(\mathbf{z}) \leq W_n^{\text{loc}}(\mathbf{z})$$

for all $n \geq n_0 + 1$ and all $\mathbf{z}$ satisfying $\mathbf{z}/n \in \text{int } \mathcal{K}_{\mathcal{B}}^1$.

F.2.1 Proof of Lemma F.3

Let $n_0 \geq 2$ be a fixed integer exceeding all finite minimum-horizon thresholds introduced thus far. Fix $n \geq n_0 + 1$ and a reduced state $\mathbf{z}$ with $\mathbf{x} \triangleq \mathbf{z}/n \in \text{int } \mathcal{K}_{\mathcal{B}}^1$. For each action $a \in \mathcal{A}$, define

$$\mathbf{x}'_a \triangleq \frac{\mathbf{z} - B_{a,\mathcal{B}}}{n-1}$$

as the normalized inventory of binding resources in the next period after taking action a .

Choose a full normalized state $\boldsymbol{\rho}$ such that $\boldsymbol{\rho}_{\mathcal{B}} = \mathbf{x}$ and $\rho_s = \mu_s(\mathbf{x}) + 3\eta$ for all $s \in \mathcal{S}$, where η is defined in Appendix C.1, and set $\mathbf{c} = n\boldsymbol{\rho}$. If $\mathcal{S} = \emptyset$, simply take $\mathbf{c} = \mathbf{z}$. Then $\boldsymbol{\rho} \in \mathcal{D}_1$, and Lemma C.2 ensures that every action is feasible and that $\mathbf{x}'_a \in \text{int } \mathcal{K}_{\mathcal{B}}^2$ for all $a \in \mathcal{A}$. Moreover, by Lemma C.2 and (45),

$$Q_n(\mathbf{c}) = \bar{Q}_n(\mathbf{z}), \quad Q_{n-1}(\mathbf{c} - B_a) = \bar{Q}_{n-1}(\mathbf{z} - B_{a,\mathcal{B}}).$$

Therefore, Lemma D.6 yields,

$$\left| \bar{Q}_n(\mathbf{z}) - \bar{T}_n^{\text{F}} \bar{Q}_{n-1}(\mathbf{z}) \right| \leq a_n, \quad n \geq n_0. \quad (46)$$

Finally, for a continuation function u and an action a , we define

$$\begin{aligned} D_a u(\mathbf{z}) &\triangleq u(\mathbf{z}) - u(\mathbf{z} - B_{a,\mathcal{B}}), \\ (\bar{P}_n^{\text{F}} u)(\mathbf{z}) &\triangleq \mathbb{E}[u(\mathbf{z} - B_{i,\mathcal{B}})], \\ (\bar{T}_n^{\text{F}} u)(\mathbf{z}) &\triangleq \mathbb{E}[R_i + u(\mathbf{z} - B_{i,\mathcal{B}})]. \end{aligned}$$

Hence, $\bar{P}_n^{\text{F}}$ contains only the continuation value, whereas $\bar{T}_n^{\text{F}}$ also includes the current reward. In addition,

$$R_a + u(\mathbf{z} - B_{a,\mathcal{B}}) = u(\mathbf{z}) + R_a - D_a u(\mathbf{z}).$$

Thus, when the current state is $\mathbf{z}$, the continuation value is u , and the opportunity $\boldsymbol{\theta}$ is received, the optimal online policy selects an action that maximizes $R_a - D_a u(\mathbf{z})$, which we call the Bellman-action score.

Step 1: Bounding the Continuation-Score Perturbation For each action a , define the score perturbation

$$\varepsilon_{n,a}(\mathbf{z}) \triangleq D_a(\bar{Q}_{n-1} + L_{n-1})(\mathbf{z}) - \boldsymbol{\lambda}(\mathbf{x})^{\text{T}} B_{a,\mathcal{B}}. \quad (47)$$

To interpret $\varepsilon_{n,a}$, recall that the dual-price score is $S_a(\mathbf{x}) = R_a - \boldsymbol{\lambda}(\mathbf{x})^{\text{T}} B_{a,\mathcal{B}}$. Hence, the Bellman-action score associated with the continuation value $u = \bar{Q}_{n-1} + L_{n-1}$ can be written as

$$R_a - D_a(\bar{Q}_{n-1} + L_{n-1})(\mathbf{z}) = S_a(\mathbf{x}) - \varepsilon_{n,a}(\mathbf{z}). \quad (48)$$

As a result, $\varepsilon_{n,a}(\mathbf{z})$ measures the perturbation from the dual-price score to the Bellman-action score.

Since $\bar{Q}_{n-1} = G_{n-1} - A_{n-1}$, from part 4 of Lemma F.2 and (20) we have

$$D_a \bar{Q}_{n-1}(\mathbf{z}) = m_{n,a}(\mathbf{x}) + A_{n-1}(\mathbf{x}'_a) - A_{n-1}(\mathbf{x}'_0).$$

Therefore, from (47),

$$\begin{aligned} \varepsilon_{n,i}(\mathbf{z}) - \varepsilon_{n,a}(\mathbf{z}) = & \underbrace{\left\{ \left(m_{n,i}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{i,\mathcal{B}} \right) - \left(m_{n,a}(\mathbf{x}) - \boldsymbol{\lambda}(\mathbf{x})^\top B_{a,\mathcal{B}} \right) \right\}}_{(a)} \\ & + \underbrace{\left(A_{n-1}(\mathbf{x}'_i) - A_{n-1}(\mathbf{x}'_a) \right)}_{(b)} + \underbrace{\left(D_i L_{n-1}(\mathbf{z}) - D_a L_{n-1}(\mathbf{z}) \right)}_{(c)}. \end{aligned}$$

Lemma C.4 bounds term (a) in absolute value by $C_1 \frac{d_{ia}^\mathcal{B}}{n}$ for a constant $C_1 < \infty$. For term (b), the global gradient bound in Lemma D.5 yields

$$|A_{n-1}(\mathbf{x}'_i) - A_{n-1}(\mathbf{x}'_a)| \leq \sup_{\mathbf{y}} \|\nabla A_{n-1}(\mathbf{y})\|_\infty \frac{d_{ia}^\mathcal{B}}{n-1} \leq C_2 \theta_n d_{ia}^\mathcal{B},$$

for some constant $C_2 < \infty$, where

$$\theta_n \triangleq \frac{\ln n}{n^{1-\xi_\star(1-\gamma)}}.$$

The first inequality follows from the mean-value theorem applied on the segment joining $\mathbf{x}'_i$ and $\mathbf{x}'_a$ and $\|\mathbf{x}'_i - \mathbf{x}'_a\|_1 = \frac{d_{ia}^\mathcal{B}}{n-1}$. Finally, for term (c), note that

$$D_i L_{n-1}(\mathbf{z}) - D_a L_{n-1}(\mathbf{z}) = M \mathbf{H}_{n-1} [\chi(\mathbf{x}'_a) - \chi(\mathbf{x}'_i)].$$

Since $\nabla \chi$ is bounded, another application of the mean-value theorem bounds this difference in absolute value by $C_3 \mathbf{H}_n \frac{d_{ia}^\mathcal{B}}{n}$ for some constant $C_3 < \infty$. Combining the three bounds yields

$$|\varepsilon_{n,i}(\mathbf{z}) - \varepsilon_{n,a}(\mathbf{z})| \leq C_4 \left(\frac{1}{n} + \theta_n + \frac{\mathbf{H}_n}{n} \right) d_{ia}^\mathcal{B} \leq C_5 \theta_n d_{ia}^\mathcal{B}$$

for some constants $C_4, C_5 < \infty$. Here, the last inequality uses $\xi_\star(1-\gamma) \geq 0$ and $\mathbf{H}_n \leq \ln n + 1$.

Step 2: Bounding the Action Deviation Gain Let $i = a^\mathbf{F}(\mathbf{x})$ denote the action that maximizes $S_a(\mathbf{x})$. Let $u = \bar{Q}_{n-1} + L_{n-1}$, and let j maximize $S_a(\mathbf{x}) - \varepsilon_{n,a}(\mathbf{z})$. Using (48), the difference between the continuation values under the optimal and fluid actions in the current period is

$$\bar{T}^* u(\mathbf{z}) - \bar{T}_n^\mathbf{F} u(\mathbf{z}) = \mathbb{E}[(S_j(\mathbf{x}) - \varepsilon_{n,j}(\mathbf{z})) - (S_i(\mathbf{x}) - \varepsilon_{n,i}(\mathbf{z}))].$$

Apply Lemma C.5 on $\mathcal{K}_\mathcal{B}^2$ with $e_a = \varepsilon_{n,a}(\mathbf{z})$ and $t = C_5 \theta_n$, and increase the fixed integer n_0 if necessary, so that $C_5 \theta_n \leq t_\star$ for all $n \geq n_0$; we obtain, for all $n \geq n_0$,

$$0 \leq \bar{T}^*(\bar{Q}_{n-1} + L_{n-1})(\mathbf{z}) - \bar{T}_n^\mathbf{F}(\bar{Q}_{n-1} + L_{n-1})(\mathbf{z}) \leq C_6 \theta_n^{1+\beta_\star} \quad (49)$$

for some constant $C_6 < \infty$. Thus, (49) controls the expected gain from optimizing the continuation value rather than following the fluid policy.

Step 3: Summable Residual and Accumulated Correction Let $Y = B_{a^F(\mathbf{x}), \mathcal{B}}$ denote the consumption of binding resources under action $a^F(\mathbf{x})$ and let

$$\Delta \triangleq \mathbf{x}'_{a^F(\mathbf{x})} - \mathbf{x} = \frac{\mathbf{x} - Y}{n - 1}.$$

Since $\mathbf{x} \triangleq \mathbf{z}/n \in \text{int } \mathcal{K}_{\mathcal{B}}^1$, Assumption 3.2 implies $\mathbb{E}[Y] = \mathbf{x}$, and hence $\mathbb{E}[\Delta] = \mathbf{0}$. Moreover, bounded consumption and compactness of $\mathcal{K}_{\mathcal{B}}^1$ yield $\|\Delta\| \leq C_7/n$ for some constant $C_7 < \infty$. From (28),

$$\chi(\mathbf{x} + \Delta) - \chi(\mathbf{x}) = \nabla \chi(\mathbf{x})^\top \Delta + \int_0^1 (1-t) \Delta^\top \nabla^2 \chi(\mathbf{x} + t\Delta) \Delta dt.$$

Since $\mathbb{E}[Y] = \mathbf{x}$ and the second-order derivatives of χ are bounded, taking expectations gives

$$|\mathbb{E}[\chi(\mathbf{x} + \Delta)] - \chi(\mathbf{x})| \leq C_8/n^2$$

for some constant $C_8 < \infty$. Thus,

$$\begin{aligned} L_n(\mathbf{z}) - \bar{P}_n^F L_{n-1}(\mathbf{z}) &= \left(L_n(\mathbf{z}) - M \mathbf{H}_{n-1} \chi(\mathbf{x}) \right) + \left(M \mathbf{H}_{n-1} \chi(\mathbf{x}) - \bar{P}_n^F L_{n-1}(\mathbf{z}) \right) \\ &= \frac{M \chi(\mathbf{x})}{n} + M \mathbf{H}_{n-1} (\chi(\mathbf{x}) - \mathbb{E}[\chi(\mathbf{x} + \Delta)]) \geq -C_9 \frac{H_n}{n^2} \end{aligned}$$

for some constant $C_9 < \infty$. As a result, for all $n \geq n_0$,

$$\bar{T}_n^F(\bar{Q}_{n-1} + L_{n-1})(\mathbf{z}) = \bar{T}_n^F \bar{Q}_{n-1}(\mathbf{z}) + \bar{P}_n^F L_{n-1}(\mathbf{z}) \leq \bar{Q}_n(\mathbf{z}) + L_n(\mathbf{z}) + a_n + C_9 \frac{H_n}{n^2}, \quad (50)$$

where the inequality follows from the proceeding inequality and (46).

Choose $C = \max\{C_6, C_9\}$ and set

$$b_n \triangleq a_n + C \left(\theta_n^{1+\beta_\star} + \frac{H_n}{n^2} \right).$$

From (49) and (50), we obtain, for all $n \geq n_0$,

$$\bar{T}^*(\bar{Q}_{n-1} + L_{n-1})(\mathbf{z}) \leq \bar{Q}_n(\mathbf{z}) + L_n(\mathbf{z}) + b_n. \quad (51)$$

Each term in b_n is nonnegative and summable. The sequence (a_n) is summable by Lemma D.6. For the second term, $\xi_\star = \beta_\star/(1 + \beta_\star)$ gives

$$\theta_n^{1+\beta_\star} = \frac{(\ln n)^{1+\beta_\star}}{n^{(1+\beta_\star)\{1-\xi_\star(1-\gamma)\}}} = \frac{(\ln n)^{1+\beta_\star}}{n^{1+\beta_\star\gamma}}.$$

$\sum_{n \geq n_0} \theta_n^{1+\beta_\star}$ is finite because $\beta_\star\gamma > 0$. Finally, $H_n/n^2 \leq (1 + \ln n)/n^2$ is also summable. Hence $\sum_{n \geq n_0} b_n < \infty$. Define

$$\zeta_n = \begin{cases} 0, & n < n_0 \\ \sum_{j=n_0}^n b_j, & n \geq n_0. \end{cases}$$

Then $\zeta_n - \zeta_{n-1} = b_n \geq 0$ for every $n \geq n_0$, and $\sup_n \zeta_n \leq \sum_{j \geq n_0} b_j < \infty$. By the monotonicity and translation invariance properties of the Bellman operator $\bar{T}^*$, we have, for all $n \geq n_0$ and

$$\mathbf{z}/n \in \text{int } \mathcal{K}_{\mathcal{B}}^1,$$

$$\bar{T}^* W_{n-1}^{\text{loc}}(\mathbf{z}) = \bar{T}^*(\bar{Q}_{n-1} + L_{n-1})(\mathbf{z}) + \zeta_{n-1} \leq \bar{Q}_n(\mathbf{z}) + L_n(\mathbf{z}) + b_n + \zeta_{n-1} = W_n^{\text{loc}}(\mathbf{z}),$$

where the inequality follows from (51). This proves the local supersolution inequality in Lemma F.3.

F.3 Global Patching and Proof of Proposition 6.3

We now combine the local supersolution $(W_n^{\text{loc}}(\mathbf{z}))$ with the global supersolution (G_n) by taking a pointwise minimum. The cutoff term L_n in $(W_n^{\text{loc}}(\mathbf{z}))$ (see Lemma F.3) ensures that $(W_n^{\text{loc}}(\mathbf{z}))$ is smaller, and thus can be selected by the minimum, only inside the region $\text{int } \mathcal{K}_{\mathcal{B}}^1$, where its Bellman inequality holds. This ensures that the minimum of (G_n) and (W_n^{loc}) is also a global supersolution. Lemma F.4 formalizes this argument.

Lemma F.4 (Local-to-Global Patching). *Let n_0 denote the integer specified in Lemma F.3 and define*

$$\bar{W}_n(\mathbf{z}) \triangleq \min \{G_n(\mathbf{z}), W_n^{\text{loc}}(\mathbf{z})\}.$$

For every $n \geq n_0$, the local branch lies above the global branch outside the region $\text{int } \mathcal{K}_{\mathcal{B}}^1$, i.e.,

$$W_n^{\text{loc}}(\mathbf{z}) \geq G_n(\mathbf{z}) \quad \text{whenever } \frac{\mathbf{z}}{n} \notin \text{int } \mathcal{K}_{\mathcal{B}}^1.$$

Consequently, $\bar{W}_n(\mathbf{z})$ is a global supersolution from horizon n_0 onward:

$$\bar{T}^* \bar{W}_{n-1}(\mathbf{z}) \leq \bar{W}_n(\mathbf{z}) \quad n \geq n_0 + 1$$

for every reduced state $\mathbf{z}$.

Proof. We first show that the local branch $(W_n^{\text{loc}}(\mathbf{z}))$ is larger, and therefore will never be selected, outside the region $\text{int } \mathcal{K}_{\mathcal{B}}^1$. Indeed, if $\mathbf{z}/n \notin \text{int } \mathcal{K}_{\mathcal{B}}^1$, then $\chi(\mathbf{z}/n) = 1$, and

$$W_n^{\text{loc}}(\mathbf{z}) - G_n(\mathbf{z}) = -A_n(\mathbf{z}/n) + M\mathbf{H}_n + \zeta_n \geq -C_A\mathbf{H}_n + M\mathbf{H}_n \geq 0,$$

where the first inequality follows from the global bound $A_n \leq C_A\mathbf{H}_n$ and the second inequality uses $M \geq C_A$. Therefore, $W_n^{\text{loc}}(\mathbf{z}) < G_n(\mathbf{z})$ implies $\mathbf{z}/n \in \text{int } \mathcal{K}_{\mathcal{B}}^1$.

We next show that $(\bar{W}_n)$ is a global supersolution from horizon n_0 onward. Let $D = \text{int } \mathcal{K}_{\mathcal{B}}^1$. From part 2 of Lemma F.2, for every $n \geq n_0 + 1$,

$$\bar{T}^* G_{n-1}(\mathbf{z}) \leq G_n(\mathbf{z}) \quad \text{for every } \mathbf{z}.$$

Moreover, by Lemma F.3, for every $n \geq n_0 + 1$,

$$\bar{T}^* W_{n-1}^{\text{loc}}(\mathbf{z}) \leq W_n^{\text{loc}}(\mathbf{z}) \quad \text{whenever } \frac{\mathbf{z}}{n} \in D.$$

Fix $n \geq n_0 + 1$ and a reduced state $\mathbf{z}$. We distinguish two cases according to the branch selected at the current state.

If $\bar{W}_n(\mathbf{z}) = G_n(\mathbf{z})$, by the monotonicity of the Bellman operator $\bar{T}^*$,

$$\bar{T}^* \bar{W}_{n-1}(\mathbf{z}) \leq \bar{T}^* G_{n-1}(\mathbf{z}) \leq G_n(\mathbf{z}) = \bar{W}_n(\mathbf{z}).$$

If $\bar{W}_n(\mathbf{z}) = W_n^{\text{loc}}(\mathbf{z}) < G_n(\mathbf{z})$, then $\mathbf{z}/n \in D$; the local supersolution inequality thus implies

$$\bar{T}^* \bar{W}_{n-1}(\mathbf{z}) \leq \bar{T}^* W_{n-1}^{\text{loc}}(\mathbf{z}) \leq W_n^{\text{loc}}(\mathbf{z}) = \bar{W}_n(\mathbf{z}).$$

The two cases prove that $\bar{T}^* \bar{W}_{n-1}(\mathbf{z}) \leq \bar{W}_n(\mathbf{z})$ for every $\mathbf{z}$. $\square$

F.3.1 Proof of Proposition 6.3

Now we are ready to prove Proposition 6.3. From Lemma F.4, $(\bar{W}_n)$ is a global supersolution from horizon n_0 onward. If there exists a constant $C < \infty$ such that $\bar{V}_{n_0}^{\text{OPT}} \leq W_{n_0} + C$, Then the same inequality holds at every later horizon. Indeed, if it holds at horizon $n-1$, then by the monotonicity and translation invariance properties of the Bellman operator $\bar{T}^*$, we have

$$\bar{V}_n^{\text{OPT}} = \bar{T}^* \bar{V}_{n-1}^{\text{OPT}} \leq \bar{T}^* (\bar{W}_{n-1} + C) = \bar{T}^* \bar{W}_{n-1} + C \leq \bar{W}_n + C. \quad (52)$$

Thus, C bounds the difference $\bar{V}_n^{\text{OPT}} - \bar{W}_n$ uniformly from above for all $n \geq n_0$.

We now find such a constant C that bounds the difference $\bar{V}_{n_0}^{\text{OPT}} - W_{n_0}$ at stage n_0 . Note that for any $\mathbf{z} \in \mathbb{R}_+^{|\mathcal{B}|}$,

$$\begin{aligned} \bar{V}_{n_0}^{\text{OPT}}(\mathbf{z}) - \bar{W}_{n_0}(\mathbf{z}) &\leq G_{n_0}(\mathbf{z}) - \bar{W}_{n_0}(\mathbf{z}) = (G_{n_0}(\mathbf{z}) - W_{n_0}^{\text{loc}}(\mathbf{z}))^+ \\ &= (A_{n_0}(\mathbf{z}/n_0) - L_{n_0}(\mathbf{z}) - \zeta_{n_0})^+ \\ &\leq A_{n_0}(\mathbf{z}/n_0) \leq C_A \mathbf{H}_{n_0}. \end{aligned}$$

The first inequality follows from part 3 of Lemma F.2, the first equality from the definition of $\bar{W}_n$, and the second equality from the definition of W_n^{loc} . Thus, set $C_{\text{init}} \triangleq C_A \mathbf{H}_{n_0}$; from (52) we have

$$\bar{V}_n^{\text{OPT}}(\mathbf{z}) \leq \bar{W}_n(\mathbf{z}) + C_{\text{init}}, \quad n \geq n_0. \quad (53)$$

Finally, fix $\mathbf{c}/n \in \mathcal{K}$. Then $\mathbf{c}_{\mathcal{B}}/n \in \mathcal{K}_{\mathcal{B}}$, and $\chi(\mathbf{c}_{\mathcal{B}}/n) = 0$ by the definition of χ . Define

$$C_{\text{OPT}} \triangleq C_{\text{init}} + \sup_{j \geq n_0} \zeta_j < \infty.$$

We have

$$\bar{V}_n^{\text{OPT}}(\mathbf{c}_{\mathcal{B}}) \leq W_n^{\text{loc}}(\mathbf{c}_{\mathcal{B}}) + C_{\text{init}} = \bar{Q}_n(\mathbf{c}_{\mathcal{B}}) + \zeta_n + C_{\text{init}} \leq Q_n(\mathbf{c}) + C_{\text{OPT}}.$$

Here, the first inequality follows from (53) and $\bar{W}_n \leq W_n^{\text{loc}}$, the equality follows from the definition of W_n^{loc} and the fact that $L_n(\mathbf{c}_{\mathcal{B}}) = M \mathbf{H}_n \chi(\mathbf{c}_{\mathcal{B}}/n) = 0$, and the second inequality follows from the definition of C_{OPT} and (45).

Combining this bound with Lemma F.1 yields

$$V_n^{\text{OPT}}(\mathbf{c}) \leq \bar{V}_n^{\text{OPT}}(\mathbf{c}_{\mathcal{B}}) \leq Q_n(\mathbf{c}) + C_{\text{OPT}}.$$

This proves Proposition 6.3 for $n \geq n_0$. For $1 \leq n < n_0$, the convention in Appendix D.1.3 gives $A_n = 0$ and $Q_n = F_n$. Therefore,

$$V_n^{\text{OPT}}(\mathbf{c}) \leq F_n(\mathbf{c}) = Q_n(\mathbf{c}) \leq Q_n(\mathbf{c}) + C_{\text{OPT}}.$$

This completes the proof.